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Review article

Deep learning models for cloud, edge, fog, and IoT computing paradigms: Survey, recent advances, and future directions

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ABSTRACT

In recent times, the machine learning (ML) community has recognized the deep learning (DL) computing model as the Gold Standard. DL has gradually become the most widely used computational approach in the field of machine learning, achieving remarkable results in various complex cognitive tasks that are comparable to, or even surpassing human performance. One of the key benefits of DL is its ability to learn from vast amounts of data. In recent years, the DL field has witnessed rapid expansion and has found successful applications in various conventional areas. Significantly, DL has outperformed established ML techniques in multiple domains, such as cloud computing, robotics, cybersecurity, and several others. Nowadays, cloud computing has become crucial owing to the constant growth of the IoT network. It remains the finest approach for putting sophisticated computational applications into use, stressing the huge data processing. Nevertheless, the cloud falls short because of the crucial limitations of cutting-edge IoT applications that produce enormous amounts of data and necessitate a quick reaction time with increased privacy. The latest trend is to adopt a decentralized distributed architecture and transfer processing and storage resources to the network edge. This eliminates the bottleneck of cloud computing as it places data processing and analytics closer to the consumer. Machine learning (ML) is being increasingly utilized at the network edge to strengthen computer programs, specifically by reducing latency and energy consumption while enhancing resource management and security. To achieve optimal outcomes in terms of efficiency, space, reliability, and safety with minimal power usage, intensive research is needed to develop and apply machine learning algorithms. This comprehensive examination of prevalent computing paradigms underscores recent advancements resulting from the integration of machine learning and emerging computing models, while also addressing the underlying open research issues along with potential future directions. Because it is thought to open up new opportunities for both interdisciplinary research and commercial applications, we present a thorough assessment of the most recent works involving the convergence of deep learning with various computing paradigms, including cloud, fog, edge, and IoT, in this contribution. We also draw attention to the main issues and possible future lines of research. We hope this survey will spur additional study and contributions in this exciting area.

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1. Introduction

In recent decades, computing paradigms have made significant advancements. The most well-known type of computing is perhaps cloud computing, which was intended to use "computing as a utility" to facilitate the explosive proliferation of new Internet services [1]. The Internet of Things (IoT) ultimately made big data feasible by laying the first stone for massive data generation [2]. Before the widespread adoption of the Internet of Things, cloud computing was a popular research area [1]. Cloud data centers are responsible for data analysis, archiving, and decision-making in cloud-based deployments. The growth in data volume and speed has made it difficult to efficiently transfer big data from IoT devices to cloud data centers due to bandwidth limitations. Additionally, this approach falls short of the low latency specifications of time-sensitive applications. It might also pose privacy problems. Since its inception, the definition of IoT has expanded to include digital connectivity of things and devices that may obtain and share information across platforms for additional value [3]. An increase in capacity, a decrease in communication costs, and incredible technological advancements all go hand in hand with the expansion of IoT. IoT mandates information exchange among interdisciplinary platforms in addition to device data management.

A study by the World Economic Forum and Accenture found that IoT could create $14.2 trillion in economic value by 2030, with the largest share of value coming from improvements in operational efficiency [4]. The vast amount of data obtained from multiple smart devices necessitates collaboration to provide value. Collaborative IoT allows for the collaboration of sensors, gateways, and services at different levels across diverse domains and settings, resulting in improved human experiences and enhanced corporate operations. The scale and complexity of the IoT ecosystem are vast, containing a variety of heterogeneous devices that span multiple IoT architecture layers. According to a report by Statista, the number of IoT devices installed worldwide is expected to reach 75.44 billion by 2025, up from 30.73 billion in 2020 [5]. IoT systems require resilient service operability because they are a part of critical infrastructure [6]. IoT applications are employed in a myriad of industries, like healthcare, smart appliances, smart cities, industrial automation, and more. Small, resource-constrained smart objects make up IoT devices, making them incapable of completing sophisticated activities without the need to offload them to distant cloud servers [7]. IoT devices must rely on cloud data centers since they have limited storage and computing capacity [8]. As a result, there will be an increase in latency, and IoT devices will be unable to manage time-sensitive real-time applications due to inconsistent internet connectivity. Consequently, the IoT revolution has led to recent research focusing on decentralized approaches.

The concept of edge computing was developed to offer cloud computing capabilities at the network edge, to address challenges that cannot be solved by cloud computing alone, such as latency, bandwidth, and connectivity issues [9]. As a result, multiple edge computing solutions have been suggested, including Mobile Cloud Computing (MCC) and Mobile Edge Computing (MEC) [10,11].

Fog computing is a highly advanced edge computing technology that has emerged as one of the most promising solutions. A report by Grand View Research found that the global fog computing market size was valued at $16.3 billion in 2019 and is expected to grow at a CAGR of 55.4% from 2020 to 2027 [12]. By

gradually moving resources from the cloud to mobile devices, fog computing aims to act as an all-encompassing framework [13]. Because it actively works to integrate the cloud and IoT, it is not just a cloud extension. IoT devices also require the need for sustainable/green computing, which helps with energy conservation. Energy-constrained IoT devices necessitate the development of energy-aware solutions in the future [14]. The carbon footprint must be reduced in tandem with technological advancement to prevent environmental degradation and global warming [15].

Research on edge computing is still in its nascent stages, and new perspectives on these paradigms that are often found in the literature require thorough investigation [16]. The Global Edge Computing Market by Component, Application Area, and Region, 2020–2027 report by Research and Markets predicts that the global edge computing market will grow at a CAGR of 26.4% between 2020 and 2027, with the market size expected to reach $43.4 billion by 2027 [17]. Numerous new applications are currently being developed for various mobile and wireless platforms, such as "smartphones" [18], "sensor networks" [19], "unmanned aerial vehicles (UAV)" [20], "cognitive smart systems" [21], and others, have benefited from the growth of the Internet of Things (IoT) technology. We may usually use a workflow model, which has five parts: question formulation, data gathering, data processing, visualization, and evaluation [22], to create efficient IoT applications. Traditional techniques typically involve combining technical expertise with machine learning to solve classification or regression problems, such as "predicting traffic using support vector machine (SVM) [23]", "tracking cars using Kalman filter and ridge regression [24]", and "estimating delivery time using Gaussian mixture models". However, as we enter the "Big Data" era, these traditional methodologies become insufficient to handle the vast, explosive, and irregular data collected from a diverse array of widespread and heterogeneous IoT data sources. The effectiveness of conventional systems largely relies on the user's preliminary knowledge of particular disciplines. The majority of learning methods used in these systems often make use of shallow architectures, which have relatively little modeling and representational capacity.

Researchers have been able to train machine learning models with much better potency thanks to the quick growth of deep learning and hardware development. This has significant advantages for a variety of applications, such as "crowdsourced delivery [25], network caching [26], energy management [27], and edge computing [28]". The benefits of applying deep learning in various applications of cloud paradigm applications are illustrated below, where the merits of deep learning over conventional ML techniques are underlined -

- Deep learning (DL) involves more complicated neural network topologies, which may characterize more challenging situations and derive more convoluted hidden properties (such as temporal and/or spatial connections). Deep learning possesses more potent abilities in extrapolating the complex interaction of large amounts of raw datasets in diverse computer applications than those old basic learning approaches.
- The vast and valuable data resource can be fully leveraged through deep learning. When working with massive amounts of data, simple learning models can easily overfit, whereas deep learning-based models generally exhibit superior performance. The complexity and specific structures of learning models, such as convolutional architectures, often dictate their ability to handle data.
- DL is an end-to-end learning technique that can autonomously learn to extract meaningful features from unprocessed data without requiring manual and time-consuming feature engineering.

The following are some highlights of this survey's contribution:

- A thorough explanation of different computing models is provided, focusing on cloud/fog/edge computing, and how these paradigms relate to others of a similar nature, like mist, cloudlet, MEC, etc.
- The reasons behind the development of edge/fog computing and associated models are explained in great depth.
- The study compares the cloud, edge, and fog computing paradigms and highlights the importance of ML convergence for fog/edge computing.
- The difficulties and potential possibilities for computing paradigms research are compiled.

The structure of this review is presented in Fig. 1.

2. Related surveys

In this study, we evaluated various review/survey papers, and Table 1 gives an overview of these papers.

Despite significant progress in recent years, the domain of incorporating deep learning into various computing paradigm applications is still in its infancy. There are a few studies in the literature that examine deep learning applications across the three domains of cloud, fog, and edge. All of these survey reports, however, only cover a small portion of the overall subject matter. There is currently a lack of a survey that thoroughly examines deep learning for many computing paradigm applications. As a result, we think now is the ideal time to assess current studies and propose fresh avenues for research. This is expanded upon in subsequent portions of the manuscript.

3. Deep learning

A subdivision of ML known as DL (Fig. 2) is modeled after how the human brain processes data. Artificial neural networks are commonly used to implement DL. Each layer of an ANN offers a unique input data interpretation. In contrast, classical ML methods require several sequential processes such as "pre-processing, feature extraction, careful feature selection, learning, and classification" to accomplish a classification task. The choice of features has a big impact on how well ML techniques work as well. Inaccurate class differentiation may be caused by incorrect feature selection. In contrast to classical ML techniques, DL enables the automation of feature-set learning for various applications [47,48]. DL enables simultaneous learning and classification (as shown in Fig. 3), which is different from the sequential processes of "pre-processing, feature extraction, cautious feature selection, learning, and classification" required in classical ML techniques.

Deep learning is laborious and time-intensive during the model training phase. It also demands heavy computation resources, which was one of the key challenges in the past. Building deep models rather than using conventional approaches enables the analysis of complex problems and the processing of data. Specialized hardware, such as potent GPUs, has been developed. Due to the increasing volume and complexity of data in the big data era, deep learning has emerged as a promising method for processing and modeling data [49].

Machine learning (ML) has recently been applied to a wide range of applications such as mining texts, spam detecting spam, classifying images, and retrieval of multimedia [50–55]. Deep learning (DL), one of the several ML methods, is frequently used in these applications [56–58]. DL is also known as representation learning (RL). Due to both the unpredictably rising availability of data and the incredible advancements in hardware technologies, such as High-Performance Computing (HPC), new and innovative research in the deep and distributed learning domains continue to be published [59].

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Fig. 1. Organization of the paper.

Although Deep Learning (DL) is attained from traditional neural networks (NN), it performs noticeably better than its forerunners. Moreover, DL uses both transformations and graph technologies at once to construct multi-layer learning models. Newly developed deep learning approaches have shown remarkable success in various applications such as "natural language processing (NLP)", "audio and speech processing", and "visual data processing", among others [60–63].

The efficiency of a machine learning algorithm is highly dependent on the quality of the data it is fed. A good data representation has been shown to perform significantly better than a subpar one. Thus, feature engineering has been a prominent research area in ML for many years, and numerous research papers have been based on this concept. This method attempts to create features out of raw data. Also, it is very industry-specific and frequently necessitates a significant amount of human effort.

Table 1
Comparative analysis of the survey papers considered in this research study.

Source	Year	Proposal	Cloud computing	Fog computing	Edge computing
[29]	2016	The study looks at the advancements of IoT, its significance in contemporary society, as well as any challenges that may arise.	✓	✗	✓
[30]	2017	The review describes the architecture and attributes of fog computing and compares them to other computing models. It also enumerates the main tools that support application and deployment.	✓	✓	✓
[31]	2017	The paper identifies areas in research that require further attention and suggests a classification system based on a detailed examination of the attributes of fog computing and the challenges associated with its structure, service, and security.	✓	✓	✓
[32]	2017	The article provides an in-depth analysis of modern IoT technologies, system architecture, security issues, and challenges that arise during the practical implementation of integrating IoT with fog/edge computing.	✗	✓	✓
[33]	2017	An extensive breakdown of modern MEC is provided, with comparisons to MCC and resource management from a communication standpoint.	✓	✗	✓
[34]	2018	The survey article gives an outline of fog computing, including its architecture and related technologies, while also examining the challenges associated with integrating fog computing with IoT systems in real-world scenarios.	✓	✓	✓
[35]	2018	Fog computing architectures, algorithms, related ideas, and their differences are exhaustively reviewed, along with obstacles and future research objectives.	✗	✓	✗
[36]	2018	The article delves into the fundamentals of fog computing and provides an overview of its architecture. It also highlights the importance of service and resource allocation in addressing issues related to latency, bandwidth, and energy consumption.	✗	✓	✗
[37]	2018	The comprehensive review investigates topics including IoT, cloud computing, mobile computing, and similar ideas to clarify emerging paradigms and technology.	✓	✓	✓
[38]	2018	This paper examines the current status of fog computing, including its attributes, structure, and advantages, and argues that fog computing is an essential element of the Internet of Things.	✗	✓	✗
[39]	2018	The paper describes edge computing technologies as well as the difficulties and worries that come with distributed settings when moving services from cloud-based platforms that are centralized to edge-based platforms that are decentralized.	✗	✓	✓
[2]	2019	This comprehensive survey explores fog computing in comparison to other related computing models and provides a classification of research topics and underlying problems, as well as identifying future trends in the field.	✓	✓	✓
[40]	2019	This review article explores recent developments in machine learning (ML) approaches and highlights the precision, resource allocation, and security aspects of fog computing, as well as its significance in edge computing.	✗	✓	✓
[11]	2019	This work provides an introduction and understanding of cloud/edge/fog computing models for beginners by discussing the fundamental features and design of each model. Additionally, the article emphasizes the significance of fog computing as it acts as a bridge that links cloud computing, edge computing, and IoT.	✓	✓	✓
[41]	2019	The paper emphasizes the significant prerequisites, and constraints, and highlights unresolved difficulties for cloud and cutting-edge edge computing.	✓	✓	✓
[42]	2020	The article analyzes edge computing research and provides a summary of the main ideas, technology, architecture, privacy, and security.	✓	✗	✓
[43]	2020	By giving a taxonomy and analysis from the perspective of fog's architectural design, the review covers current computing paradigms and places special emphasis on fog computing research topics.	✓	✓	✓
[44]	2020	This research examines cutting-edge fog computing systems, provides tips for creating and operating robust fog systems, and highlights current research concerns and emerging trends.	✗	✓	✗
[45]	2020	In this study, a taxonomy of real-world performance indicators that can be used to evaluate cloud, fog, and edge computing paradigms is presented.	✓	✓	✓
[46]	2020	This paper delves into the principles, organization, protocols, tools, and the IoT-fog Cloud of Things to explore the IoT-fog-cloud ecosystems. Additionally, a taxonomy of emerging issues is provided. The paper concludes that in fog computing scenarios that involve latency-sensitive and resource-constrained systems, ML and AI will be valuable tools.	✓	✓	✓

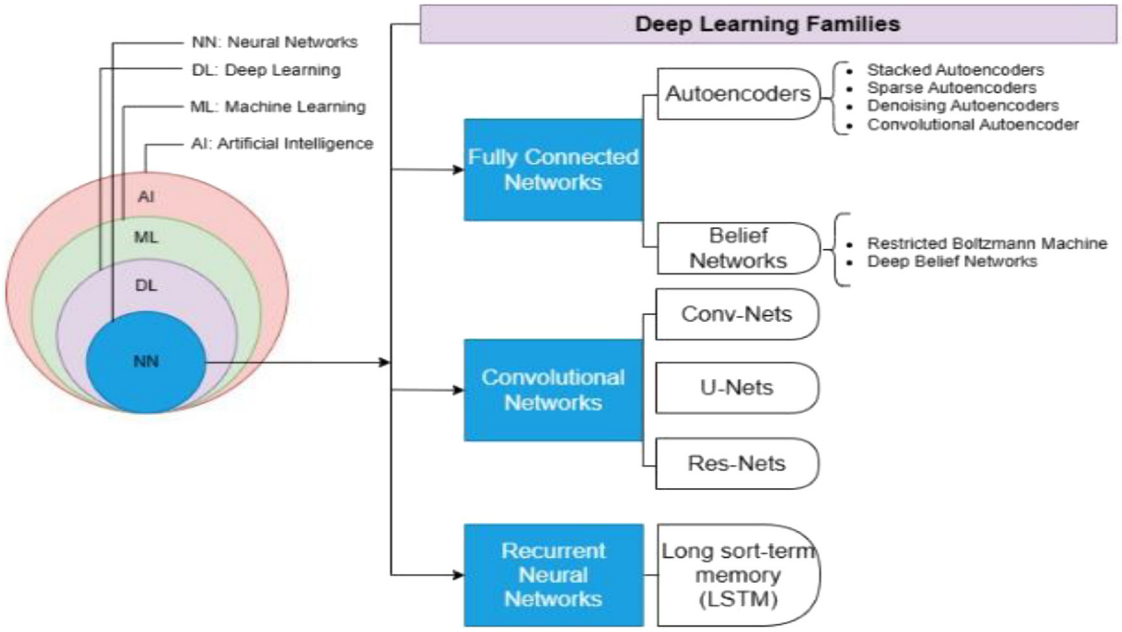


Fig. 2. Families of deep learning.

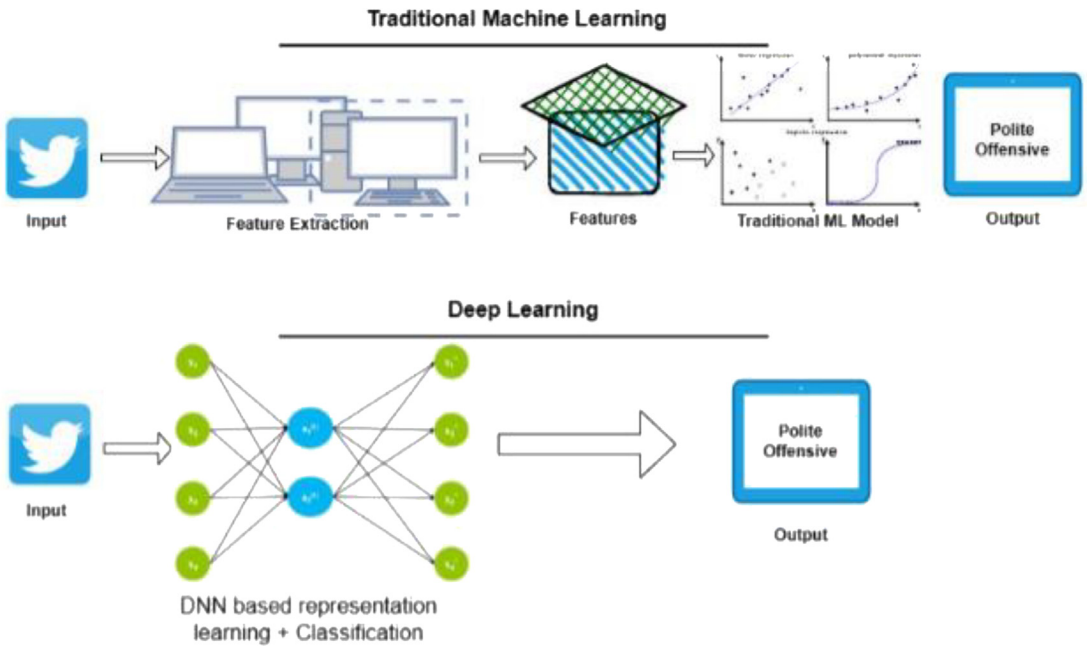


Fig. 3. Deep learning vs. traditional machine learning.

In the domain of computer vision, for instance, many feature types such as the "bag of words (BoW) [64]", "scale-invariant feature transform (SIFT)", and "histogram of oriented gradients (HOG) [65]" have been introduced and compared. After a unique feature is implemented and proven to work successfully, it develops into a new area of research that is investigated for several years.

DL algorithms are designed to automatically extract features, which eliminates the need for manual feature engineering and reduces the required domain expertise [66]. Deep learning algorithms utilize a data representation architecture that consists of multiple layers, where the first layers extract low-level features, and subsequent layers extract higher-level features. This methodology encourages researchers to extract the most distinguishing features with minimal effort. It should be noted that this kind of architecture, which mimics the action that takes place in central sensory regions of the human brain, was originally inspired by artificial intelligence (AI). The human brain can automatically extract data representations from various scenes. In particular, the acquired scene information serves as the input and the classified objects are the process output [67]. This method replicates how the human brain functions. Hence, it highlights the primary advantage of DL.

The success of Deep Learning (DL) in Machine Learning (ML) has made it a popular research area. This work provides an

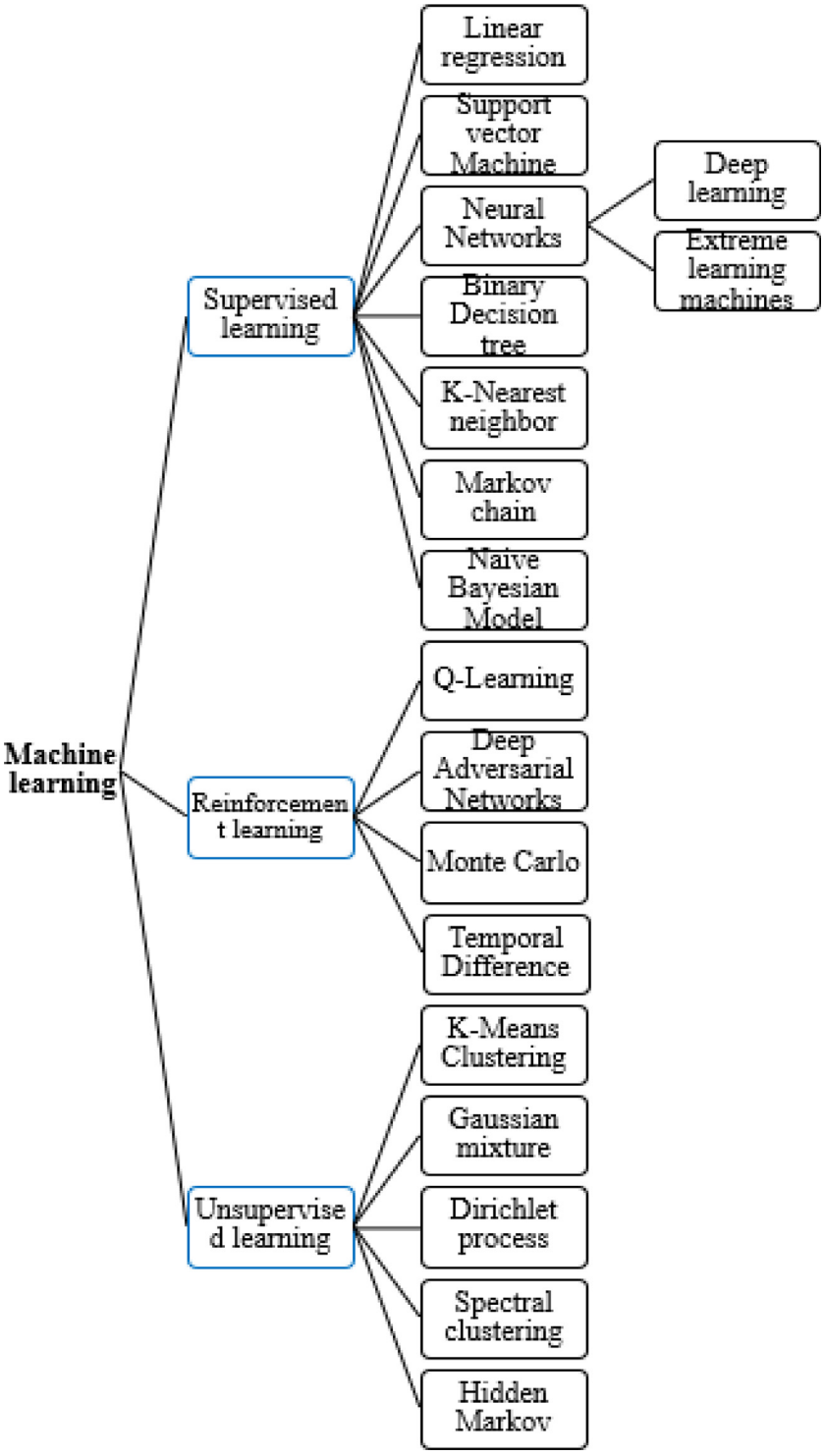


Fig. 4. Classification of machine learning.

overview of deep learning (DL), which is currently an extensively researched trend in the field of machine learning (ML). The article covers various aspects of DL, including its important principles, structures, constraints, applications, computational techniques, and progression. One of the most well-known and widely applied DL networks is the convolutional neural network (CNN) [68,69]. CNN is to blame for the current popularity of DL. The primary benefit of CNN over its forerunners is that it naturally recognizes the important aspects without any human supervision, making it the most popular.

3.1. An insight into deep learning focusing on machine learning

Based on the learning methodology employed, machine learning can be classified into one of three categories: in the system's training as shown in Fig. 4. There are three types of learning methods: supervised, unsupervised, and reinforcement learning [70] (Durodola, 2022). In the supervised learning method, the system is first put up as a model with the ability to learn but is underdeveloped or 'ignorant' until it is taught. The training procedure entails giving the model a collection of input and

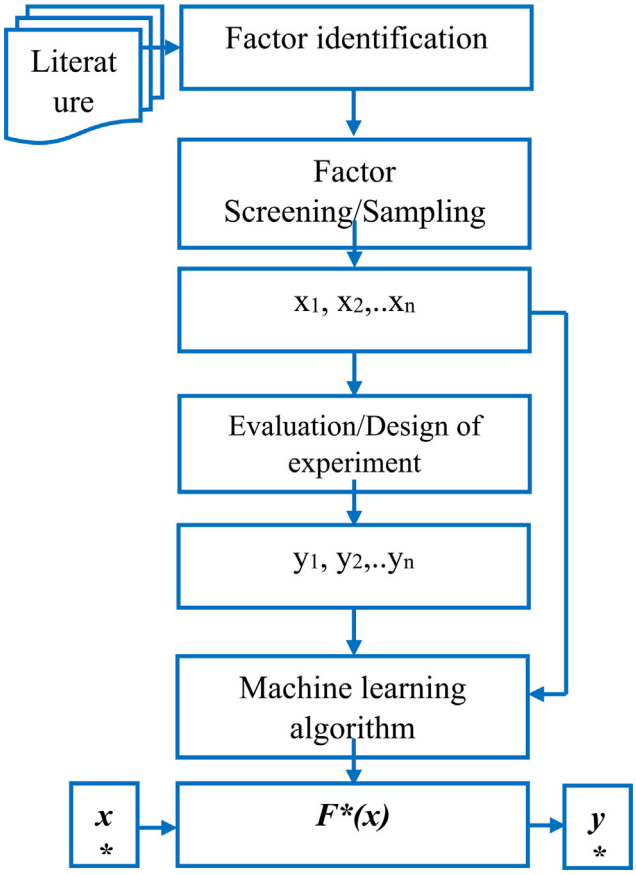


Fig. 5. System diagram of supervised learning framework.

output data, such as *x* and y, for which an association or link must be established. During training, supervision refers to the presentation of the proper output data that matches the input data to the model. As previously said, the training phase entails repeated or adaptive adjustments to the model until it has learned to correctly associate or build a link between the input and output data to a fair degree of accuracy. Once the model has been trained, it is evaluated using a set of input and output data that it has never encountered before. The model's performance is measured by how well or correctly it predicts the outputs that correspond to the test inputs. The degree of nonlinearity in the interactions between the input and output data determines the accuracy that can be achieved.

The field of supervised learning can be isolated further into linear regression, support vector machines, neural networks, decision trees, nearest neighbor, Markov chain, and naive Bayesian models. In general, the supervised machine learning structure comprises factor identification through literature search [71,72] (Le, 2021; Ghanem & Elgazzar, 2021), experience, experiment, factor screening/sampling to produce arbitrary input vectors x, then a framework that produces yields *y* for a given input vector *x* and finally, a learning machine that forecast obscure qualities y^* at test point x^* utilizing training data x, y. Fig. 5 outlines the supervised learning structure.

Aizenberg et al. [73] were the first to coin the phrase "deep learning" in 2000. Deep learning (DL) has gotten a lot of attention in the recent decade and has become a prominent technology in the artificial intelligence sector. Machine learning has a subclass called DL [74]. DL is a type of machine learning that is more advanced than traditional machine learning. A Deep Neural Network (DNN) is made up of multilayers and deeper layers [75,76]. A neural network with more inputs and more complicated neural layers is known as DL. Based on the literature review, DL can be classified into three types i.e., supervised (discriminative), unsupervised (generative), and hybrid. Supervised learning is used as a feature of extraction. These features will be made available to simple ML methods for tasks like classification and detection. Unsupervised learning is used for the extraction of the complete mode's best features [77,78]. To enhance the training of deep neural networks, the hybrid uses generative feature learning models [79,80]. Deep learning's fundamental job in an application is to make decisions, categorizations, and forecasts. Learning features and automated extracting features are important aspects of DL [81]. DL has been employed in various applications., including "speech recognition, natural language processing, computer vision, image processing, intrusion detection, and more [82]". A taxonomy of DL is shown in Fig. 7.

Some optimization techniques that can be employed for deep learning models to enable real-time inference in resource -constrained devices are discussed in Fig. 6. Weight pruning is the process of removing weights from a deep learning model that are less than a certain threshold value. Because the pruned weights are set to zero, the pruned deep learning model has fewer valid weights. The number of computations necessary during the inference is reduced as a result. Additionally, after reducing the weights, compression techniques can be used to reduce the amount of storage required.

The successful weight compression strategies are compressed sparse raw (CSR) and compressed sparse column (CSC). These strategies make use of the sparse nature of the data and can save 20%–30% of memory space [83]. Filter decomposition can be used to reduce the computational complexity and memory requirements of CNN. Filter deconstruction is the process of replacing a bigger filter with a succession of smaller filters and applying them in sequential order [84]. Convolutions based on the Fast Fourier Transform (FFT) can minimize the computational complexity of convolutional layers, especially when the filter sizes are large [85]. The goal of knowledge distillation is to convert the complex model's knowledge into a basic mode [86]. The projected score from the complicated models can be used to train the simple shallow models, according to the knowledge distillation technique. Simple models can achieve more accuracy in this training than training with the dataset's real class labels. Quantization techniques can be used on the parameters of deep learning models to reduce computational complexity and storage requirements. Quantization is the conversion of data into a smaller set of intervals [87].

We will conclude this section with a quote from the AI pioneer Geofrey Hinton: "Deep learning is going to be able to do everything".

In what instances should deep learning be used?

Machine intelligence has proven to be useful in various contexts and has even outperformed human experts in some cases [88–91]. Therefore, DL could potentially provide solutions to the following problems:

- Instances in which there are no human experts on hand.
- Instances in which people are unable to justify the decisions they took based on their knowledge such as speech recognition, medical judgments, and language comprehension.
- Instances when the solution to the issue changes over time such as prediction of prices, stock preferences, forecasting of the weather, and tracking.
- Instances when solutions must be modified in light of particular circumstances (personalization, biometrics).

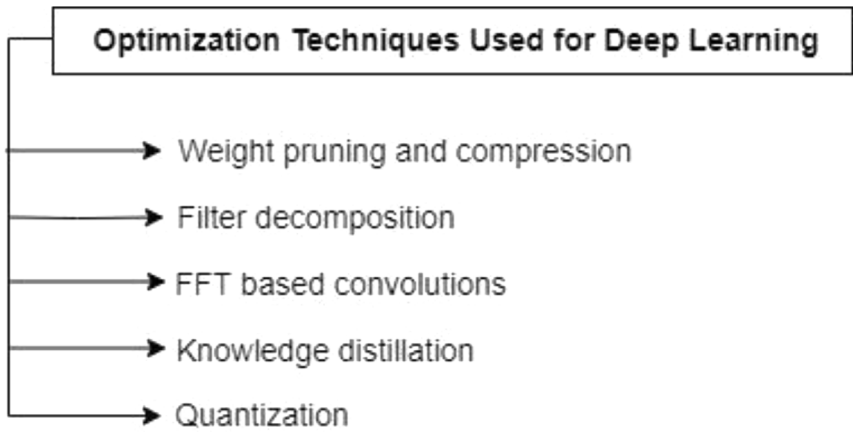


Fig. 6. Optimization techniques for DL.

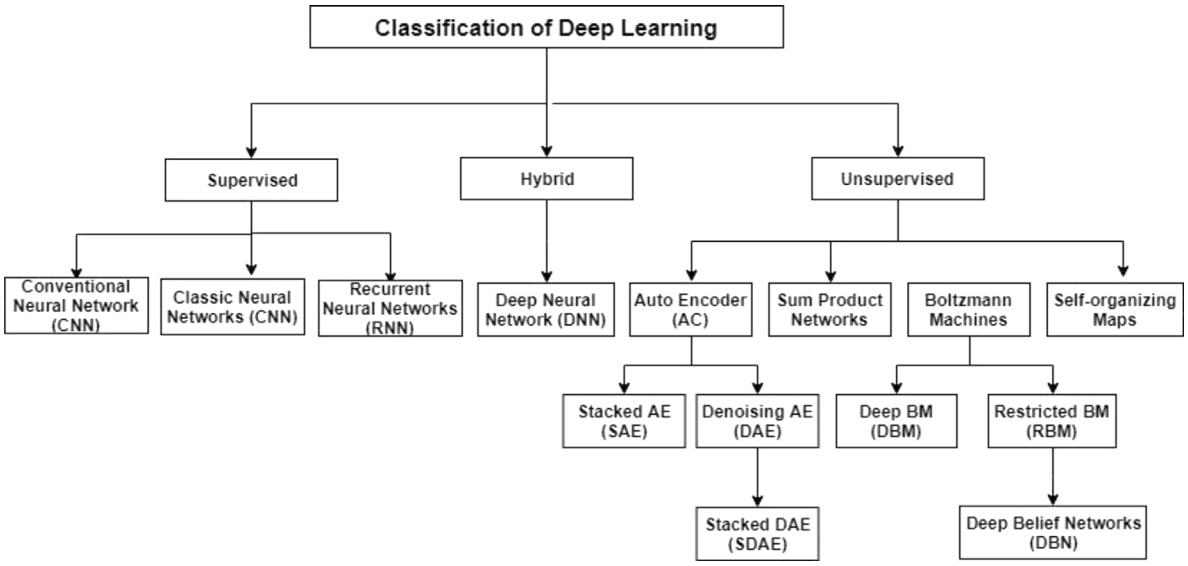


Fig. 7. A taxonomy of general DL.

- Instances in which the problem's size is extraordinarily enormous and goes beyond what our limited thinking skills can handle such as emotion recognition, YouTube ad recommendation, and ranking of websites.

Why should we use deep learning?

This can be determined by several performance characteristics, including:

1. **Universal Learning Approach:** DL is often referred to as a universal learning approach due to its ability to be applied across a wide range of application fields.
2. **Durability:** DL approaches are generally durable as they do not rely on precisely constructed features. Instead, they automatically learn the optimal features for the given task, while remaining robust to typical variations in the input data [92].
3. **Generalization:** DL techniques such as transfer learning (TL) can be used for other applications or data types, particularly in situations where there is limited data available. Further details on TL will be discussed in the following section.
4. **Scalability:** DL has excellent scalability, as demonstrated by "Microsoft's ResNet" [93], which has 1202 layers and is widely used in supercomputing. Additionally, "the Lawrence Livermore National Laboratory (LLNL)", a major organization dedicated to developing network frameworks that can support thousands of nodes, has followed a similar approach [94].

3.2. Categorization of deep learning techniques

There are three primary types of DL approaches: "unsupervised, semi-supervised (partially supervised), and supervised". Deep reinforcement learning (DRL), which is frequently categorized as a partially supervised (and occasionally an unsupervised) learning technique, is another learning strategy.

3.2.1. Deep supervised learning

This approach requires labeled data to work with. The environment contains a set of inputs and corresponding outputs $(a_t, b_t) \sim \rho$ when contemplating such a technique. If the input is a_t, the smart agent will assume $b^t = f(a_t)$ and return i (b^t, b_t) as a loss value. Supervised DL techniques require labeled data, where the network learns from inputs and their corresponding outputs. The agent adjusts its network parameters iteratively to improve the accuracy of the predicted outputs. Common examples of supervised DL techniques are "deep neural networks (DNNs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs)", which incorporate approaches such as "gated recurrent units (GRUs) and long–short-term memory (LSTM)". This approach allows for efficient data collection and output generation based on existing knowledge. The downside of this strategy is that when the training set lacks samples that

should be in a class, the decision boundary may be overstrained. In general, this method of learning with good performance is simpler than others.

3.2.2. Deep semi-supervised learning

Partially supervised learning is a technique that utilizes semi-labeled datasets for the learning process. It is often used in combination with other DL techniques such as generative adversarial networks (GANs) and deep reinforcement learning (DRL). RNNs including "gated recurrent units (GRUs) and long–short-term memory (LSTM)" are also commonly used for partially supervised learning. The method's ability to require less labeled data is one of its advantages. Nevertheless, one drawback of this method is that it could result in incorrect choices as a result of irrelevant input features found in training data. Due to the difficulties of collecting a large labeled corpus of text documents, semi-supervised learning is frequently utilized for this task. The text document classifier is a well-known example of a semi-supervised learning application.

3.2.3. Deep unsupervised learning

Deep unsupervised learning is a learning method that does not rely on labeled data for training. Instead, it uses algorithms to identify significant features or inner representations in the input data, which can help uncover previously unknown relationships or structures. There are various unsupervised learning techniques, including "generative networks, dimensionality reduction, and clustering", that are commonly used to achieve this. Among the recently developed members of the deep learning family, "Restricted Boltzmann machines, auto-encoders, and GANs" have shown promising results in solving nonlinear dimensionality reduction and clustering problems. Recurrent neural networks (RNNs), which utilize techniques such as GRUs and LSTMs, are also widely used in unsupervised learning applications. However, unsupervised learning has some limitations, including its inability to provide accurate data sorting information and its high computational complexity. Clustering is one of the most commonly used unsupervised learning techniques [95].

3.2.4. Deep reinforcement learning

Reinforcement Learning (RL) involves interaction with the environment, as opposed to supervised learning which relies on presented sample data. In 2013, Google DeepMind created this technique, which led to the development of many enhanced strategies based on reinforcement learning [96]. For example, if the input environment samples: $a_t \sim \rho$ agent predicts: $b^t = f(a_t)$ and the received cost of the agent is: $c_t \sim \rho\left(c_t \middle| a_t, b^t\right)$, the unknown probability distribution is shown as ρ, the environment then asks the agent a question. It returns a noisy score as a result. Semi-supervised learning is a term used to describe this process. Several supervised and unsupervised techniques have been developed based on this principle. Because the reinforcement learning technique does not have a straightforward loss function, it is far more difficult to perform than typical supervised techniques. [58,97] highlight two key differences between supervised and reinforcement learning: Firstly, reinforcement learning does not have complete access to the function and requires optimization through interaction with the environment. Secondly, the state being interacted with is dependent on the environment, where the input a_t is based on preceding actions. The type of reinforcement learning required to solve a given task depends on the complexity of the problem at hand. Deep reinforcement learning (DRL) is well-suited for problems with multiple parameters, while derivative-free reinforcement learning is suitable for situations with few parameters. RL has various applications, including business strategy planning and robots for industrial automation. RL is a subfield of machine learning that focuses on executing appropriate actions to maximize reward in a given circumstance. It is all about finding out how to behave in a specific scenario such that the idea of cumulative reward is maximized. RL differs from supervised learning in that it does not need the display of labeled input/output pairs or the explicit correction of sub-optimal behavior. Instead, striking a balance between exploration and exploitation should be the top objective. RL varies from supervised learning in that supervised learning contains an answer key, allowing the model to be trained with the right answer, but reinforcement learning does not, and instead depends on the reinforcement agent to select how to finish the job. In the absence of a training dataset, it is bound to learn from its experience.

The fundamental disadvantage of RL is that parameters can affect learning speed. The following are the key reasons for using RL:

- It aids you in determining which activity yields the highest benefit over time.
- It aids you in determining which situations necessitate action.
- It also allows it to choose the optimum strategy for achieving high rewards.
- RL also includes a reward function for the learning agent.

RL is ineffective in a variety of situations, including:

- When there is enough data to use supervised learning techniques to tackle the problem.
- When dealing with large environments or complex tasks, it is a computationally intensive technique and can be time-consuming.

NNs have a lengthy history, and numerous individuals have contributed to their evolution throughout the years. We aimed for an unbiased presentation in the Fig. 8, focusing only on the most notable contributions. Given the recent surge in interest in DL, it is not unexpected that crediting major advancements has been a contentious issue. Based on the literature review [73,98–115], numerous findings corresponding to their accuracy are given in Table 2.

Year-wise analysis of DL publications

From 2007 through June 2021, Fig. 9 shows the number of DL articles from the ScienceDirect database per year. There is a steady increase in the number of publications, which may be described as exponential. Fig. 10 shows the number of DL articles with content type has increased in the Springer database. Fig. 11 depicts Springer's published articles by discipline/subdiscipline. Fig. 12 illustrates the content kinds of published papers in the IEEE digital library database. Fig. 13 depicts the year-by-year distribution of published publications (Conferences/Journals/Books/Magazines/Standards) in the IEEE Xplore database (2007–2021).

3.3. Deep learning architectures by category

The deep learning method is the most significant branch of machine learning. This is a diagram that illustrates the many deep learning categories depending on computer paradigms. As depicted in Fig. 14, deep learning can be classified into several categories, which are further discussed in the following sections.

3.3.1. Convolutional Neural Networks (CNN)

A CNN is a type of feed-forward neural network that consists of convolutional, hidden, pooling, and fully connected layers. Neurons in the network represent the individual elements, and the input can be stored in an array of two or three dimensions.

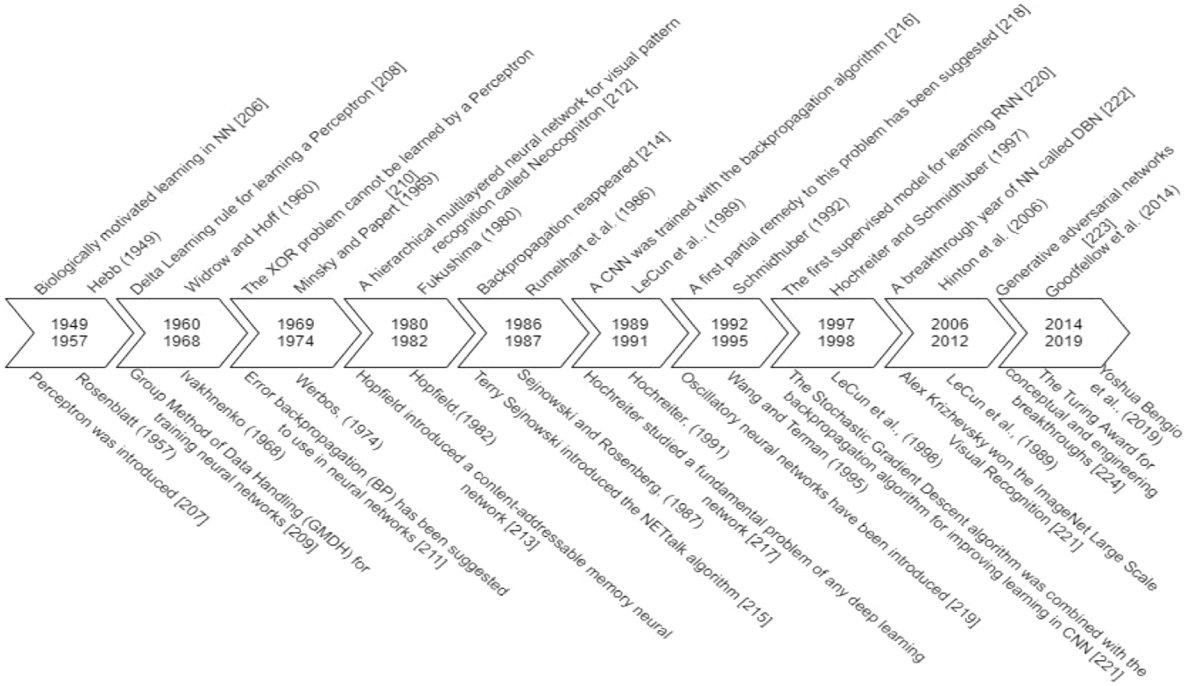


Fig. 8. A timeline of key development of NN.

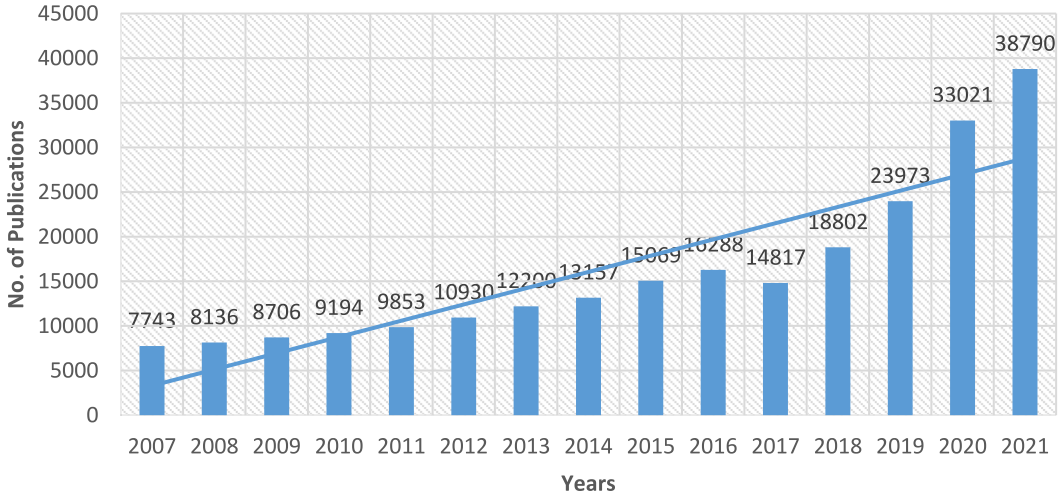


Fig. 9. The number of DL papers is increasing, according to the ScienceDirect database (2007–2021).

The convolutional layer is typically considered the primary unit of the CNN, where the initial input is a convolutional kernel that represents the network's weights, and the receptive field is a smaller window used for convolution [132,133]. The input calculation yields the feature map. Fig. 15 depicts the convolutional neural network's fundamental structure. There are several types of CNN models, including the "Acoustic CNN model", "Single CNN", "Multi-CNN", "Variants of CNN", and "Restricted Weight Sharing CNN".

(a) Single CNN

To detect malicious activity, the deep security approach utilizes a supervised training method with a single CNN [134]. For secure image recognition, a different approach is used which involves multiple locally connected CNN layers instead of weight sharing, and three-dimensional alignments are performed in the ninth layer of the CNN [135]. The training set is enhanced using a web-scale bootstrapping technique. To improve the performance of cybersecurity threat detection, various techniques such as "discriminative feature learning, feature fusion, designing loss functions, novel learning systems, and precise activation" can be employed.

(b) Multi-CNN

Multi-CNN, which employs several CNNs, is used to evaluate deep features. It mostly serves to efficiently combine the characteristics. The many inputs and diverse environmental factors are used to extract the deep characteristics. Each CNN receives additional training from other CNNs, which improves performance [136]. The fundamental function of a multi-CNN is feature extraction from several regions and aspects. As a result, collecting and computing data demands significant effort. With this method, the characteristics of the various malware can be retrieved.

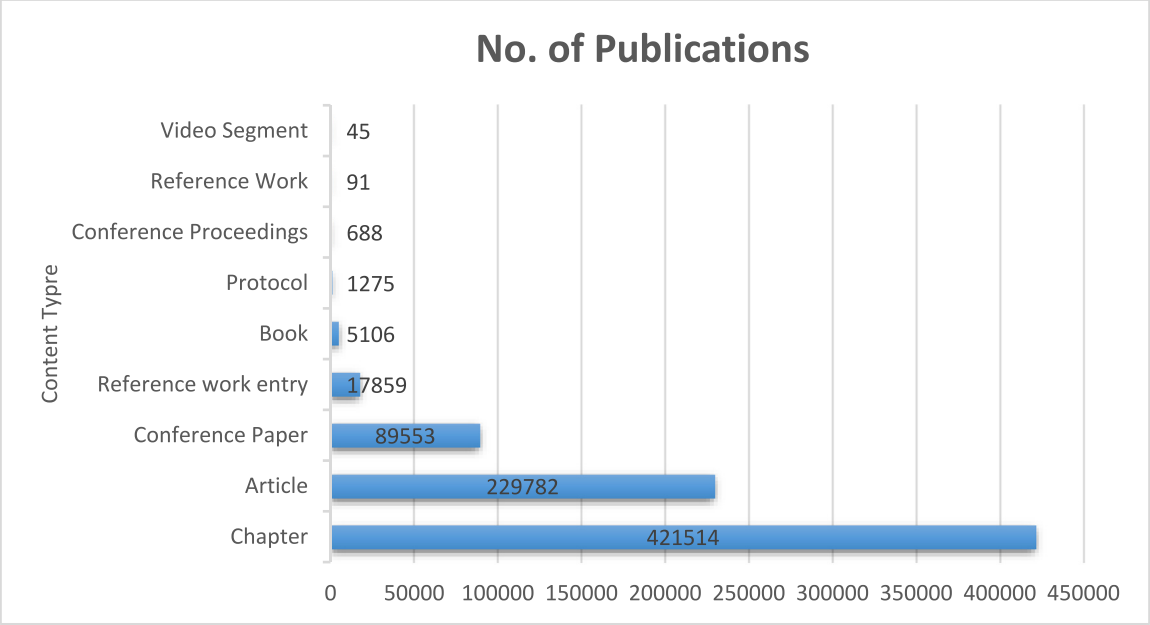


Fig. 10. Springer database shows an increase in the number of DL publications with content type.

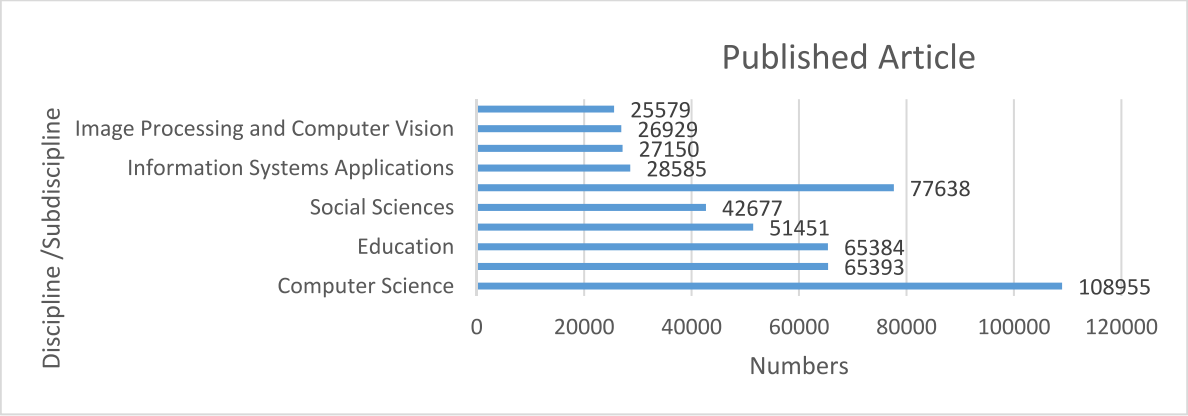


Fig. 11. Published articles in Springer based on discipline/subdiscipline.

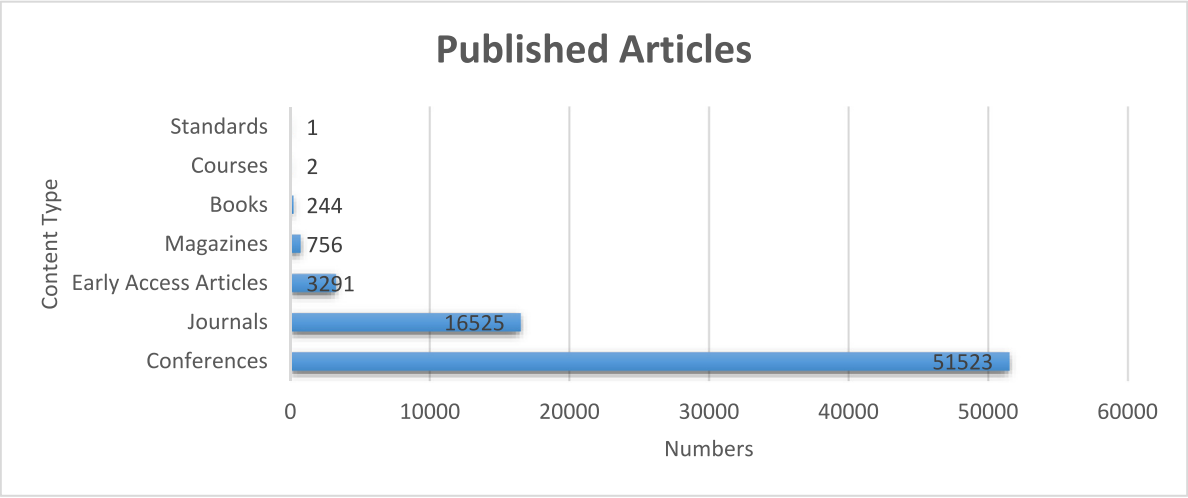


Fig. 12. Published articles in IEEE digital library based on content types.

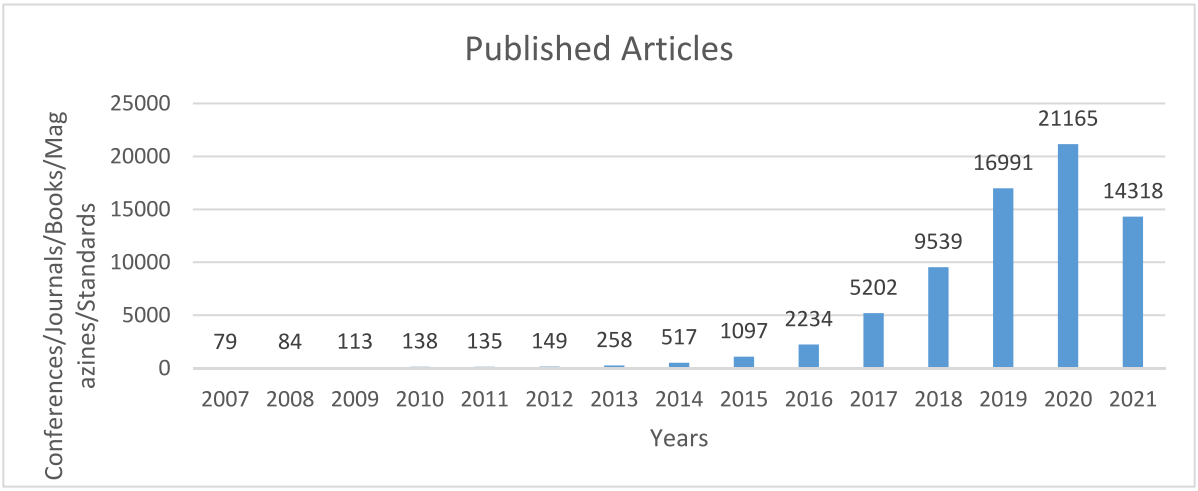


Fig. 13. Published articles (Conferences/Journals/Books/Magazines/Standards) in IEEE digital library (2007–2021).

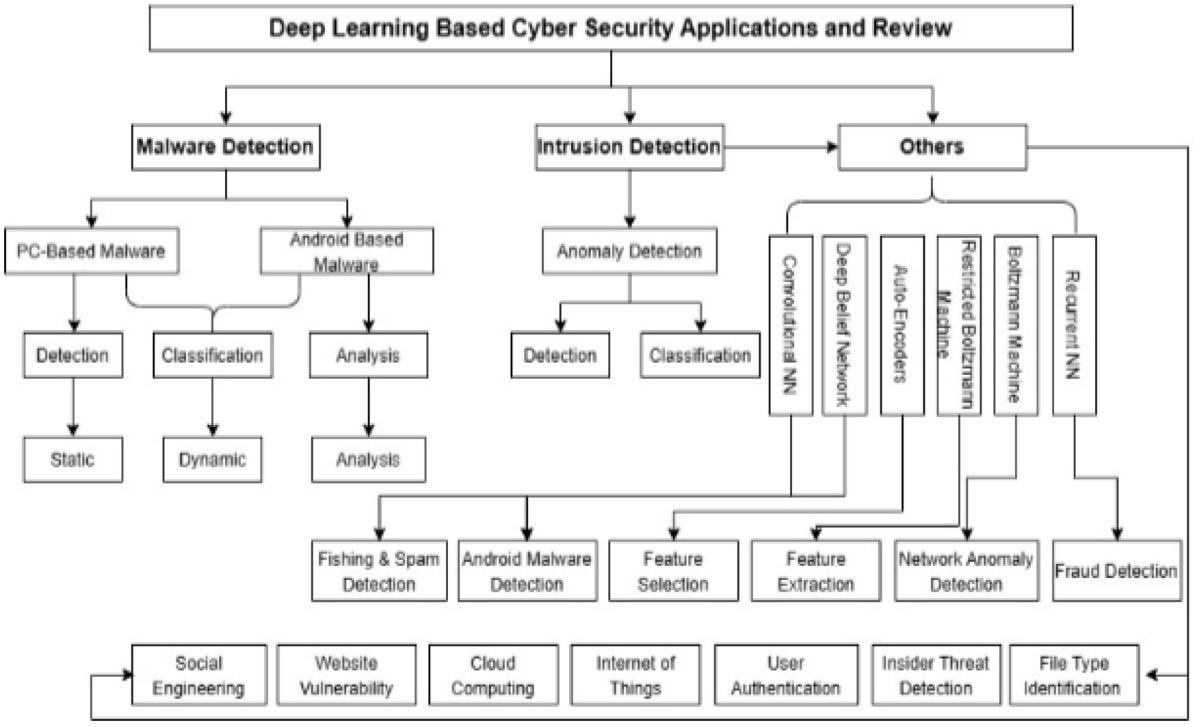


Fig. 14. Deep learning for cyber security.

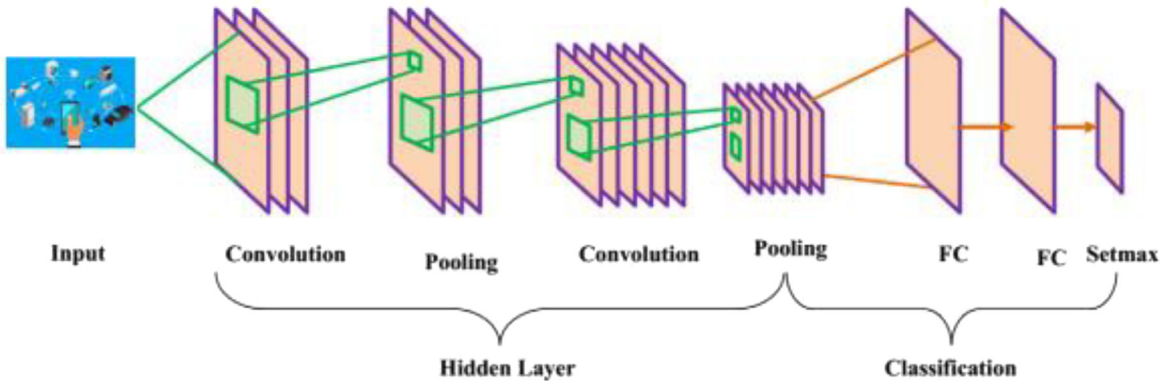


Fig. 15. Structure of convolutional neural network.

Table 2
Important findings of DL based on the accuracy.

References	Important findings	Accuracy
Schwartz F. et al. [98]	To simplify the model and accelerate its development, the number of dimensions was kept as low as possible.	98.2162%
B. Alsughayyir et al. [99]	In [99], AE was given a classification task. The Min–Max normalization is employed during the data preprocessing stage.	91.28%
Farahnakian and Heikkonen [100]	One proposal for feature learning is the DAE, which uses a "softmax classifier" in the last hidden layer.	94.71%
Yeom et al. [101]	A random forest method was utilized in the AE deep learning technique for feature extraction and categorization. By isolating some traits, this strategy decreases time and complexity.	98%
Sara et al. [102]	Long–Short Term Memory (LSTM) was shown utilizing four neural networks. Gates are used to manipulating memory in the cells in this investigation.	85%
Al-Emadi Al. et al. [103]	Deep learning RNN and CNN were used to create an intelligent system for detecting cyber-attacks.	97%
Kasongo and Yanxia Sun [104]	The authors suggested a wireless IDS classifier based on "Deep Long–Short-Term Memory (DLSTM)".	99.51%
Peng. X. et al. [105]	A network IDS based on a DL algorithm was proposed. "Deep Belief Network (DBN)" and "Back Propagation (BP)" neural network classifiers are utilized with this method for feature extraction.	95.45%
Wei et al. [106]	Authors proposed a deep belief network-based optimization algorithm "Particle swarm optimization (PSO)", "genetic algorithm optimization back propagation (BP)", and an artificial fish swarm method were employed in this work.	83.86%
Dai and Pan [107]	Authors suggested an IDS for classification based on enhancing "Deep Belief Networks (DBN)" and "Extreme Learning Machines (ELM)".	97.82%
Xiao X. and et al. [108]	One of the deep learning classification techniques, "Convolutional Neural Network (CNN–IDS)", was presented by the authors. They concentrated on the implications of dimension reduction on classification techniques and model training duration.	94.0%
Lin et al. [109]	For extracting features, CNNs with five layers network were proposed.	97.53%
Yong and Bo [110]	A CNN was shown. The Batch normalization approach increased CNN's training phase speed by reducing data complexity and increasing model speed.	94.11%
Zeng et al. [111]	The author presented a method for classification and identification using multiple DL models. Three different deep learning models, including" CNN", "LSTM", and an "autoencoder", were utilized to extract features from various perspectives.	–
Ludwig [112]	An ensemble approach using an "autoencoder", a "deep belief neural network", a "deep neural network", and an "extreme learning machine" was proposed for the classification challenge.	93%
Malik et al. [113]	The authors suggested a particular kind of hybrid deep learning known as cuda-enabled for effective and prompt detection of multi-vector threats and attacks.	98.6%
Zhang et al. [114]	To create an IDS, the authors proposed an ensemble Bayesian CNN. The proposed systems are evaluated using both the NSL-KDD and UNSW-NB15 data sets.	99.3271%
Atefi and H. et al. [115]	The authors provided a hybrid classification technique for IDS that combines DL and Binary Algorithms (BA).	99.002%
Santhosh and A. M [73]	The author presented a cloud-based real-time IDS that uses deep learning and ML algorithms.	83.87%

(c) CNN variations
The following is how the many types of variants utilized by CNN to improve the system's performance are derived. Hence, down-sampling is used to lower the computational effort and parameters [137]. Kernel functions are widely used to create and alter CNN layouts.

(d) CNN's acoustic model
The Acoustic CNN method utilizes shared weight, receptive field, and spatial sampling data to improve performance [138]. This method also incorporates convolutional and max pooling layers to facilitate fast task adaptation in a new environment. The time axis is utilized for data convolution without any validations. When weight sharing is implemented, local correlations are captured and the CNN equivalent variances are collected, resulting in improved algorithm efficiency.

(e) CNN's minimal weight sharing
Pre-training is a process that can help improve the performance of CNNs by adding a small number of weights [139]. Weight sharing is used in the neurons to execute a similar feature mapping. During this process, similar pooling layers are also coupled with the traditional layers [58]. Neurons are used to calculate the features, and the number of parameters is rapidly growing. Weight training produces the superior initial value, and the pooling layer can then sub-sample the learned values.

3.3.2. Autoencoder (AE)

An Autoencoder (AE) is a type of neural network that typically consists of one or more hidden layers that are connected to input and output layers. The input and output capacities of the AE are the same, but the data transfer between them occurs over a shorter channel. By using neural networks, AEs can overcome the challenges associated with transfer and unsupervised learning. Autoencoders are used to do task discovery and analysis based on those qualities [140]. The encoder and decoder are the two main elements that make up the autoencoder. The encoder takes the inputs and passes them through the new model. During the autoencoder's training process, the reconstructed errors are minimized once the encoder sends a code to the generator. Fig. 16 shows the basic design of an autoencoder as well as how little

Table 3
A look at how DL algorithms are used in the real world.

Source	Practical applications	DL type	Explanation
[116]	Text	LSTM	Using a long–short-term memory network, we were able to retrieve drug-drug interactions from biomedical texts.
[117]	Images	CNN	Deep neural networks are being used to classify skin cancer at the level of dermatologists.
[118]	Text	LSTM	Deep sentence embedding and information retrieval analysis employing long–short-term memory networks.
[119]	Images	CNN	A retrospective multi-cohort radiomics study used deep learning to predict lung cancer prognosis.
[120]	Audio	B-LSTM	Framewise phoneme classification is performed with bidirectional LSTM and other neural network topologies.
[121]	Text	CNN	Text classification using character-level convolutional networks.
[122]	Text	LSTM	DeepCare is a predictive medicine model based on deep dynamic memory.
[123]	Text	CNN	Text classification with RCNNs
[124]	Images	SD-AE	With the aid of SDAE, meaningful representations are learned in deep networks employing a local denoising criterion.
[125]	Genomics	DBN	A comparison is made between the effectiveness of deep belief networks and support vector machines in categorizing gene expression data from complex disorders.
[126]	Text	AE	For learning graph representations, deep neural networks are used.
[127]	Audio	C-DBN	Convolutional deep belief networks are used for unsupervised feature learning for audio classification.
[128]	Text	AE	For words, there is an AE.
[129]	Audio	DBN	Deep belief networks are used to model acoustics. For words, there is an autoencoder.
[130]	Text	DBN	DBN and softmax regression are used to classify text.
[131]	Classification of digits	Priors on Belief Networks with Complementary Priors (PBNCP)	For digit classification, the PBNCP algorithm is used.

We demonstrate various examples of varied applications incorporating photos, audio, text, and genomics data in Table 3.

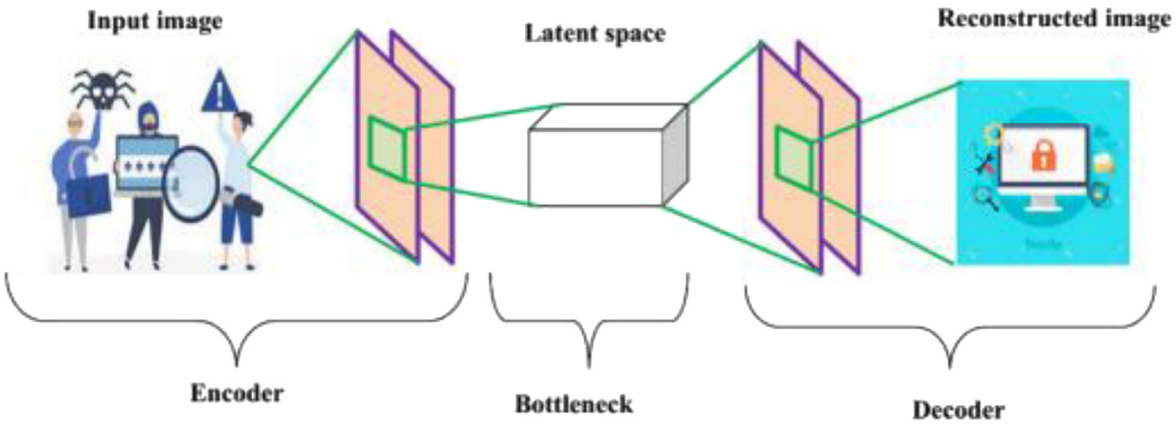


Fig. 16. Organization of auto encoder.

input and output variance there is. The primary three categories of autoencoders are stacked, denoising, and variational.

(a) Stacked Auto Encoder (SAE)
One type of unsupervised learning autoencoder is the stacked encoder, which consists of three fundamental levels: the input layer, the output layer, and the hidden layers [141]. The input data is mapped and reconstructed using a representation of the hidden layer. The training procedure involves using the greedy layer selection method, where each layer is trained separately in a hierarchical manner. The supervised learning process is the last feature of the bottom layer. Both the autoencoder and decoder processes are run in the forward and backward directions, respectively. The higher-order feature provides vector input as a result. In the end, the softmax is employed to successfully address classification problems.

(b) Denoising Auto Encoder (DAE)
The autoencoder procedure uses similar input and output data [142]. The training process now includes noise, which has the added benefit of providing denoising power. All convolutional kernels can be trained using the denoising

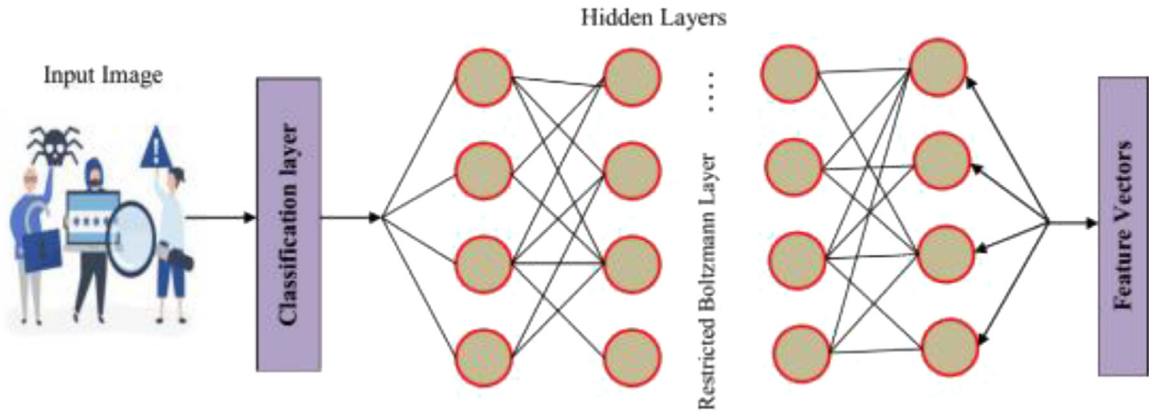


Fig. 17. Structure of deep belief network (DBN).

autoencoder, and the auto-coder is also trained. The best feature and denoise input are acquired for classification.

(c) Variational Auto Encoder (VAE)
2013 saw the introduction of the significant generative representation of the variational encoder. Backpropagation has produced short training times and semi-supervised learning [143]. It is appropriate for the use of sensor failure detection, intrusion system security, and IoT device security, respectively [144]. A variational autoencoder effectively implements the probabilistic learning model. It performs various tasks such as visualization, recognition, representation, and denoising.

3.3.3. Deep Belief Network (DBN)

Deep belief networks consist of hidden and stochastic layers, which are implemented using stochastic variables with direct acyclic graphs [145]. The generative and discriminative DBN is applied to the greedy selection theory, but it faces challenges such as unobserved variables and learning difficulties [146]. During the training process, posterior distribution can make the process complex, hence binary values of 0 and 1 are used as units in the stochastic model. The organizational structure of DBN is shown in Fig. 17, and its different classes are discussed in the following section.

(a) Deep Boltzmann Machine (DBM)
In DBM, the limited Boltzmann machine is utilized in stacked layer form with graphical, unsupervised, generative, and probabilistic representation. DBM is used to identify the data's latent features, which have a unidirectional link [147]. The approach for efficient learning and parametric inference is carried out using the greedy layer. Unsupervised representation is used to identify the optimum parameters.

(b) Restricted Boltzmann Machine (RBM)
The restricted Boltzmann machine is the form of deep belief neural network that is most widely used. A stochastic neural network consists of edges and a stochastic binary unit [148]. So, during the Boltzmann machine learning process, the difficulties of scalability and practicability are raised. Each hidden layer is connected to the others by the neurons from the RBM, resulting in a bipartite chart [149]. An efficient algorithm is used to conduct the procedure and restrict a similar layer connection. RBM is used for feature learning and dimensionality reduction.

(c) Deep Restricted Boltzmann Machine (DRBM)
Deep-restricted Boltzmann machines combine deep belief and the Boltzmann network to increase power [150]. Using DRBM, the greater quality features are extracted. The DRBM imposes stringent limitations on each layer. Each layer can be organized using a deep and restricted network combination [151]. Each data set undergoes computations for the gradient and free energy during the training process.

3.3.4. Recurrent Neural Network (RNN)

The RNN is connected to one or more feedback connections, and these connections operate as loop activators. The enabled network is used to carry out the temporal and sequence learning processes [152]. A fundamental component of RNNs is the additional loop with multi-layer Perceptron, which has a lower memory. Potential connections exist between the activation of stochastic function and neurons [153]. Learning, activation, and architectural functions are all carried out using the same gradient descent process. With the annealing idea, the repeated features are gathered. The architecture of the recurrent neural network (RNN) is illustrated in Fig. 18.

(a) Bidirectional RNN (BRNN)
Bidirectional RNN is used to overcome the constraints of conventional RNN [154]. The BRNN is trained at a predetermined time using previous and future input. There are progressive and backward states in the structure. The forward state's output has no input link for the backward state [155]. Such networks require time and effectively minimize objective functions.

(b) Long–Short-Term Memory (LSTM)
Long–short-term memory is used to represent the RNN elaboration factor. Every computing gate unit uses the inputs, 0 and 1 [156]. The LSTM, which is made up of many neurons, is used to store the feedback data. The memory gates in this instance are written, read, and forgotten [58]. These gates are used to control the memory cells. The vanishing problem has an impact on the backpropagation training stage.

(c) Acoustic model of RNN (ACNN)
The long vector data and temporal variability constitute the acoustic model. Consequently, CNN [157] does not collect statistics on long-term reliance. These networks are easily impacted by vanishing issues. Backpropagation allows for the implementation of the exponential error and decay function. The undesirable weight arrangement is quickly adopted. The feature learning and dimensionality minimization processes are completed fast.

(d) Gated recurrent unit
The gated recurrent unit is used to anticipate the long-term dependency during the RNN training process (GRU). Aside from a few tweaks in LSTM, the hidden layer with the

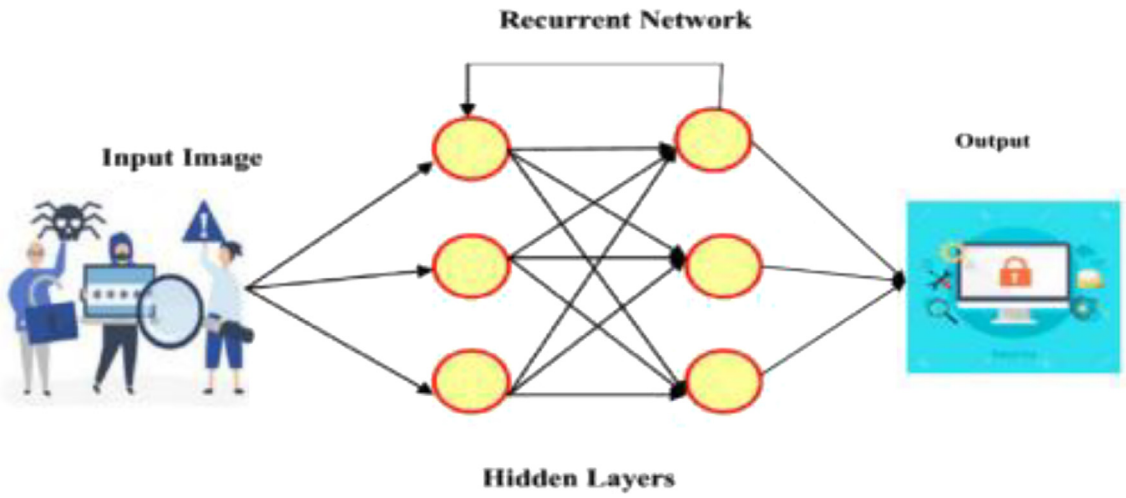


Fig. 18. Structure of recurrent neural network.

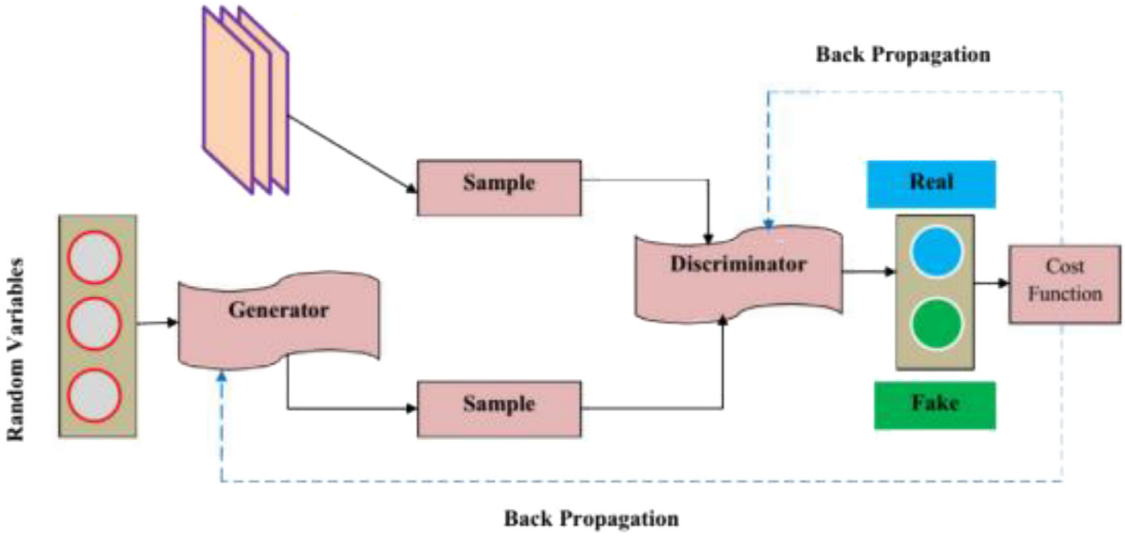


Fig. 19. Structure of generative adversarial network.

state-to-state transmission is altered to produce GRU [158]. Hence, the primary motivation is the expression of simplicity and appeal. As a result, GRU just needs a few parameters, and network operations can be carried out with ease.

3.3.5. Generative Adversarial Network (GAN)

One of the main GAN models is the generator and discriminator. The discriminator model is used to define the tasks, and the discriminator is used to generate realistic output [138, 159]. With GAN, the genuine and artificial inputs, and outputs are appropriately identified. High-quality and artificial data are produced [160]. The updated data gleaned from the previous network. The discriminating mechanism distinguishes between authentic and fraudulent data in a latent network. As a result, the main objective function of GAN is the minimax theory [161]. Fig. 19 depicts how the GAN's discriminator and generator functions are expressed.

3.3.6. Deep Reinforcement Learning (RIL)

Mnih et al. published the first description of deep mind reinforcement learning in 2015 [162]. The core principle of the machine learning function is to maximize cumulative rewards. Deep neural networks are used to reduce the size of the high-dimensional input data [163]. Multi-layer perceptrons [164] are used to implement the values of Q-functions. Fig. 20 describes the deep reinforcement learning process.

(a) Multi-task reinforcement (MTR)
It takes time to learn and a certain number of trajectories to fully develop the powerful reinforcement characteristic [146]. Using the multi-task reinforcement learning strategy, the performances are noticeably improved [165].

(b) Multi-agent reinforcement (MAR)
The combination of optimal rewards is used to operate the multiple-agent reinforcement. As a result, MAR allows for the quick implementation of many task arrays [166].

(c) Asynchronous reinforcement (AR)
Deep reinforcement learning's longer period is a significant issue [167]. The asynchronous architecture is commonly used to improve the efficiency of deep reinforcement learning. In this architecture, the central learning process involves the main server parameter, which is responsible

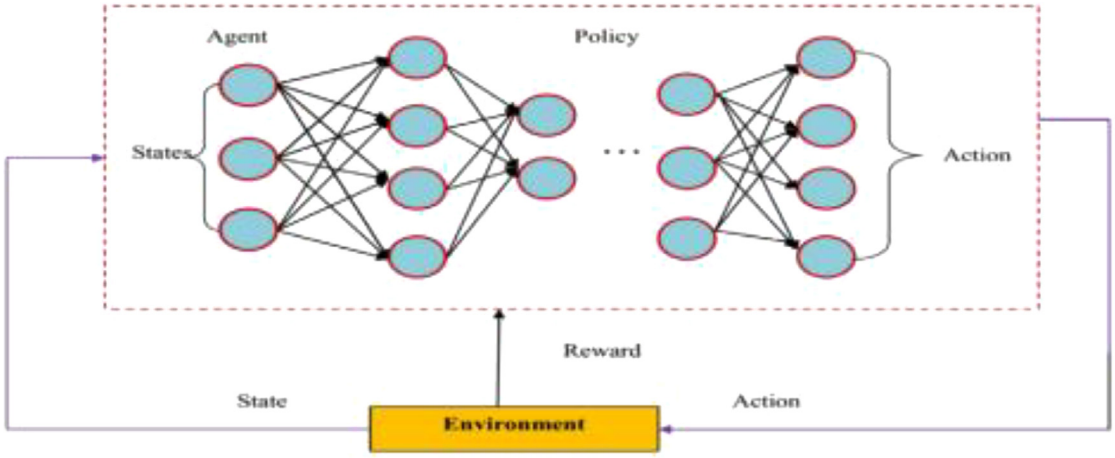


Fig. 20. Organization of deep reinforcement learning network.

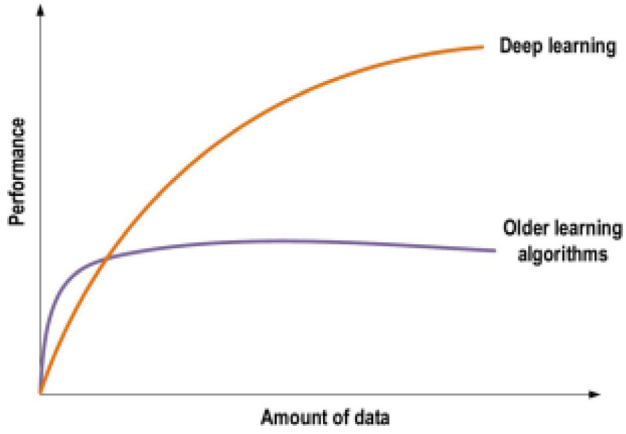


Fig. 21. Performance of DL about the amount of data.

for coordinating and synchronizing the training of multiple agents in a distributed system [168].

(d) Q-learning Reinforcement (QR)
The main learning set for the Q-learning procedure in this case is temporal difference (TD) learning. Moreover, none of your environmental expertise is ever necessary [169]. A traversed state is used to capture the best action solution in each state [170]. The neural network's incremental operation updates the Q factor, and the prior state values are stored in the memory space.

3.4. Limitations of deep learning and their alternative solutions

DL often poses many challenges that need to be considered. Given below are some of the most challenging problems, followed by their potential alternatives-

1. **Training Data**
DL, which includes representation learning, requires a vast amount of data, making it incredibly data-hungry [171, 172]. Accumulating more data can result in a well-behaved performance model, but there may not always be enough data available to use DL directly [173]. To overcome this problem, three approaches are proposed. The first approach involves using transfer learning once related task data has been gathered to improve the model's performance [174]. The second approach involves data augmentation, which is particularly useful for enhancing image data by using operations such as picture translation, mirroring, and rotation that do not alter the image label [175]. The third option involves adding simulated data to expand the training set, which is feasible when the issue is fully grasped and simulators can be built based on the physical process. Several transfer learning strategies used in DL are summarized in [176,177]. Fig. 21 illustrates how the performance of a well-behaved model improves with an increasing amount of data.
2. **Imbalanced data**
Imbalanced data is a common issue in DL, where negative samples greatly outnumber positive ones [178–180]. This can lead to undesirable results when training a DL model. To address this problem, several techniques are used. Firstly, it is important to use the proper loss evaluation criteria and prediction results, taking into account imbalanced data. This means that the model should perform well on both small and large classes, and the area under the curve (AUC) can be used as a loss criterion [181]. More details on predicting software bugs for larger and new data sets can be found in [182]. Secondly, the model's performance with small classes can be enhanced using weighted cross-entropy loss. This approach allows for effective training in small classes while still utilizing cross-entropy loss. Furthermore, when training the model, it may be useful to either reduce the representation of large classes or increase the representation of small classes. Finally, as suggested in [183], constructing models for every hierarchical level can also balance the data, as many systems have hierarchical label space.
3. **Interpretability of data**
In some domains, DL methods are often considered as a black box, but in reality, they can be interpreted. In the field of bioinformatics [184], understanding the meaningful motifs and patterns detected by a deep learning network is particularly important. This is especially true for tasks like disease diagnosis, where simply knowing the diagnostic or predictive outcomes of a trained DL model is not sufficient. A deeper understanding of the underlying biological mechanisms and features that contribute to the disease is necessary to develop effective treatments and interventions [185]. It is also necessary to understand how to improve the certainty of the prediction outcomes by assigning a score to each component in the specific example based on

its importance. Two main techniques are used for this purpose: back-propagation-based and perturbation-based approaches [186]. Perturbation-based methods involve modifying a portion of the input, and then showcasing how this alteration impacts the output of the model [187–190]. These methods are effective in [191], and different interpretations of the model's interpretability can be represented in various scenarios.

4. **Uncertainty scaling**
   DL approaches are commonly used for making predictions, and it is important to obtain not only the final prediction label but also the confidence score for each query from the model [192]. The confidence score indicates how certain the model is in its prediction, which is critical in preventing inaccurate and misleading predictions. Assessing the precision of machine learning-based disease detection and automated clinical judgments is crucial in healthcare and similar fields, and uncertainty scaling plays a vital role in this process [193,194]. In some cases, DL models may produce overconfident predictions due to incorrect scaling of the probability score obtained from the SoftMax output of direct-DL models [195]. To output the probability score in the appropriate scale, various methods have been developed, such as "Bayesian Binning into Quantiles (BBQ)" [196], "isotonic regression" [197], "histogram binning" [198], and "Platt scaling" [199]. Recently, temperature scaling for DL approaches has been introduced and has shown better performance than other strategies.
5. **Catastrophic forgetting**
   Extending a DL model with new data can interfere with previously learned information. For example, if a model is trained to categorize 1000 types of flowers and a new variety of flowers is added, improving the model solely with the new class can cause a decline in its performance with the older classes [200,201]. This is a common scenario in many fields, including biology, where data are continuously collected and updated. To address this issue, one possible solution is to train a completely new model from scratch, incorporating both old and new data into the training process, but this is computationally expensive and can result in an unstable representation of the starting data. To address this issue, three ML strategies based on neurophysiological theories are available that do not cause catastrophic forgetting [202,203]. The first strategy involves regularizations like EWC [200]. The second strategy uses rehearsal training methods and dynamic neural network architectures such as iCaRL [204,205]. The third strategy is based on "dual-memory learning systems" [206]. Additional information can be found in [207–209].
6. **Model compression**
   Because of their extreme complexity and vast number of parameters, DL models require a lot of memory and processing power to produce well-trained models capable of being used productively [210,211]. The fields of environmental science and healthcare are two of those that are deemed to be data-intensive. These requirements limit the use of DL in devices with little computational power, particularly in the healthcare industry. The issue necessitates further calculation because there are many different ways to assess human health, and the degree of data variability has increased considerably [212]. To address the computing challenges related to DL, new hardware-based parallel processing systems such as FPGAs and GPUs [213–215] have been created. Several DL model compression strategies have also recently been proposed to lessen the computational challenges the models face from the outset. Four categories can be used to group these strategies. First-class models feature fewer redundant parameters, which do not significantly affect model performance. Parameter pruning is the name of the class that comprises the well-known deep compression technique [216]. Knowledge distillation is the process of training a smaller model from the larger model's distilled knowledge in the second class [217,218]. Compact convolution filters are employed in the third class to lower the parameter count [219]. Low-rank factorization is used to estimate the information parameters for preservation in the final class [220]. These classes stand for the most representative methods for model compression. It has given a more in-depth explanation of the subject in [210].
7. **Overfitting**
   DL models have numerous parameters that are intricately connected, which makes them highly susceptible to data overfitting during the training phase [221,222]. This problem is not specific to a particular field but affects various domains. Therefore, it is crucial to consider and address this issue when proposing DL approaches. Recent studies have shown that the inherent bias of the training process in DL can help overcome overfitting issues [223–225]. However, it is still necessary to develop methods to deal with the overfitting problem. Research has identified three groups of DL algorithms that can alleviate overfitting. The first group, which includes well-known techniques such as "weight decay" [226], "batch normalization" [227], and "dropout" [228], works on both the model architecture and parameters. Weight decay [226] is a widely used regularizer in almost all ML algorithms. The second group focuses on model inputs, such as data augmentation and corruption [175,229]. Insufficient training data can result in the learned distribution deviating from the actual distribution, and data augmentation can be used to mitigate this issue by expanding the available training data. On the other hand, marginalized data corruption is a technique that exclusively leverages data augmentation to enhance the accuracy of the model's solutions.
8. **Vanishing gradient problem**
   The vanishing gradient problem is a common issue that arises when ANNs are trained using gradient-based learning techniques, particularly during the backpropagation stage [230–232]. In each training iteration, the weights of the network are updated by taking into account the current weight values and the partial derivative of the error function. Sometimes the gradient in a DL model can become too small, making it difficult to update the weights and even causing the network to stop functioning altogether. One of the causes of this problem is the use of activation functions such as the sigmoid function, which compresses a large input space into a smaller output space. This compression can lead to a small derivative, making it difficult to update the weights and contributing to the vanishing gradient issue.
9. **Exploding gradient problem**
   The gradient problem is the opposite of the vanishing problem, where large error gradients accumulate during backpropagation [233–235]. This can lead to very large weight updates, making the system unstable and unable to learn efficiently. During backpropagation, the gradient can grow exponentially by repeatedly multiplying gradients as one moves backward in the network. This can cause weight values to become extremely large, even leading to overflows and NaN values. To address this issue, several potential solutions have been proposed such as-
   1. Using various regularization methods for weights.
   2. Redesigning the network model's architecture.

10. Underspecification

In 2020, Google computer scientists discovered a new issue known as "under specification" that has been found in machine learning (ML) models, especially deep learning (DL) models, when used in real-world applications like medical genomics, natural language processing, computer vision, and medical imaging. This problem leads to unexpectedly poor performance of the models [236]. Underspecification is responsible for the poor performance of these models. Even small modifications can lead a model to an entirely different solution and result in different predictions in deployment domains. There are several ways to address under specification, one of which involves creating "stress tests" to evaluate a model's performance on actual data and identify any potential issues. However, a thorough understanding of the process is required, or else the model may operate incorrectly.

The team stated that "Designing stress tests that are well-matched to applied requirements, and that provide good "coverage" of potential failure modes is a major challenge".

4. Networks computing paradigms

4.1. Cloud computing overview

The notion of cloud computing was first pioneered by Google's CEO during the search engine conference in August 2006, according to the literature [237]. Since then, there have been significant advancements in computing and information technology, including the development of the Internet backbone, widespread use of wireless Internet access, and innovations in high-performance and flexible computing for data centers and the web. These developments have paved the way for the shift to the cloud computing age, as reported by the International Data Corporation (IDC) in their third-quarter report of 2016. Sales of cloud computing products such as servers, public cloud storage, and private clouds increased by 8.1% year over year to reach $8.4 billion. The report also predicts that public cloud spending will increase at a rate seven times faster than overall IT spending, reaching $53.1 billion by 2019 and $203.4 billion globally. Companies like Amazon, Google, Microsoft, and others are working quickly to create cloud computing platforms and expand their facilities to accommodate a huge number of customers. The success of these enterprises inspires many more organizations, like Media Temple, Mosso, Joyent, and others, to move their operations to the cloud. Three delivery models – SaaS, PaaS, and IaaS – are used in cloud computing to provide consumers with infrastructure, platforms, applications, and software as a service [238]. SaaS allows users to run apps on a cloud service provider's infrastructure. Consumers can access these apps through web browsers. Clients can use hardware, operating systems, storage, and networks to develop and test new applications or lease their current ones utilizing PaaS service delivery models. Customers who use IaaS have access to computer resources such as networking, storage, and other resources, enabling them to run any application [239].

Cloud computing, according to [Mell and Grance, 2011], must supply the essential characteristics as follows [241] (as shown by Fig. 22):

On-demand self-service: Customers can provision any computing resources they need, such as server time and storage space, whenever they need them, without requiring human intervention.

Broad network access: Computing services are accessible across the network using standard procedures that enable diverse thin and thick client platforms (e.g., smartphones, tablets, and laptops) to access them.

Resources pooling: Because computing resources are shared among several users, cloud computing is a multi-tenant paradigm. As a result, these resources must be dynamically assigned and reassigned in response to consumer demand.

Rapid elasticity: Computing resources must be easily acquired or released to scale and respond to demand quickly, making them appear boundless and appropriate at any time from the consumers' perspective.

Measured service: By providing tools that allow cloud customers to oversee and supervise the use of their resources, the cloud ensures transparency for both the supplier and the consumers.

4.1.1. Cloud service models

Cloud service providers use three main models to offer their services (as shown in Fig. 23). It includes "software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS) [242]".

(a) SaaS (Software as a Service): is also known as a service or application cloud. It refers to a model of delivering software as a service via the Internet to many users. The software can be accessed from different client devices by using a web browser.

(b) Platform as a Service (PaaS): PaaS (Platform-as-a-Service) is a cloud computing service model that offers a platform for the creation, deployment, and management of applications. In this model, the cloud service provider takes care of the underlying infrastructure, which includes the network, servers, operating systems, and storage. This allows users to focus on developing their applications without worrying about the technical details. The application-hosting environment requires users to simultaneously offer specified restrictions for installed applications and prospective configuration options [243]. The PaaS paradigm achieves so at a cheaper cost than the IaaS strategy while still providing greater safety and client access.

(c) Infrastructure as a Service (IaaS): As the name implies, it offers customers resources (storage space, servers, etc.), enabling them to have more power over their software development environment. Physical capital is also required for IaaS. The vendor, on the other hand, takes care of all hardware needs for virtualization, storage, servers, and networking. Infrastructure, to put it simply, is offered in this form of activity.

Customers do not have control over CC. As a result, there are several threat models, such as "data breaches"; "misconfiguration and inadequate change control"; "lack of cloud security architecture and strategy"; "insufficient identity", "credential, access, and key management"; "account hijacking"; "insider threat"; "insecure interfaces and APIs"; "weak control plane"; "meta-structure and appli-structure"; "limited cloud usage visibility"; "abuse and nefarious use of cloud services" (as shown by Table 4). Our biggest fear is insider attacks, in which cloud service providers (like AWS, Microsoft Azure, Google Cloud, and Alibaba) may deceive or eavesdrop on clients' confidential information. Table 5 explains the threat assessment analysis of cloud computing [244].

4.1.2. Cloud deployment models

Cloud computing serves as an online network-based machine, which communicates shared computations of data and assets on request to computers and various gadgets. It is an uncommon way to provide access to common computing resources (for example, servers, storage, operating system, software, and management) on-demand that can be managed and accessed immediately with reduced control. Cloud storage and capacity agreements offer IT organization's consumers the ability to store and process their information from third-party data centers located anywhere on the planet. It relies upon asset sharing for achieving stability and economic scale. This is also the product of developing and obtaining current technologies and exemplar models. Its goal is to provide consumers with the ability to take advantage of major advances in cloud computing deprived of the

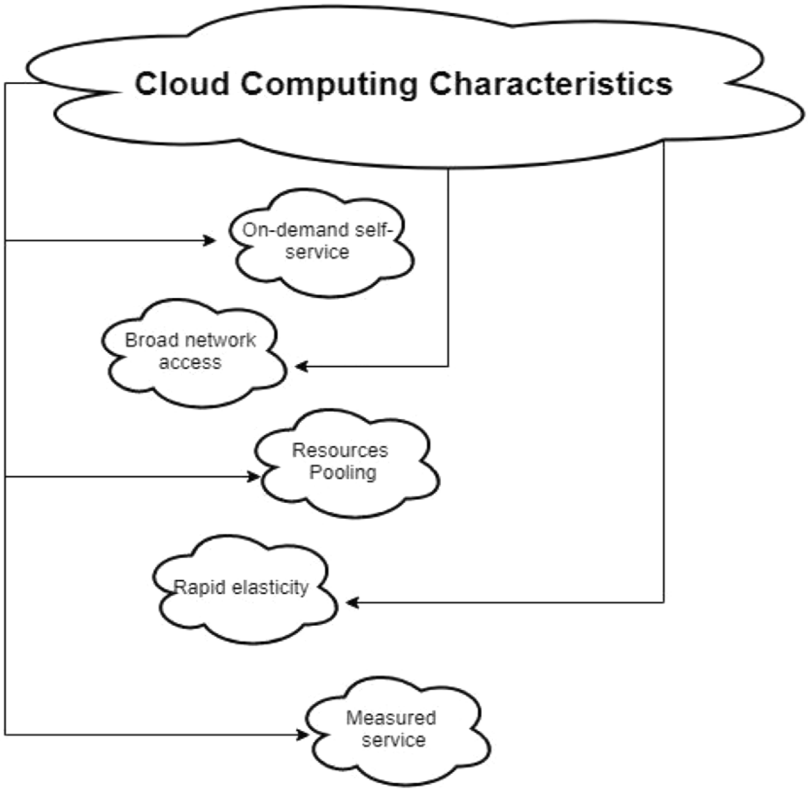


Fig. 22. Essential characteristics of cloud computing [240].

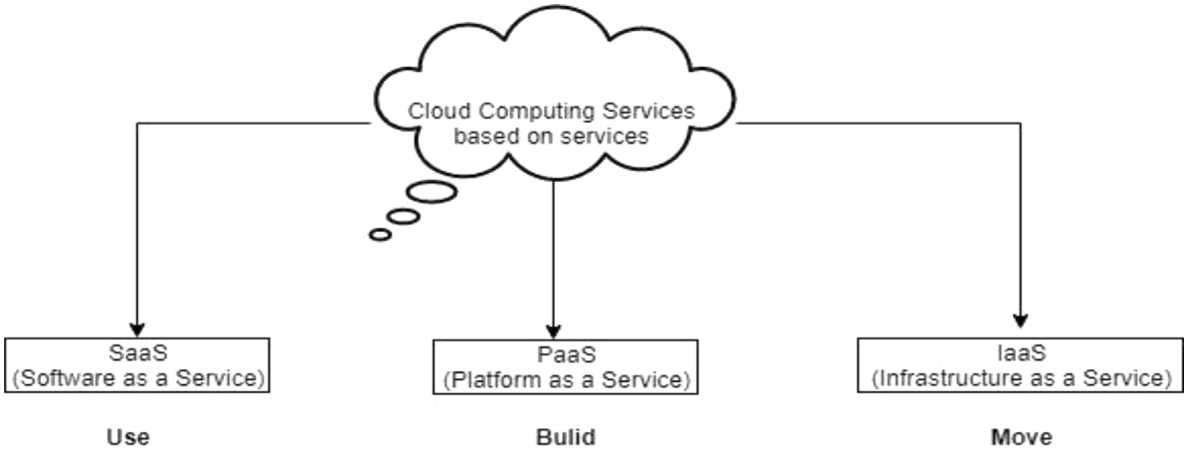


Fig. 23. Cloud computing services.

Table 4
Cloud security Alliance's Egregious.

Security issues and rank	Security responsibility	Cloud service model
Data Breaches	Customer and CSP	SaaS, PaaS, IaaS
Misconfiguration and Inadequate Change Control	Customer	SaaS, PaaS, IaaS
Lack of Cloud Security Architecture and Strategy	Customer	PaaS, IaaS
Insufficient Identity, Credentials, Access, and Key Management	Customer	SaaS, PaaS, IaaS
Account Hijacking	Customer and CSP	SaaS, PaaS, IaaS
Insider Threat	Customer	SaaS, PaaS, IaaS
Insecure Interfaces and APIs	Customer and CSP	SaaS, PaaS, IaaS
Weak Control Plane	Customer	SaaS, PaaS, IaaS
Meta-structure and Appli-structure Failures	Customer and CSP	SaaS, PaaS, IaaS
Limited Cloud Usage Visibility	Customer and CSP	SaaS, PaaS, IaaS
Abuse and Dubious Use of Cloud Services	Customer and CSP	SaaS, PaaS, IaaS

Table 5
The threat assessment of the cloud.

Threat	Spoofing identity	Tampering with data	Repudiation	Information disclosure	Denial of service	Elevation of privilege
Inadequate Due diligence	YES	YES	YES	YES	YES	YES
Data Breaches	NO	NO	NO	YES	NO	NO
Cloud Service abuses	NO	NO	NO	NO	YES	NO
Broken authentication and credentials	YES	YES	YES	YES	YES	YES
Hacked interfaces/ APIs	NO	YES	YES	YES	NO	NO
Exploited system vulnerabilities	YES	YES	YES	YES	YES	YES
Shared Technology	NO	NO	NO	YES	NO	YES
Account hijacking	YES	YES	YES	YES	YES	YES
Malicious insiders	YES	YES	NO	YES	NO	NO
Advanced persistent threat	NO	NO	NO	YES	NO	YES
DoS attacks	NO	NO	NO	NO	YES	NO
Constant data loss	NO	NO	YES	NO	YES	NO

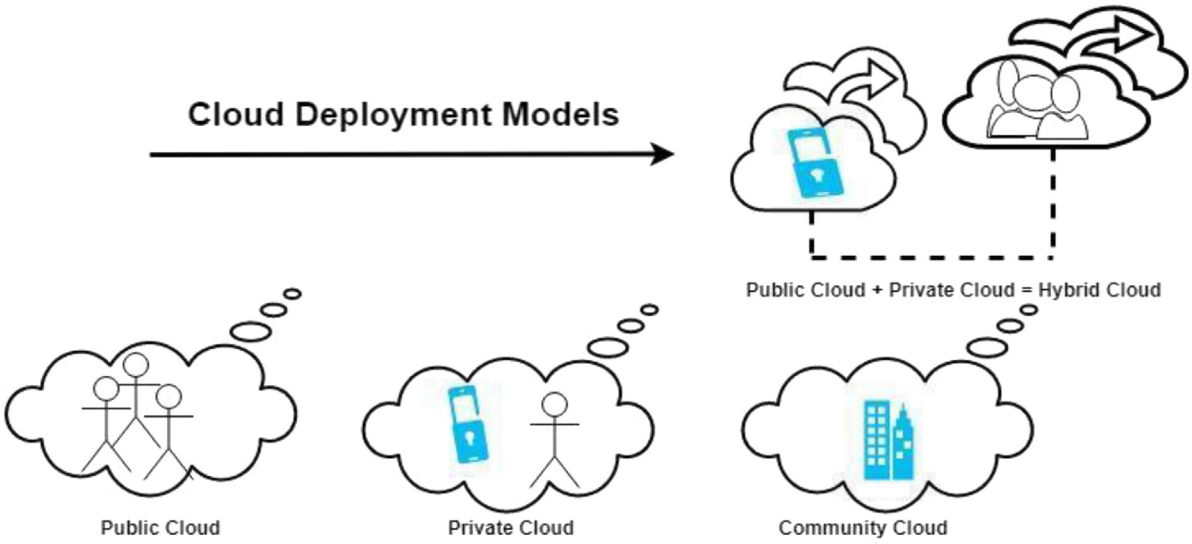


Fig. 24. Different cloud deployment models.

actual requirement for profound and extensive cloud knowledge. Within the presented segment, we deliberate the crucial features inside cloud frameworks deployed by CSPs. These features comprise a comprehensive assortment of facilities that can be availed across the internet. Virtualization plays a key part in cloud deployment. Multi-tenant ecosystems have several users or customers, and they might not take in or share each other's data [245]. Cloud storage is conserved, administered, and backed up at remote locations; and is made accessible on a network where clients can retrieve data. The hypervisor is a significant component of virtualization and allows the execution of many Virtual Machines (VMs) upon a solitary hardware host. Hypervisor administers and manages the diverse operating systems which execute on a shared physical system. NIST is again split into four implementation models based on the consumer cloud's suitability and exact intent. A survey by Flexera found that 92% of enterprises have a multi-cloud strategy in place, with an average of 2.2 public clouds and 2.2 private clouds being used per organization [246]. Various cloud deployment models are given in Fig. 24. Fig. 25 shows the advantages/disadvantages with an example.

(a) Public Cloud: In this kind of cloud, the infrastructure is accessible to everyone. It can be controlled, owned, and operated by a business, educator, government agency, or any combination of these. It benefits everyone who wants to use computational resources. The most common uses of public clouds are for developing applications, etc.

(b) Private Cloud: It is a form of cloud computing that either a single organization or numerous users may use (e.g., business units). It could be handled by a service provider off-site or on-site. Regardless of the investment they make in them, they are expensive. They are excellent for authentication and secrecy. Private cloud providers include IBM Cloud Burst, VMware, and Eucalyptus.

(c) Community Cloud: One user group should only access the cloud to address general issues (e.g., mission, security requirements, policy, etc.). Only the members of the group may use it. It can be looked up online or offline.

(d) Hybrid Cloud: Two or three separate cloud networks (private, group, or public) that are specifically linked to structured or proprietary technologies are included in the cloud framework (e.g., cloud load-balancing explosion). Public clouds are more affordable and scalable than hybrid clouds. Private cloud security and management, in contrast, are also being observed. Issues with data integrity and security plague the hybrid cloud. It is also adaptable and economical. Government organizations, like Microsoft, Amazon Online Services, and VMware, choose private clouds when dealing with classified material. Companies frequently store datasets for the broader public on public clouds. Here, we will construct a cloud.

4.1.3. Players in cloud computing

According to the reference architecture, there are five primary players, each with specific duties and obligations (as shown in Fig. 26). Among the players are cloud users, cloud providers, cloud auditors, cloud brokers, and cloud carriers [247,248].

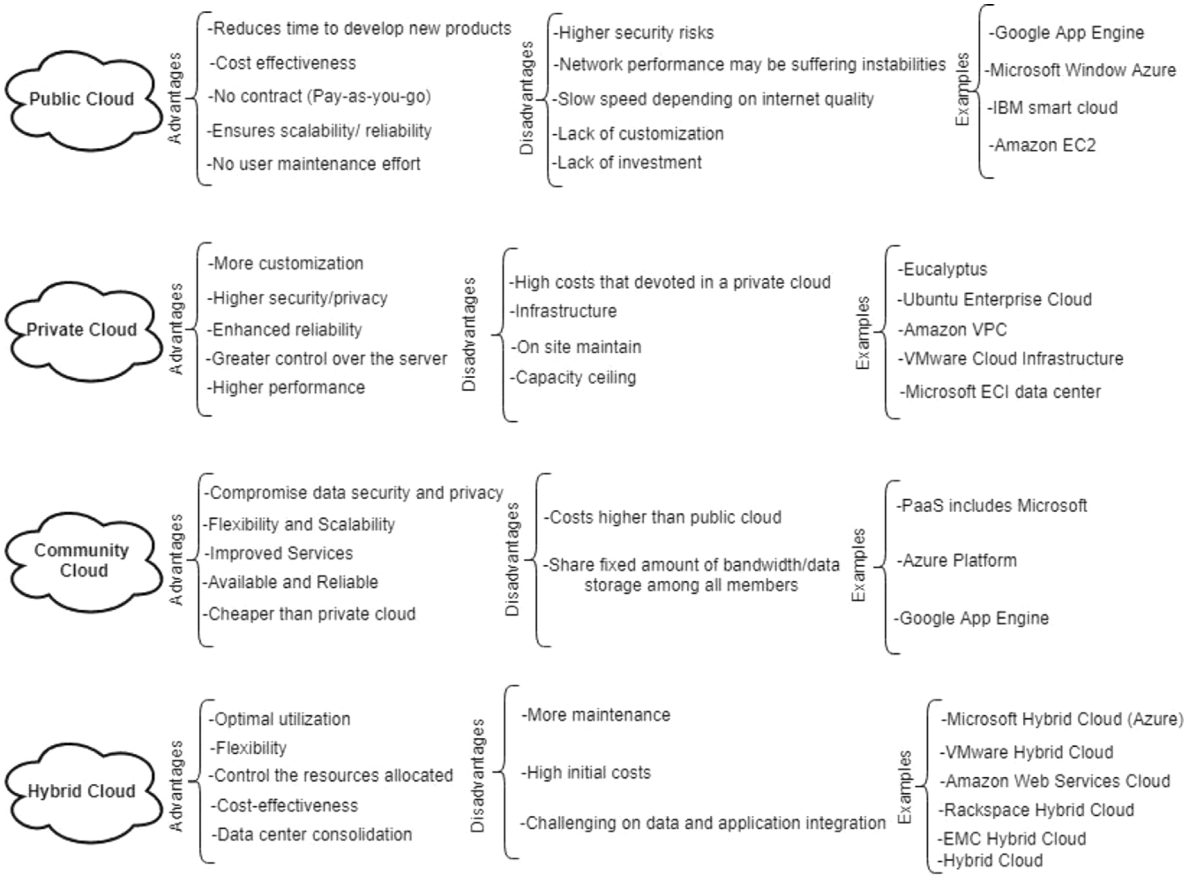


Fig. 25. Cloud deployment models-advantages and disadvantages.

Cloud Consumer: Who (person/organization) employs the services offered by cloud providers and maintains commercial partnerships with them

Cloud Provider: A (person/organization/entity) that provides cloud services ensures that a service is accessible to the concerned people.

Cloud Broker: A cloud broker is a third-party entity that can provide an assessment of cloud service performance, information system operations, and security for cloud deployment on an individual basis.

Cloud Carrier: An arbitrator that offers connectivity and transmission for cloud services is referred to as a cloud carrier (from CSP to Cloud Consumers).

4.1.4. Security issues in cloud environment- An unsolved mystery

The main obstacles to cloud computing are reliability and security. The risk of security breaches emerges due to the possibility of one client being able to access the data of another client, leaving it open to theft, manipulation, and unauthorized modification. To increase cloud security, various methods like encryption, authentication, and authorization are employed. Security hazards related to cloud computing can be categorized into two sections: risks faced by cloud users and those faced by cloud service providers. These include potential risks such as data leakage, hacking, breaches, malicious insider attacks, denial of service, and shared technology problems. Cloud service providers must address aspects of security such as authentication, authorization, and data protection, which constitute the basic security principles that must be upheld, particularly as data moves to the cloud. Trust in the cloud service provider and its services is crucial for customers to migrate to a cloud platform or stick with a legacy framework [249]. "Confidentiality, integrity, and availability" are the three main considerations for cloud system security review (CIA), which is commonly used to identify security vulnerabilities in conventional information systems [250]. Although the CIA domain is frequently used to uncover security flaws in conventional information systems, the major goal of this section is to define the security needs for contemporary cloud architecture. Fig. 27 illustrates the key elements of data security in cloud computing, along with possible risks and countermeasures [251].

4.1.4.1 Confidentiality and privacy According to a report by Gartner, the global public cloud services market is projected to grow by 18.4% in 2021, reaching a total value of $304.9 billion [252]. Ensuring confidentiality is crucial for protecting sensitive business assets from unauthorized access. In the cloud environment, client data may be stored in the same database as the data belonging to the cloud service provider (CSP), which raises concerns about unauthorized access to client information. Additionally, there is a risk of unethical or malicious representatives of the CSP accessing and tampering with confidential data. In addition to client information, virtual machines, images, and other components of the cloud infrastructure also require confidentiality.

Cloud data confidentiality can be compromised in various ways. Some common concerns about cloud data are:

- Cloud storage services often transfer data to web folders that contain customer data.
- In addition, the geographic location of the client's data can also affect its security.

According to (Dukes, 2015) [253], the security of information systems is defined as "protecting information and systems from unauthorized access, use, disclosure, disruption, modification, or

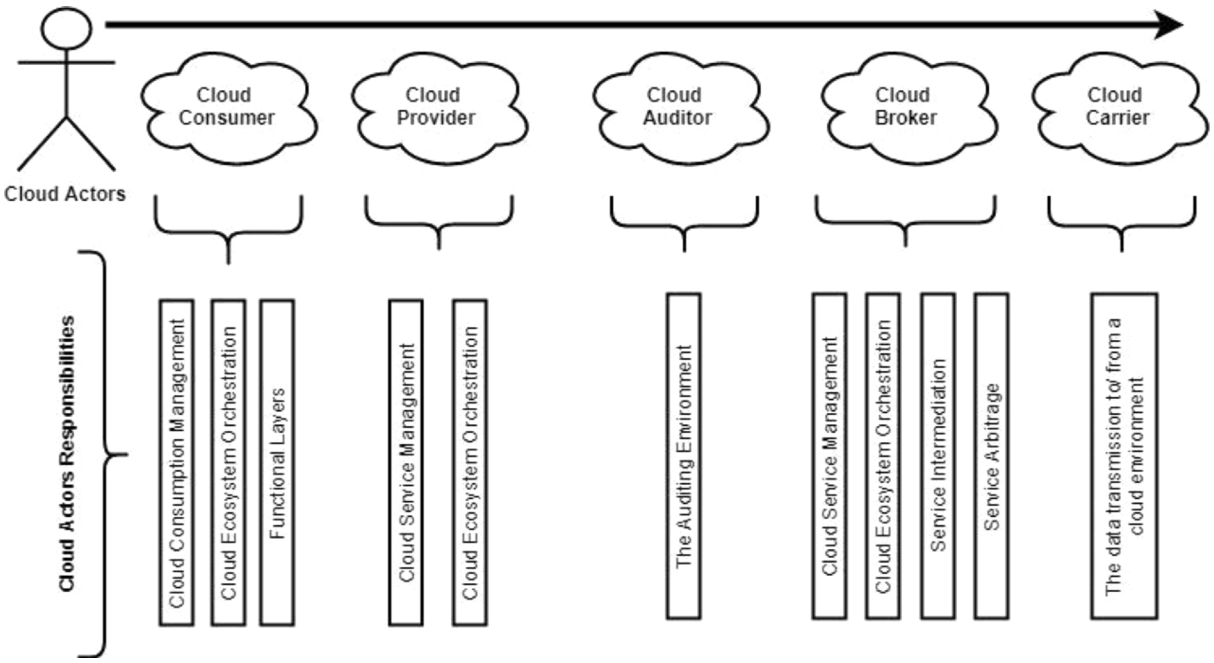


Fig. 26. Players in cloud computing and their responsibilities.

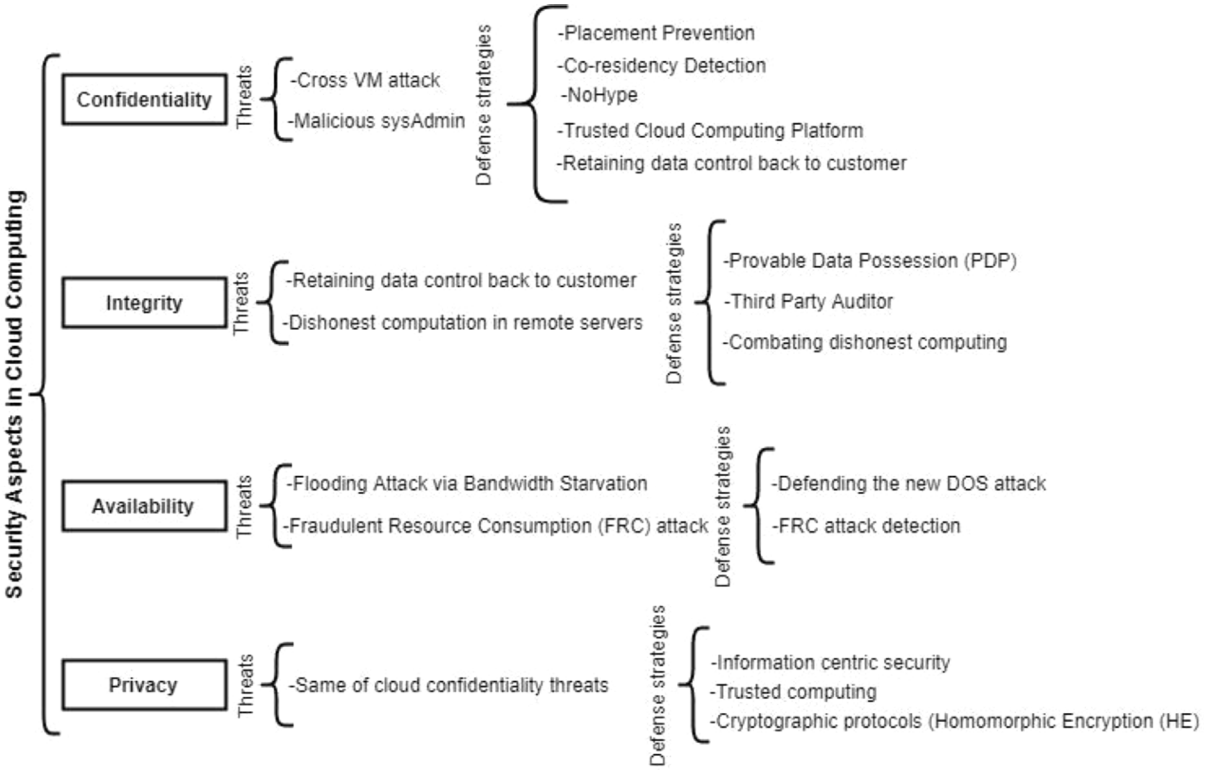


Fig. 27. The key elements of securing data in cloud computing.

destruction to maintain confidentiality, integrity, and availability". Three main security services may be distinguished from there:

4.1.4.2 Integrity Maintaining data integrity is crucial for ensuring the security of cloud assets, as it ensures that unauthorized third parties have not modified them. Ensuring the integrity of assets in a cloud environment is crucial to keep the asset to the owner's specifications. Any modification, insertion, or removal of data is considered to impact the asset's integrity. As cloud resources are accessed by users through web browsers, web-based attacks are common in cloud environments that can potentially alter user files, databases, virtualized storage, and WSDL files. The integrity of data in public cloud systems can be put at risk by various security issues:

(a) Data outsourcing: Data outsourcing can present a challenge to the credibility of cloud service providers (CSPs). When data

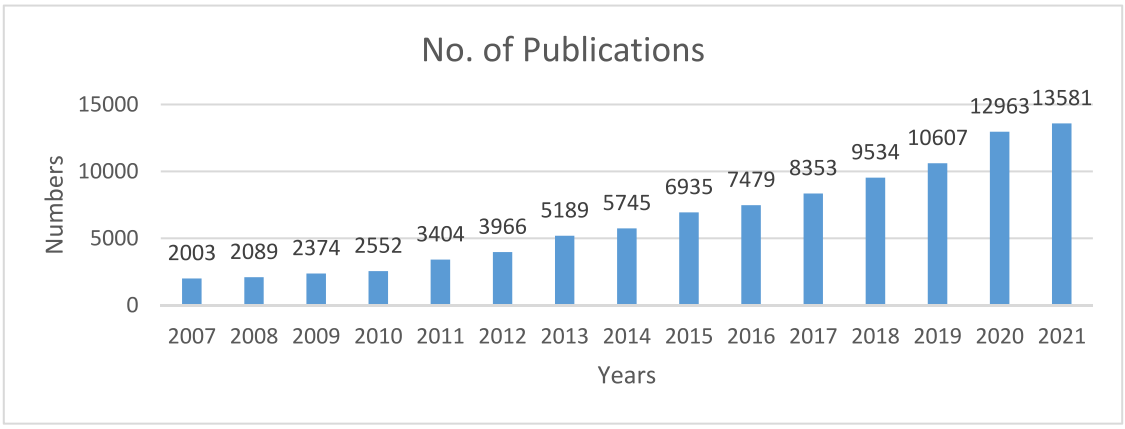


Fig. 28. No. of CC publications in the ScienceDirect database (2007–2021).

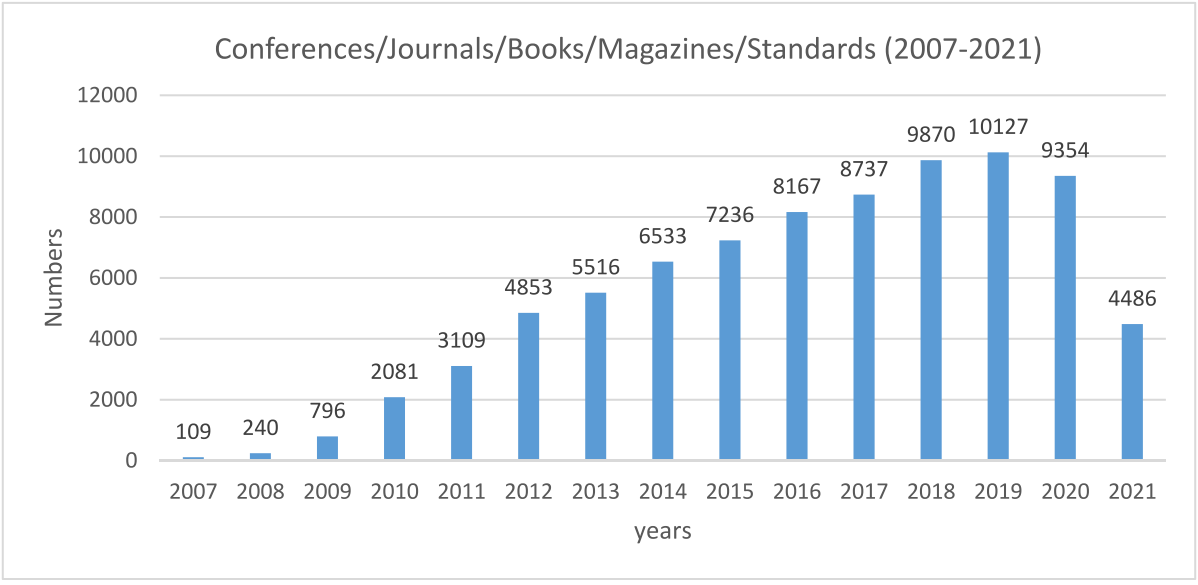


Fig. 29. No. of CC publications in IEEE digital library (2007–2021).

is outsourced to a CSP, there is a risk that the CSP may delete legitimate consumer data tuples.

(b) Insecure API: Insecure APIs sourced from obscure places have been used by web developers for a long time. The API key can allow an attacker to gain unauthorized access to an application or system.

(c) Collision Attack: A collision attack is a type of attack where an attacker can create a new file by merging multiple copies of media or other files. This new file may include "data averaging, substitution, linear data mix", and other techniques.

(d) Wrapping attack: This is a prevalent type of network attack that can happen in cloud systems. It involves replicating the text and signature while SOAP (Simple Object Access Protocol) translation is taking place in the Transport Layer Service (TLS) layer. Afterward, the attacker sends the duplicated text and signature to the server, masquerading as an authorized user.

4.1.4.3 Availability Maintaining demand is a crucial aspect of protection that a CSP must uphold. Various business organizations rely on cloud-based technology to support their customers, and any significant downtime can result in a substantial financial loss that may be difficult to recover. Typically, a service-level agreement outlines the promises made by the supplier regarding the availability and response to demand. For instance, the service level agreement may state that programs will be usable 99.999% of the time, and additional resources will be dynamically available if more than 80% of the provided support is used.

Year-wise analysis of CC publications

From 2007 to 2021, Fig. 28 shows the number of CC articles from the ScienceDirect database per year. There is a steady increase in the number of publications, which may be described as exponential. Fig. 29 shows the number of CC articles in the IEEE digital library.

4.2. Edge computing overview

Edge Computing (EC) is a modern computing approach that involves processing data at the network's edge, aiming to bring computation closer to the data source. According to a report by MarketsandMarkets, the global edge computing market is projected to grow from $3.6 billion in 2020 to $15.7 billion by 2025, at a compound annual growth rate (CAGR) of 34.1% during the forecast period [279]. Researchers, including Shi et al. M. Satyanarayanan, Zha, et al. X. Hong, and Y. Wang, have provided different definitions for EC (as shown in Table 6). EC also meets

Table 6
Edge computing is defined differently by different researchers.

Study	Definition
Shi et al. [47,254,255]	"Edge computing is a new computing mode of network edge execution. The downlink data of edge computing represents cloud service, the uplink data represents the Internet of Everything, and the edge of edge computing refers to the arbitrary computing and network resources between the data source and the path of the cloud computing center".
M. Satyanarayanan [256]	"Edge computing is a new computing model that deploys computing and storage resources (such as cloudlets, micro data centers, or fog nodes, etc.) at the edge of the network closer to mobile devices or sensors".
Zha et al. [257]	"Edge computing is a new computing model that unifies resources that are close to the user in geographical distance or network distance to provide computing, storage, and network for applications service".
X. Hong and Y. Wang [258]	"Near the edge of the network or the source of the data, an open platform that integrates core capabilities such as networking, computing, storage, applications, and provides edge intelligent services nearby to meet the industry agility key requirements in connection, real-time business, data optimization, application intelligence, security, and privacy".
IDC [259]	"Mesh network of micro data centers that process or store critical data locally and push all received data to a central data center or cloud storage repository, in a footprint of less than 100 square feet".
ETSI [260]	"Mobile edge computing provides Internet service environments and cloud computing functions at the edge of mobile networks, near radio access networks (RANs) and mobile users".
N. Abbas et al. [261]	MEC basics, application cases, infrastructure, and security and privacy concerns
W.Z Khan, H. Liu, Y. Ai et al. [262–264]	Concepts, applications, architectures, and open research challenges are all part of the MEC.
P. Porambage, G. Premsankar, W. Yu et al. [265–267]	Review of how MEC and other EC paradigms can be used in IoT applications.
R. Roman, N. Mäkitalo, S. N. Shirazi et al. [268–270]	Review and analysis of edge computing technologies' security and resiliency.
H. Wu [271]	Using multi-objective optimization, a survey of compute offloading decisions was conducted.
J. Moura and D. Hutchison [272]	MEC and the fundamentals of game theory models. A look at the contributions of game theory to wireless networks and MEC systems.
Y. Mao et al. [273]	From the standpoint of communication, a review of the basic MEC models. In MEC systems, a review of the allocation of joint communication and compute resources is performed.
T. Taleb et al. [274]	The basics of Mobile Edge Computing (MEC) are presented along with its necessary components, such as Network Function Virtualization (NFV), Software Defined Networking (SDN), and Virtual Machines (VM). The MEC reference architecture is also discussed, along with its deployment possibilities for orchestration.
P. Mach and Z. Becvar, and C. Jiang et al. [275,276]	Examining alternative MEC architectures and outsourcing computing.
S. Wang, C. Wang et al. [277,278]	Review of communication, computing, and caching convergence and integration.

the network's requirements for low latency and high bandwidth. Fig. 30 presents various aspects of EC, including their corresponding descriptions such as "Deployment location", "Real-time interaction", "Application scenarios", "Architecture", "Computational power of devices", and "Nature" [280].

4.2.1. Related previous works

EC is a type of CC that has its own set of properties. CC and EC are critical components of the intelligent Internet of Things' future evolution. Fig. 31 summarizes the key differences between CC and EC.

EC architecture is a federated network topology that introduces edge devices between terminal devices and CC to extend cloud services to the network's edge [48,281]. The "terminal layer", "edge layer", and "CC layer" are the three layers that make up cloud–edge collaboration. Fig. 32 shows the architecture of EC from different perspectives [282].

For scientists both at home and abroad, EC has become a hot topic. EC's primary technologies include various "computing offloading [275]", "mobility management [283,284]", "traffic offloading technology", "cache acceleration [285,286]", "network control [287]", and many more. Roman et al. [288] performed security analyses on a variety of mobile edge paradigms, developed a general cooperative security protection system, and provided research recommendations. Based on the literature review, encryption algorithms are classified into two categories i.e., symmetric encryption ("DES, 3DES, ADES", etc.) and asymmetric encryption ("RSA, Diffe-Hellman", etc.). However, data encrypted using typical encryption techniques have low operability, which creates significant challenges for further data processing. At the moment, "attribute-based encryption (ABE) [260]", "proxy re-encryption (PRE) [289]", and "all homomorphic encryption (FHE) [290]" are some of the most widely used data encryption techniques. The four functional requirements [291] ("Dynamic audit", "Batch audit", "Privacy protection", and "Low complexity") have been the subject of recent data integrity audit research. The detailed hot EC research content is given in Fig. 33.

Fig. 34 depicts a comparison between CC, FC, and EC [292–295]. These computing models can work together to create a more flexible and adaptive computing model that is suitable for a wide range of IoT application scenarios.

Through the EC platform, users can perform "local computing", "message forwarding", "data caching", "synchronization", and other services on edge devices securely and conveniently. Based on the literature review, the EC platforms market is divided into two categories (as shown by Fig. 35) based on their object-oriented characteristics: "EC platforms for Internet users" and "Industrial Internet of Things (IIoT)". "AWS Greengrass" [296], "Microsoft Azure IoT", "Baidu BIE", and other internet EC solutions are available. Docker [297] is a well-known container technology that enables user services to be independent of a particular host by decoupling from the host through container isolation. Kubernetes [298] is a container management system that controls the entire application life cycle, including cluster establishment and deployment, publication, extension, and upgrade of various applications. OpenStack is a collection of operating

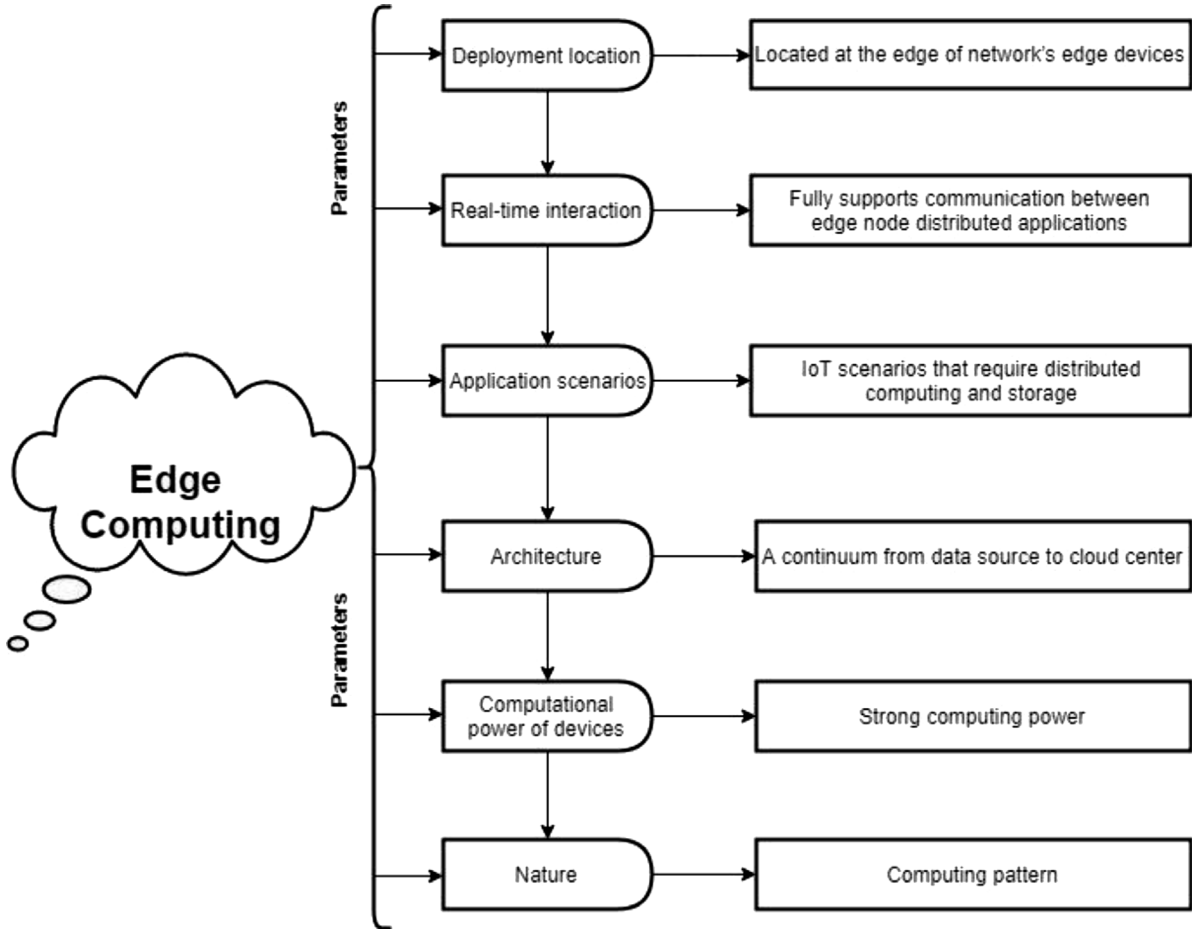


Fig. 30. Parameters affecting EC.

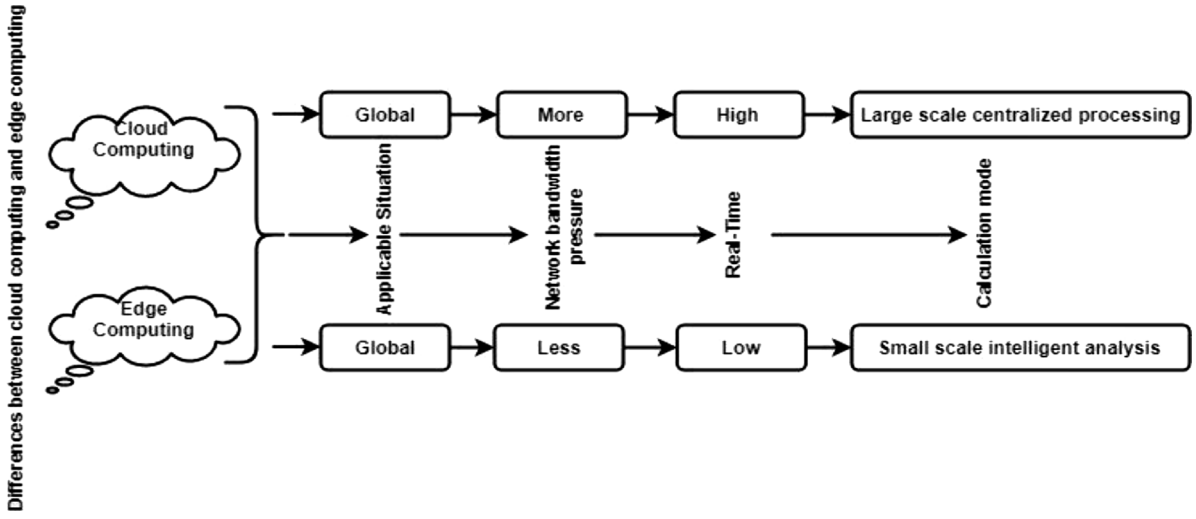


Fig. 31. Main difference between CC and EC.

tools designed for cloud deployment. It aims to help customers run virtual computing or storage services, as well as provide scalable and adaptable CC for public and private clouds, as well as large and small clouds [299]. Keystone is a fundamental OpenStack service that manages user privileges, maintains endpoints, and authenticates and authorizes OpenStack services [298,300]. In the EC platform, NFV [301] (network functions virtualization) is the most popular virtualization technology. The distributed storage management system is a crucial component of the distributed operating system, which is the core open-source component of the entire EC platform [267]. It also serves as a foundation for providing practical EC platform application needs with independent solutions. In the EC platform for Big Data, Hadoop HDFS [302] and IBM GPFS [303] are two of the most frequently used distributed

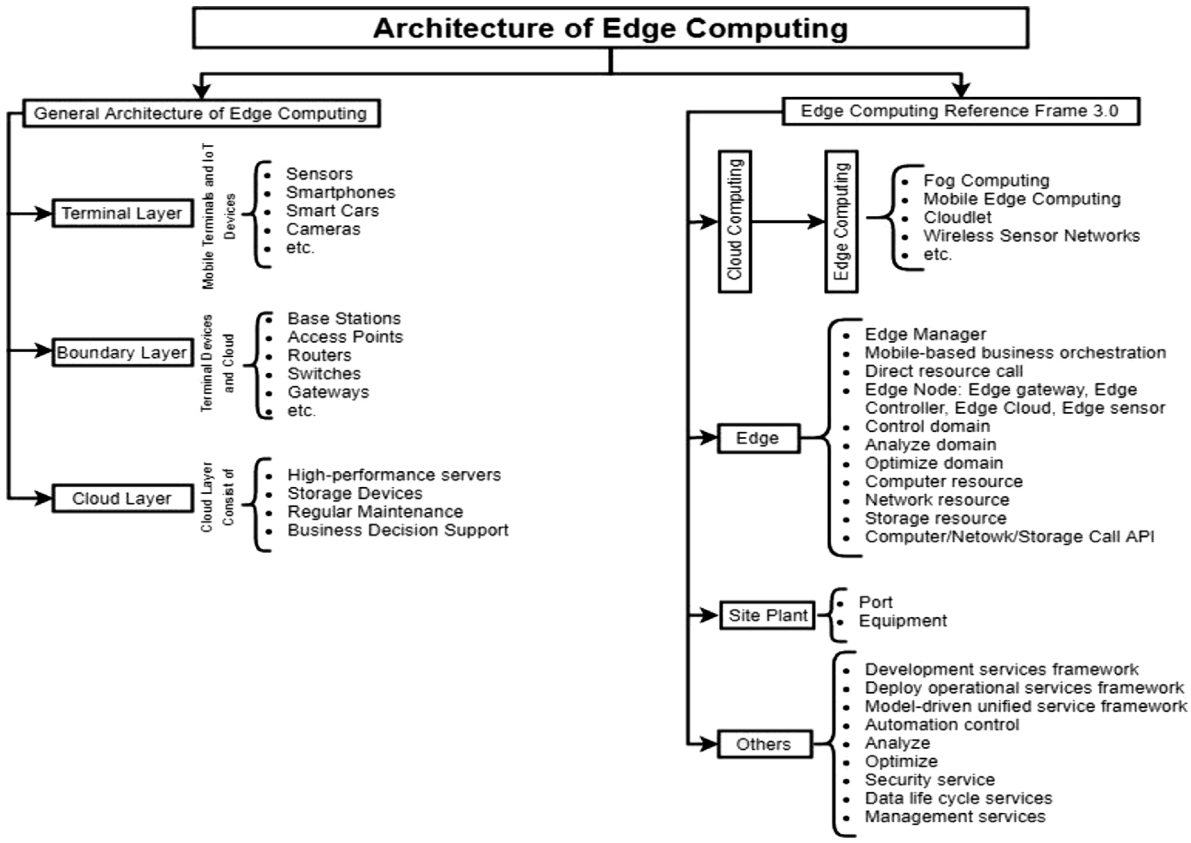


Fig. 32. Architecture of EC.

file systems. To replace the NFS file system, Facebook Haystack stores each image as a file. It is useful for image object storage and has many applications in pattern recognition and image processing [304]. Facebook's Haystack focuses on long-tail services, reducing the effective replication factor from 3.6 to 2.8 or 2.1 [305]. Lustre [306] provides "high-performing file management for heterogeneous computer clusters with high performance and open authentication, so it is usually used in supercomputers" in large-scale cluster EC platforms. Ceph is an open-source computing platform for multi-integration that has a lot of scalabilities, and many new commercial servers have been built on top of it [307–309]. "Open-Stack Cinder" [310], "Amazon EBS" [311], "Microsoft Azure" [312], and "Google Persistent Storage" [313] are all edge computing platforms for virtual block storage.

EC has several benefits, including (as shown by Fig. 36): "Increasing data security and privacy", "Better, more responsive and robust application performance", "Reducing operational costs", "Improving business efficiency and reliability", "Unlimited scalability", "Conserving network and computing resources", and "Reducing latency". Edge computing aids data decentralization and reduces reliance on the cloud. The following are some of the most common use cases for edge technology: "Autonomous Vehicles", "Healthcare Devices", "Security Solutions", "Retail Advertising", "Smart Speakers", and "Video Conferencing".

The following is a list of third-party EC platforms that customers can use to create custom edge computing applications (as shown in Fig. 37). Microsoft Azure IoT is an open EC platform that allows businesses to create their own cloud services and edge apps. Azure offers PaaS and SaaS services [312]. Baetyl [314], which comes from the Baidu open-source project, is an open EC platform for the Linux Foundation Edge. Cloudlet [315] proposes an edge computing-based approach for quick resource awareness and disaster panorama formation. The EC framework [316, 317] consists of edge devices, edge nodes situated in fire scenes, and cloud servers. A survey conducted by Forrester Research found that 53% of global IT and business decision-makers have already implemented or are planning to implement edge computing solutions to support their digital transformation initiatives [318].

Year-wise analysis of EC publications

From 2007 to 2021, Fig. 38 shows the number of EC articles from the ScienceDirect database per year. There is a steady increase in the number of publications, which may be described as exponential. Fig. 38 shows the number of EC articles in the IEEE digital library (see Fig. 39).

4.3. Fog computing overview

The term FC was coined by Cisco [319] as an alternative to traditional CC. FC, also known as fog computing, extends the boundaries of computing applications, data, and services beyond the centralized cloud and into the logical stream of the network edge. FC shares similarities with traditional CC but has limited capabilities in terms of processing, storage, and networking services, which are distributed across multiple end devices. It is an excellent choice for latency-sensitive IoT applications [320]. Even though Cisco [319] invented the phrase, FC has been described in a variety of ways by numerous researchers and groups. A general definition of FC was presented by Yi et al. [321]. It stated that; "FC is a geographically distributed computing architecture with a resource pool which consists of one or more ubiquitously connected heterogeneous devices (including edge devices) at the edge of the network and not exclusively seamlessly backed by Cloud services, to collaboratively provide elastic computation, storage and communication (and many other new services and tasks) in isolated environments to a large scale of clients in

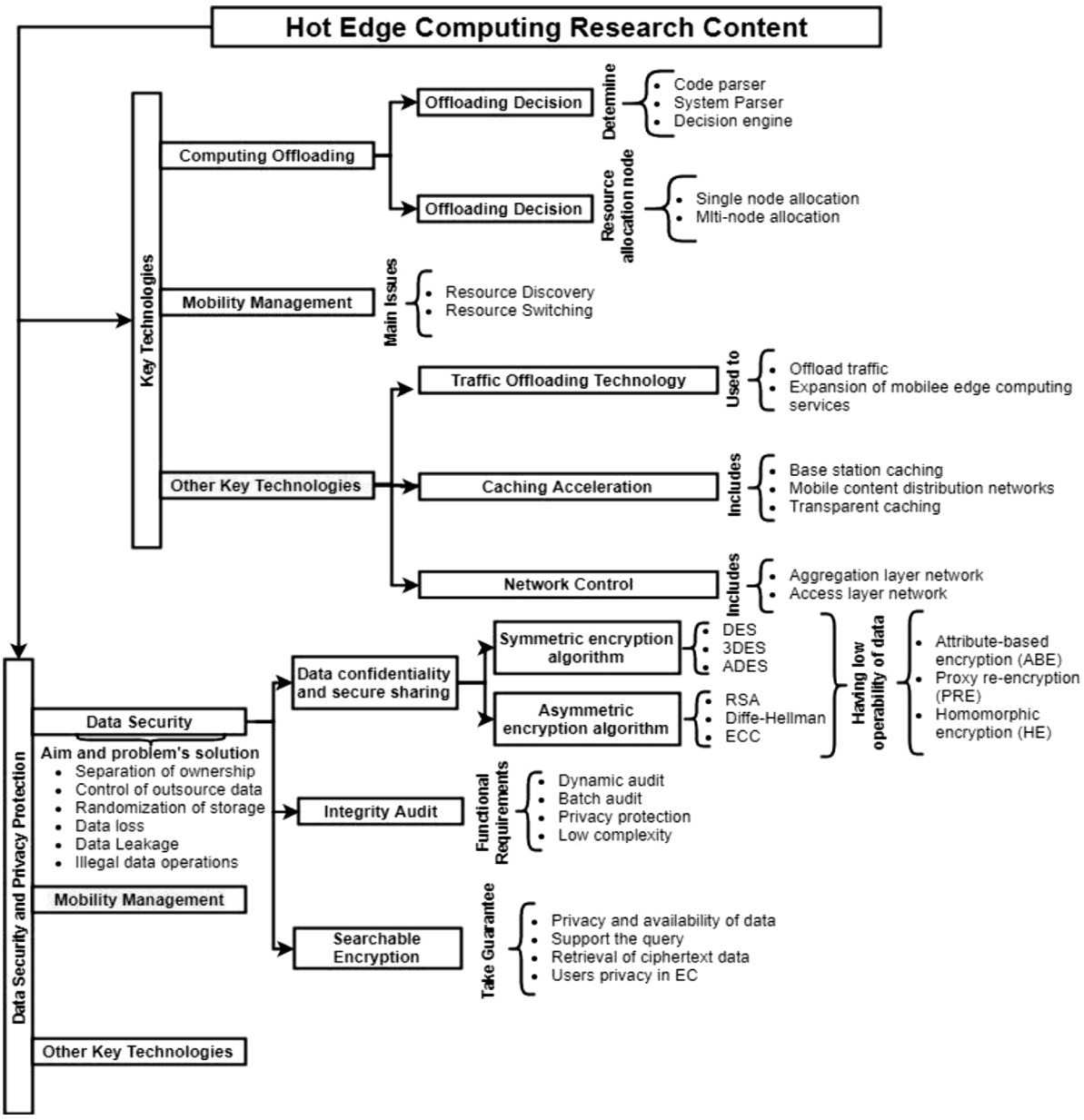


Fig. 33. EC research content.

proximity [321]". A general definition of FC was presented by Yi et al. [322]. It is stated as follows; "a scenario where a huge number of heterogeneous (wireless and sometimes autonomous) ubiquitous and decentralized devices communicate and potentially cooperate among them and with the network to perform storage and processing tasks without the intervention of third parties. These tasks can be for supporting basic network functions or new services and applications that run in a sandboxed environment. Users leasing part of their devices to host these services get incentives for doing so [322]". The OpenFog Consortium [323] defines fog computing as follows; "a system-level horizontal architecture that distributes resources and services of computing, storage, control, and networking anywhere along the continuum from Cloud to Things [323]". In today's digital and technical world, any technology's birth is determined by three "W"s (what, why, and when). We compared FC and CC in terms of different qualities and criteria in Table 7. Because of its numerous benefits, FC has become a pressing demand among consumers and businesses. Some of the benefits are: "support of real-time services", "low latency", "location awareness", "support of mobility", "geographic deployment", "low-power consumption", "less network congestion", "processing a high number of nodes", "cost-effective", "reliable" etc., [324–328].

Fig. 40 depicts a typical FC architecture. In the same diagram, multiple layers are made up of networking equipment such as "routers", "set-top boxes", "proxy servers", "base stations", and so on. To provide services to edge users, all of these devices save frequently used information. EC layer (see Fig. 30) depicts the End user Devices, with all IoT-equipped devices (known as Terminal Nodes) utilizing cloud/fog server services (IaaS, PaaS, and SaaS). The fog nodes at the FC layer are "routers", "set-top boxes", "proxy servers", "base stations", and other devices. This layer contains all cloud data centers and cloud servers, as well as storage and processing resources.

4.3.1. Related previous works

Table 8 summarizes related studies that have examined the integration of IoT and FC, highlighting each paper's contribution.

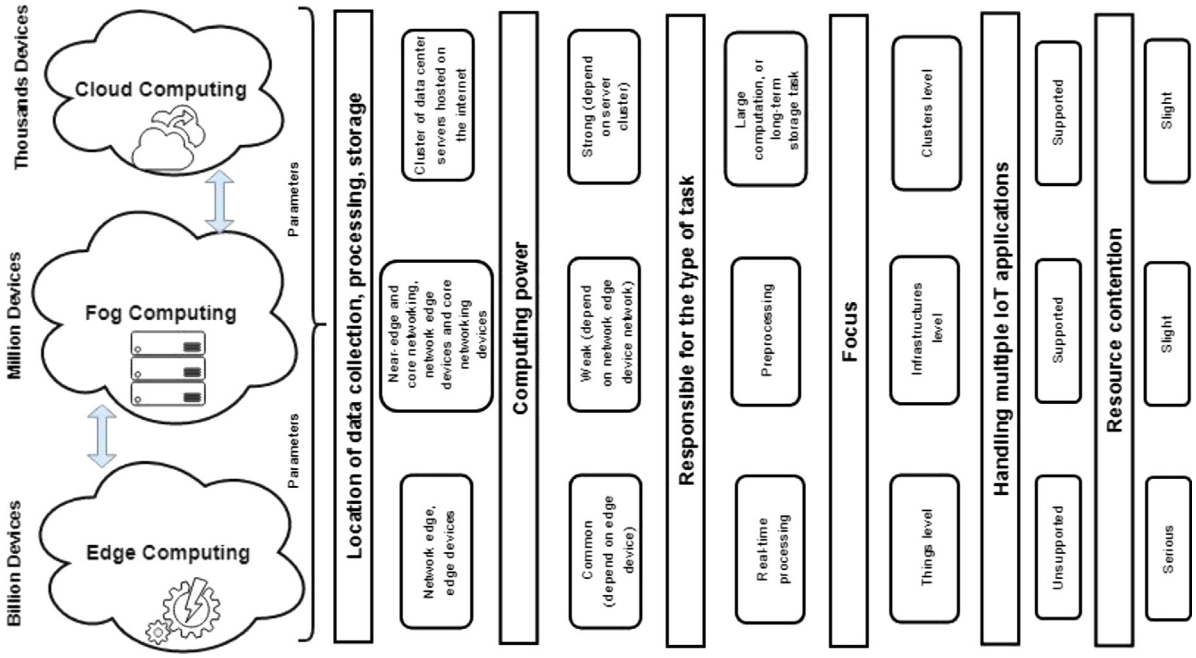


Fig. 34. CC, FC, and EC are compared.

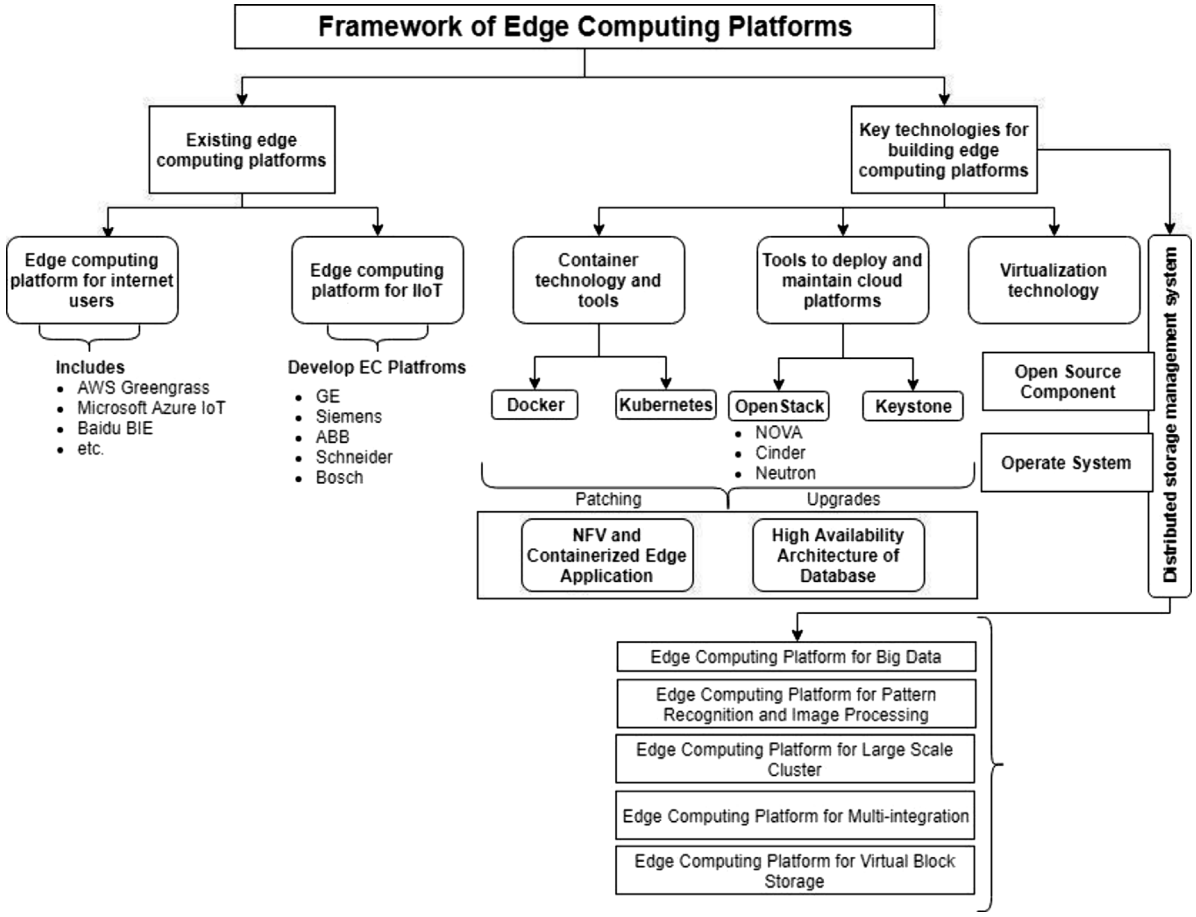


Fig. 35. Framework of EC platform.

Table 7
CC and FC comparison.

Functional requirements	Cloud computing	Fog computing
Awareness of location	No	Yes
Security	Low	High
Latency	High	Low
Server nodes	Less	More
Jitter delay	High	Low
Bandwidth of network	High	Low
Attack	Low	High
Geographical deployment	Centralized	Distributed
Server nodes location	Within the internet	Local Network (Edge nodes)
Client–server distance	Many hops	Single hop
Measurement of security	Measurable	Difficult to measure
Hardware	Computing power and scalable storage	Computing power with limited storage
Agent	Internet agent	Mobile agent
Time of response	Minutes	Milliseconds
Connectivity	Leased Line	Wireless

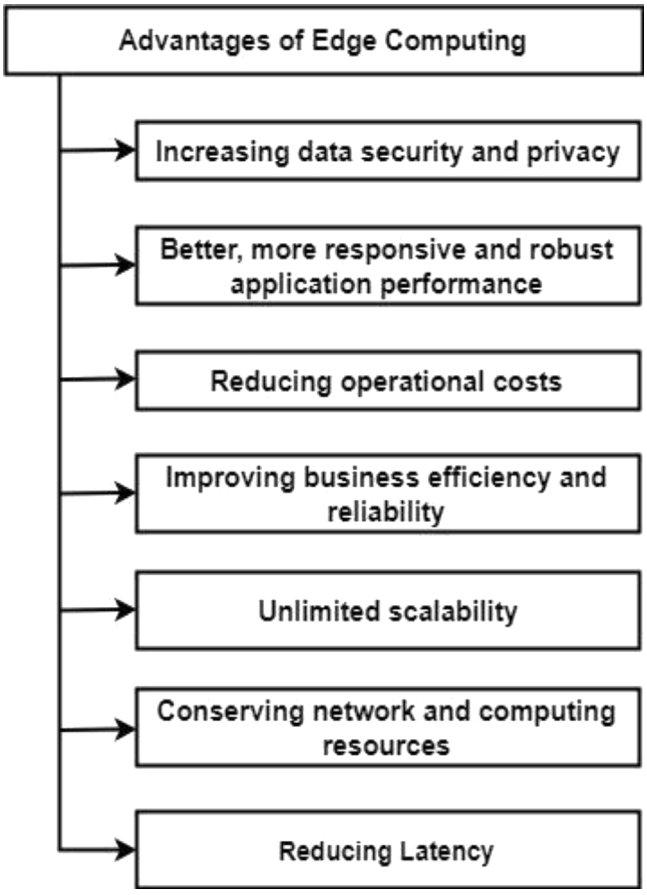


Fig. 36. Advantages of EC.

The architecture of FC (as shown in Fig. 41), according to Mukherjee et al. [329], Aazam and Huh [330,331], and Muntjir et al. [332], comprises six layers (L1, L2, …L6): "physical and virtualization", "monitoring", "pre-processing", "temporary storage", "security", and "transit layer".

4.3.2. Applications of FC based on IoT

Various FC applications are listed in Fig. 42. FC can play a critical role in a variety of IoT applications. It is anticipated by Cisco that autonomous vehicles will become a new trend in the automotive industry [362]. In the coming years, it is predicted that all new vehicles will be equipped to connect with nearby cars and the internet, allowing for communication between cars, access points, and traffic signals, resulting in exceptional user services [355]. With fog computing, which eliminates the delay related to the centralized cloud approach, lives can be saved by decreasing accidents and other incidents [363]. Intelligent traffic lights can be viewed as fog nodes that communicate with each other to send signals to nearby vehicles. Wi-Fi, 3G, smart traffic signals, and roadside equipment enhance fog interactions between cars and access points [355]. FC solves the challenges posed by the significant computation and storage demands of smart homes. Integrating smart home technology is difficult due to the various manufacturers and operating systems. FC consolidates all platforms, providing resources for smart home applications [321]. Additionally, it offers scalable storage, real-time processing, and low-latency resources [364]. Actuators in WSNs can be compared to fog nodes that perform various functions to manage sensor-equipped end devices. These WSNs need less "bandwidth", "energy", and "processing power" [355]. FC is particularly useful in the healthcare sector, where real-time processing and event reactions are crucial [365]. Combining FC, IoT, and cyber–physical systems will enable the development of "intelligent medical equipment", "smart buildings", "agricultural", and "robotic systems" [319]. AR applications are highly sensitive to latency because even a small delay in the application's response can adversely affect the user experience [361]. By merging fog and cloud servers to enable real-time applications, FC could become a major player in the AR market. Zao et al. [366] developed an augmented brain–computer interface game based on FC (ABCI) and demonstrated how a cloud server and fog technology can create a real-time game.

Table 9 presents a comparison of the studies. While CC has gained popularity, certain services and applications are unable to leverage this extensive computing paradigm due to cloud-specific challenges like "lack of mobility", "lack of location awareness", and "excessive latency". FC has emerged as a reliable infrastructure for delivering resources at the network's edge [139]. Due to its potential, FC has garnered significant attention from researchers. This section examines some of their applications and their performance across different implementations and methodologies.

Because the Fog is a CC extension, it inherits many of the Cloud's risks [382]. Furthermore, Fog nodes in general are "honest but curious" [383]. Fog nodes are considered a desirable target for a range of attacks, which is why they must be placed by trustworthy Fog vendors who are committed to providing reliable services to end-users. Fig. 43 lists the many attacks that can be launched against Fog nodes [383–386]. Several hostile assaults

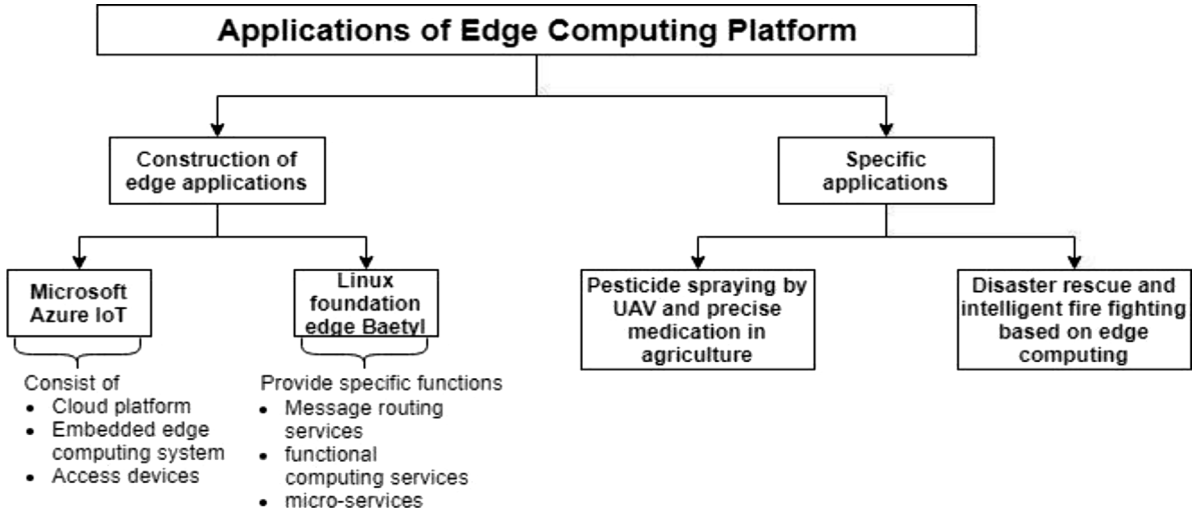


Fig. 37. Applications of EC platform.

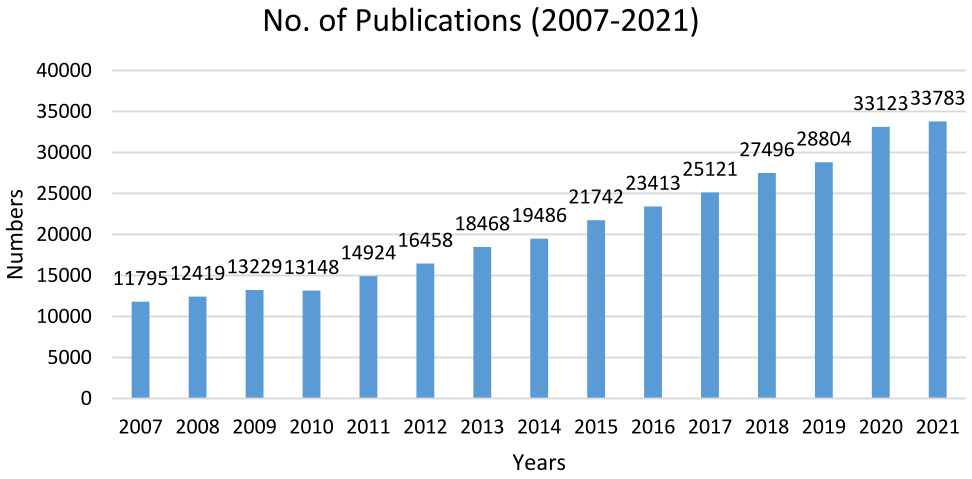


Fig. 38. Published articles of EC in the ScienceDirect database (2007–2021).

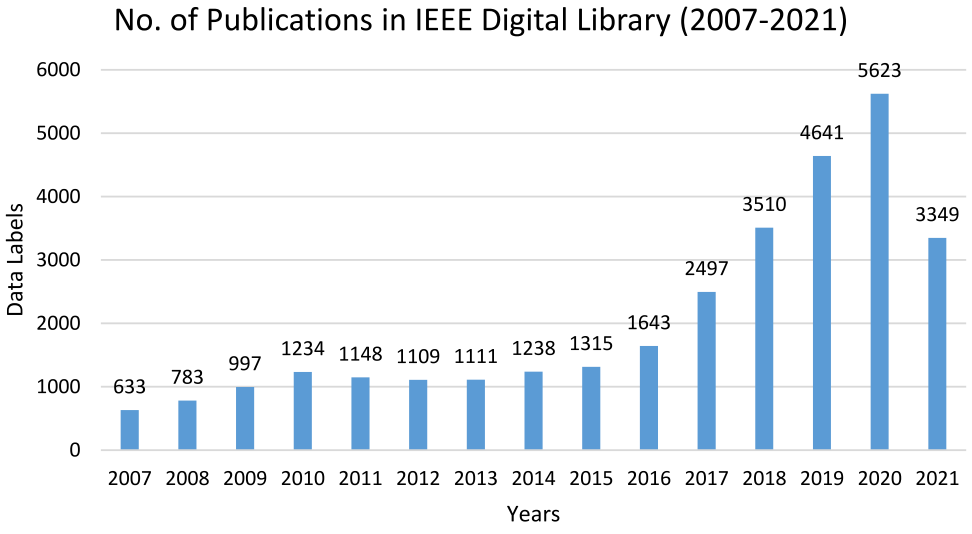


Fig. 39. Published articles of EC in IEEE digital library (2007–2021).

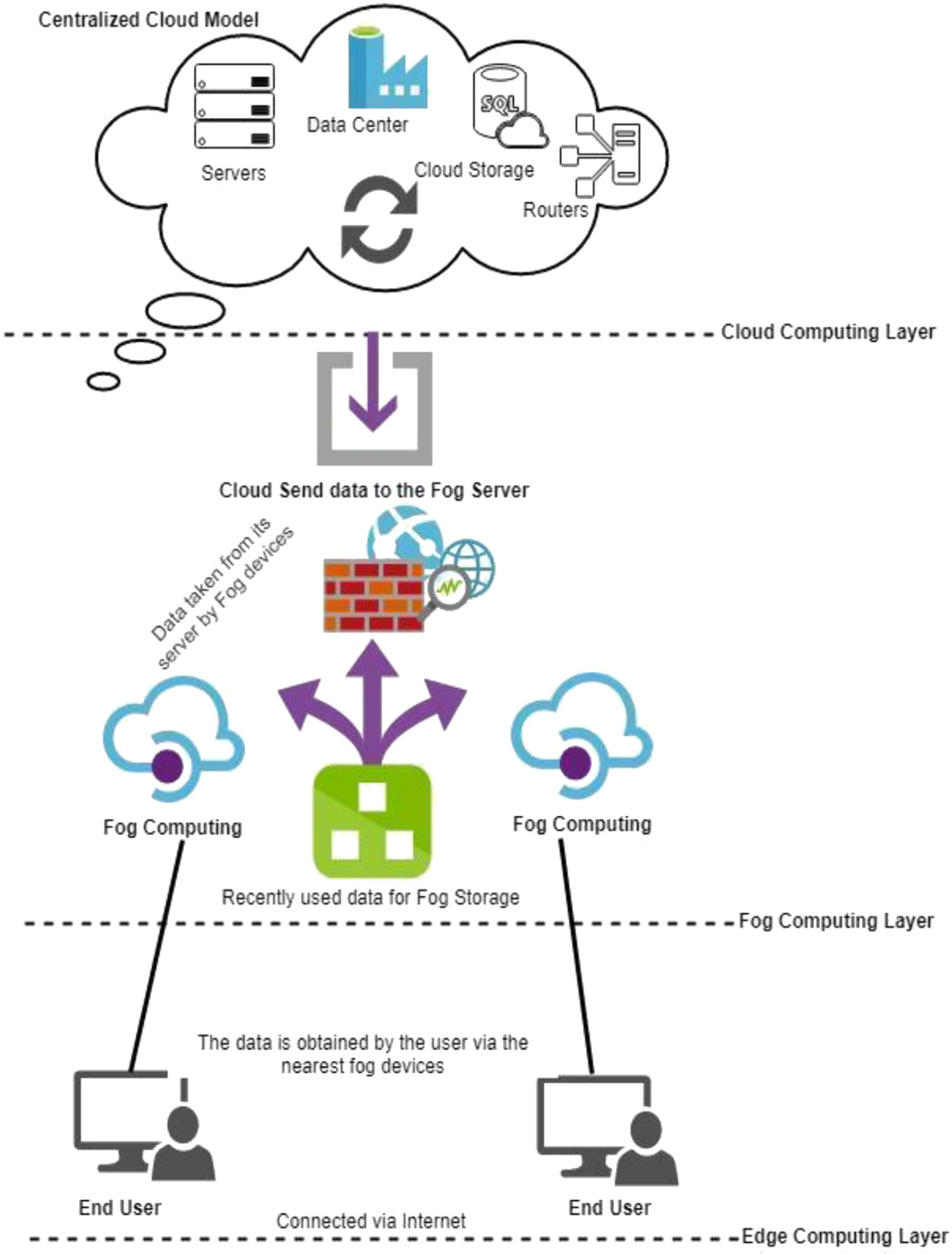


Fig. 40. Cloud-fog computing architecture in its most basic form.

(e.g., "man-in-the-middle", "authentication", "distributed denial of service (DDOS) attacks", "access control", and "fault tolerance") may be encountered by fog nodes, as well as novel security and privacy concerns [387].

The current status of security and privacy threats, as well as the challenges and requirements, of Fog, are all investigated in the study. Fig. 44 illustrates the security and privacy challenges of FC. Different security and privacy challenges exist for network services and communication ("fault tolerance", "authentication", "access control", "malicious fog node", "light-weight protocol design", "packet forwarding", "detection systems", and "trust management"); device privacy ("usage privacy", "location

Table 8
The authors looked through a list of relevant study.

Authors	Summary of author's works
Khan et al. [333]	The authors of a study focused on addressing a crucial concern, which is the security vulnerabilities in the fog computing environment. They aimed to provide a detailed comparison of EC, cloudlets, and micro-data centers and explored various architectures where the concept of FC can be applied.
Ni et al. [334]	The IoT is being proposed as a domain that can be merged with FC. The authors discovered several fog-assisted IoT applications in the sectors of "smart transportation", "smart grid", and "smart healthcare", among others. The authors also identified certain open research areas in the FC domain that could lead to new architectures that will become more important in the future.
Mukherjee et al. [335]	Under the file of FC, discussed many security concerns and issues, and proposed some strategies to overcome them. The security challenges mentioned in FC include trust, authentication, secure communications, end-user privacy, and malicious attacks. The disadvantage of this article is that it could be improved by using the concept of FC to address security concerns [336,337].
Alrawais et al. [325]	Open research topics such as "privacy", "IoT device updates", "safe and efficient protocols", "authentication", "attack detection", "location verification", and "access control" were discussed by the authors.
Dong et al. [338]	The authors examine the use of elliptic curve cryptography (ECC) and decoy technology in a security data theft attempt.
Vishwanath et al. [339]	The authors suggest the use of the AES algorithm to overcome security challenges in the fog computing environment. They look at three different types of datasets and apply the proposed encryption approach to them.
Ibrahim et al. [340]	Proposed a mechanism that offers a secure way for both fog users and fog servers to authenticate each other when communicating at the fog layer. By using a single secret key, the communication between the two parties can be fully authenticated and ensured to be secure.
Sehgal et al. [341]	The author worked on an FC security framework and designed security techniques to address security concerns in the realms of IoT, CC, and FC. A study looked at the differences between FC and CC, as well as the security and forensics gaps between the two. The authors propose connected "vehicles", "wireless sensors and actuator networks", "decentralized smart building control and IoT", and "cyber–physical systems" as some of the architectures.
Dsouza et al. [342]	For FC, the authors developed a framework called Policy Driven Security Management (PDSM). This was implemented to improve "interoperability", "provide more robust features", and "secure data management".
Stolfo et al. [228]	The authors analyze how users rely heavily on CC and FC infrastructures to store their personal and critical data, which raises concerns about data security. To address this issue, they propose and implement a novel data security method using decoy technology.
Sarkar et al. [343]	The authors conducted a thorough comparison of the FC and CC concepts. They have paid special attention to the IoT. The authors have demonstrated that the FC paradigm can assist in "reducing latency", "power consumption", and "carbon dioxide emissions using mathematical factors".
F. Bonomi et al. [319]	Suggested an FC environment to enable IoT devices with limited resources.
Mahmud et al. [344]	Developed a taxonomy for FC by considering the challenges and features associated with it.
Hong et al. [345]	Suggested a mobile fog (MF) solution that facilitates the aggregation and local processing of data for IoT applications, along with load-balancing capabilities.
S. Khan et al. [346]	Examine FC applications to find common security vulnerabilities.
F. Bonomi et al. [347]	Presented a hierarchical distributed fog architecture and illustrate its key characteristics using examples of a smart traffic light system and a wind farm.
M. Sookhak et al. [348]	Put forth a cross-layer architecture for VFC that outlines decision-making mechanisms and how diverse services are distributed among vehicles.
Stojmenovic and Wen [349]	The security and privacy vulnerabilities in the current fog computing paradigm have been demonstrated through the example of a man-in-the-middle attack.
C. Puliafito et al. [350]	The paper presented three scenarios to examine the challenges and problems associated with mobility support in a fog computing environment. The analysis focused on identifying the key issues related to mobility support in this context.
K. Lee et al. [351]	Investigated the security and privacy implications of FC integration.
A. Yousefpour et al. [352]	Proposed a paradigm for analyzing, evaluating, and modeling service latency in IoT–fog–cloud scenarios.
K. Saharan and A. Kumar [353]	Gave an overview of FC as a CC expansion.
A. Alrawais et al. [354]	Proposed a fog-based mechanism for improving the propagation of certificate revocation information among IoT devices for improved security.
N. Peter [355]	Presented an overview of FC potential in real-time applications, as well as how to overcome congestion and latency issues.
X. Hou et al. [356]	The proposed solution is an architecture called VFC (Vehicle Fog Computing) which aims to facilitate communication and computation collaboration among end-users by leveraging the resources available on each vehicle.
M. Aazam and E. Huh [331]	Suggested a model for resource management using FC. This paradigm allows for dynamic resource management and can adapt to the needs of CSPs.
O. Skarlat et al. [357]	Proposed a fog resource provisioning architecture that allows for delay-sensitive exploitation of fog-based computing resources.

(*continued on next page*)

Table 8 (*continued*).

Authors	Summary of author's works
S. Yi et al. [358]	Provided a summary of fog computing (FC) and highlighted the potential challenges that may arise when implementing FC systems.
M. Chiang and T. Zhang [359]	Presented an overview of the advantages and disadvantages of fog computing (FC), with an emphasis on the networking environment of the Internet of Things (IoT).
V. Gazis et al. [360]	Addressed operational requirements for industrial processes, an adaptive operations platform (AOP) that provides end-to-end management of fog computing has been proposed.
Dastjerdi et al. [361]	Put forth a fog computing (FC) reference architecture that processes Internet of Things (IoT) requests within the local fog environment, rather than relying on cloud resources.
Y. Shi et al. [329]	Described the salient features of FC in the healthcare sector.

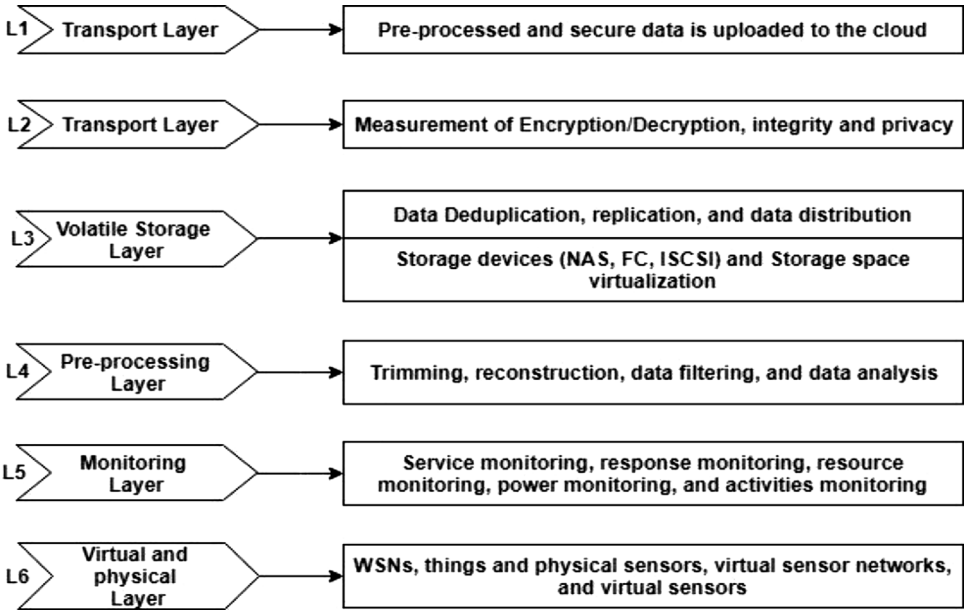


Fig. 41. FC architecture consisting of multiple layers.

privacy", "data privacy", and "identity privacy"); data processing ("forensic", "data identification, aggregation, and integrity", "verifiable computation", "data distribution", "aided computation", "data protection and search", "big data analysis", and "content distribution").

Fig. 45 depicts the new taxonomy, which was developed using security principles from the FC literature. Different security techniques for FC environment like machine learning techniques (deep learning, real-time machine, and J48 decision), computational intelligence techniques (F-IoV, Fog FISVER, Evolutionary Gsame), Cryptographic techniques (Block-chain, AES and SSL, Diffie–Hellman, Decoy, Access Control, CP-ABE, Shibboleth), and other techniques include (Bounded retrieval, Intrusion detection, VFN and SDN, Win Win Spiral model, Virtualization, DFC, Multitenant, HPCS, Aggregation scheme, Markov and honeypot, Sb-SBP). Table 10 depicts the important findings for securing the FC environment.

4.3.3. Year-wise analysis of FC publications

From 2007 to 2021, Fig. 46 shows the number of FC articles from the ScienceDirect database per year. There is an observable trend of publications related to Fog computing, showing exponential growth over time. Fig. 47, shows the number of FC articles with content type has increased in the Springer database. Fig. 48 depicts Springer's published articles by discipline/subdiscipline. Fig. 49 illustrates the content kinds of published papers in the IEEE digital library database. Fig. 50 depicts the year-by-year distribution of published publications (Conferences/Journals/Books/Magazines/Standards) in the IEEE Xplore database (2007–2021).

5. Deep learning enabling applications

Deep learning (DL) applications are in use across the globe. "Healthcare", "social network analysis", "audio and speech processing (such as detection and enhancement)", "visual data processing methods (such as multimedia data analysis and computer vision)", and "natural language processing (translation and sentence classification)" are among these applications (See Fig. 51) [401–404]. "Classification", "localization", "detection", "segmentation", and "registration" are the five categories in which these applications have been categorized. Although DL applications have different objectives, there is significant overlap in their implementation in the pipeline, as depicted in Fig. 52. Classification refers to grouping a set of data into different categories. Detection involves identifying important features in an image while taking into account the background. Bounding boxes are used in detection to enclose multiple objects, which may belong to different classes. Localization is used to identify an object that is enclosed within a single bounding box. Segmentation involves outlining the boundaries of the target object, which are also labeled in semantic segmentation. Registration involves overlaying one image (2D or 3D) onto another. Healthcare is an extensive DL application, as it is critical to human life [405–410]. DL has made significant strides in healthcare, making it an essential field

Table 9
A survey of the literature on CC and FC.

Source with year	Tools	Objective	Important findings
Guevara, Judy et al. (2021) [367]	Integer linear programming	Introduces two integer linear programming-based schedulers for scheduling work in the cloud or on fog resources.	The numerical results indicate that QoS criteria can be maintained while using established algorithms such as Random and Round Robin
Hong (2017) [368]	CloudSim	To better serve applications that are sensitive to latency, time, location, and vast size.	The proposed algorithms outperformed the baseline algorithms by at least 30.3%, 20.0%, and 89.4%, respectively, according to the numerical results.
Srirama, Satish et al. (2021) [369]	Actor Model, Akka toolkit	For Akka distributed Fog applications, the author proposes an Actor Model-based frame.	To assess the proposed actor-based architecture's applicability, a case study was conducted on a wireless sensor network, demonstrating the feasibility of creating applications on Fog networks.
Osanaiye et al. (2017) [370]	Linear regression approach	To keep application downtime to a minimum	The cloud to the edge of the network had to be extended to FC due to the low latency requirements of applications. Virtualized networks, such as CC and FC, can comprise "storage", "networking", and "computing".
Karagiannis and Schulte (2020) [371]	Java	To provide advice for choosing a suitable FC distributed architecture while taking into account the needs of fina applications.	In terms of cloud independence, the hierarchical architecture outperforms the flat architecture, resulting in a 13% reduction in communication latency.
Fu et al. (2018) [372]	CloudSim	To address difficulties in Industrial IoT such as "secure data storage", "data processing", "dynamic data gathering", and "effective data retrieval".	The suggested solution remarkably enhances the reliability and security of IoT data storage and recovery.
Abedi and Pourkiani (2020) [373]	Matlab	Distribute jobs between cloud and fog servers to reduce Internet traffic and response time.	The proposed solution was found to significantly reduce internet traffic and response time as compared to fog and cloud-based methods.
Garcia et al. (2018) [374]	MNIST	Taking advantage of local opportunities to increase the cloud model's functioning.	As a result, they explained that FC was created to supplement CC capabilities by utilizing location opportunities.
Jindal et al. (2020) [375]	FogSim	They proposed a mechanism for making the critical decision of when to shift the job from FC to CC.	If the computing capacity of Fog is insufficient to handle heavy computations, the workload can be offloaded from Fog to the cloud.
Guan et al. (2018) [376]	VANET	To provide a foundation for the development of stable FC data services.	They clarified that data management software could not be used in CC explicitly.
Ema et al. (2019) [377]	Blockchain Technology	To figure out how to deliver real-time Fog infrastructure services, with a focus on cloud storage issues.	According to their findings, traditional cloud data centers are not suitable for handling large amounts of data storage due to their high costs and time-consuming nature.
Aazam et al. (2018) [378]	CloudSim	To handle cloud IoT convergence difficulties and to conflate FC with CC.	The study reported that the suitability of Fog depends on the type and scale of the tasks being performed, as per the results of the performance evaluation and review.
Ali et al. (2019) [379]	CloudSim	The author made a distinction between FC and CC to describe the differences between both technologies.	As per the study, Fog infrastructure is more flexible and offers superior data processing capabilities compared to CC as it demands lower network bandwidth instead of transferring all the data to the cloud
Delfin et al. (2019) [380]	CISCO	To investigate the focal points for FC administrations in a few different spaces.	The researchers found that fog computing is akin to cloud computing but with the key difference of utilizing closer end users for faster processing and response times to clients.
Rabay'a et al. (2019) [381]	Peerfact-Sim.KOM	Fog P2P model, which shows that the bandwidth efficiency of CC and FC architectures is superior to the file transfer method.	The study's findings provide a solid foundation for future attempts to improve peer-to-peer FC.

of study. Therefore, we will focus on DL applications in medical image analysis to illustrate DL applications.

5.1. Deep learning enabling cloud applications

DL is becoming more accessible thanks to cloud computing services, which make it easier to manage massive datasets and train algorithms on dispersed hardware. DL is made possible by cloud services in four ways:

- Provide on-demand access to large-scale computing capability, allowing model training to be distributed across numerous machines.
- Give users access to specialized hardware, such as GPUs, FPGAs, and massively parallel high-performance computing (HPC) systems.
- Do not necessitate an initial investment—you can obtain advanced hardware, as well as big numbers of hardware,

Table 10
Important findings for securing the FC environment.

References	Important findings
Sharma et al. [388]	The authors suggested a revolutionary distributed CC platform based on blockchains with software-defined networks (SDN).
Yang et al. [389]	The authors discussed the development of a safe locating protocol with privacy in the constrained retrieval mode.
Zahra et al. [390]	The implementation of the Shibboleth protocol in the cloud IoT network was carried out by the authors to ensure secure data access and outsourcing.
Jiang et al. [391]	Authors developed novel CP-ABE strategies to protect any type of key delegation misuse.
Wang et al. [392]	In fog-based public CC, the authors suggested the anonymous and secure aggregation method (ASAS).
Xiong et al. [393]	The authors presented a mechanism that can protect against various forms of attacks from CC and FC providers.
El-Latif et al. [394]	The authors proposed a security technique based on a quantum steganography architecture to secure the flow of communications in fog cloud IoT.
Shropshire et al. [395]	The study highlights the importance of ensuring "integrity", "confidentiality", and "availability" (CIA) to address security concerns in the desired infrastructure, with a focus on identifying potential vulnerabilities before software development.
Koo and Hur [396]	The authors proposed an efficient and secure data deduplication system for Fog computing, which ensures proper ownership management.
Fu et al. [397]	The authors created a storage and retrieval architecture that combined FC and CC approaches in a secure, flexible, and efficient manner.
Singh et al. [398]	The authors proposed a cyber security framework for detecting malicious edge devices in a Fog Computing environment, which utilizes three technologies: "Two Markov models, an intrusion detection system (IDS), and a hosted virtual desktop (HVD)".
Donovan et al. [399]	The authors utilized Fog Computing (FC) to develop real-time embedded machine learning for Industrial 4.0 design.
Sood and Mahajan [400]	Sood and Mahajan presented a system that uses IoT sensors, FC, and CC to distinguish and identify users affected with mosquito-borne diseases (MBDs) and to keep track of them.

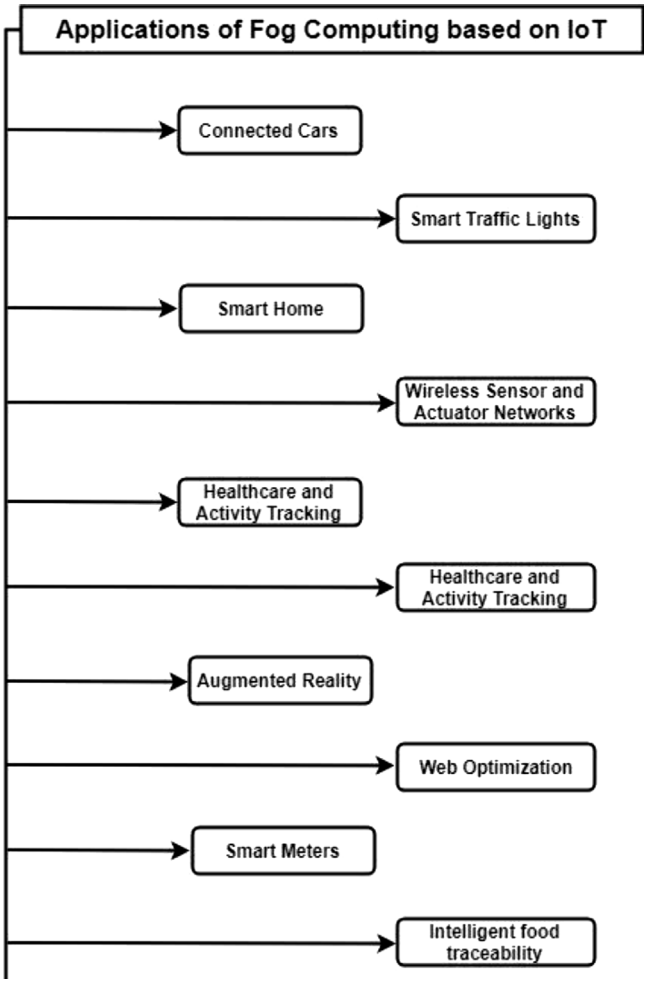


Fig. 42. Applications of FC with IoT.

without purchasing it. You simply have to pay for the time you use.
- Assist with DL workflow management—cloud services offer advanced tools for managing datasets and algorithms, as well as training and deploying models to production.

DL workloads can be executed in a "do it yourself" model in each of these clouds. This entails choosing pre-installed deep learning infrastructure machine images and operating them in an infrastructure as a service (IaaS) paradigm, such as Amazon EC2 instances or Google Compute Engine VMs. All of the cloud providers we will look at below offer DL compute instances with specialized hardware like "graphics processing units (GPUs)", "field-programmable gate arrays (FPGAs)", and "TensorFlow Processing Units (TPU)". Following training, these models could be used in embedded devices to analyze data using the learned weights. The models must be optimized based on the embedded hardware and application needs in such cases.

We will look at each cloud's platform as a service (PaaS) offering for DL users in the sections below. These PaaS products include both the hardware and software services needed to manage DL pipelines, from data import to production deployment and real-world inference.

5.2. Deep learning enabling fog applications

The sheer amount of data generated by IoT devices makes it impractical for traditional cloud computing models to satisfy IoT application needs [411]. Such massive data uploads to the cloud would result in network overhead, bandwidth usage, and latency problems [412]. As a result, service providers are now turning towards edge computing and fog computing to decrease data transfer costs, reduce network delays [413], and ensure security and privacy [414]. Edge hardware, sensors, and actuators used in IoT systems must meet requirements for latency, bandwidth, and security [411]. Fog computing involves bringing computing and storage resources nearer to the edge of a network, which can eliminate bottlenecks in networking and processing [412]. This approach enables fast processing near the data source [415]. Fog computing's ability to support low-latency communication networks and its inherent complexity and dynamism make it

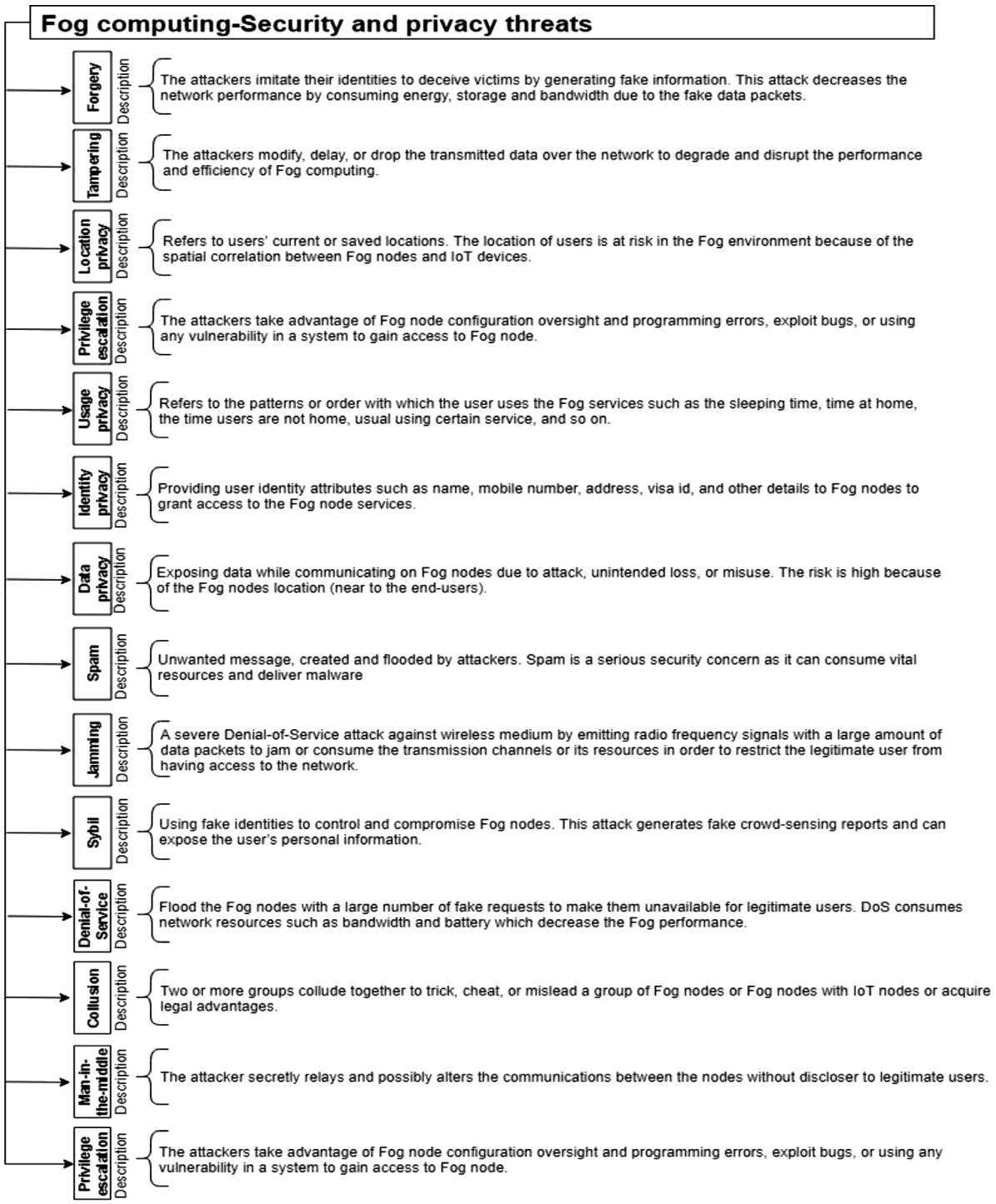


Fig. 43. Privacy threats in FC.

conducive to performing complex processing tasks in favorable environments. Fog computing provides benefits to society in various areas such as Industry 4.0, healthcare, autonomous cars, and intelligent buildings, among others, as illustrated in Fig. 53. However, fog computing still encounters security and performance issues. To address these difficulties, there is a growing trend towards the use of machine learning (ML), which is a subset of artificial intelligence (AI). There has been a recent surge in interest in utilizing machine learning (ML) to enhance the performance of fog computing applications and provide effective services such as improved reliability, latency minimization, energy demand, security, confidentiality, resource management, and traffic management [40,416,417]. By incorporating machine learning (ML), resource management in fog computing can be optimized to improve computation, decision-making, resource allocation, and delay prediction. Given the decentralized nature of fog computing, the use of ML algorithms enables accurate real-time data processing and analysis while controlling network overhead and communication traffic. The successful combination of fog computing, machine learning (ML), and security measures for devices, networks, and data has resulted in powerful services for end-users and upper-layer applications. ML is a

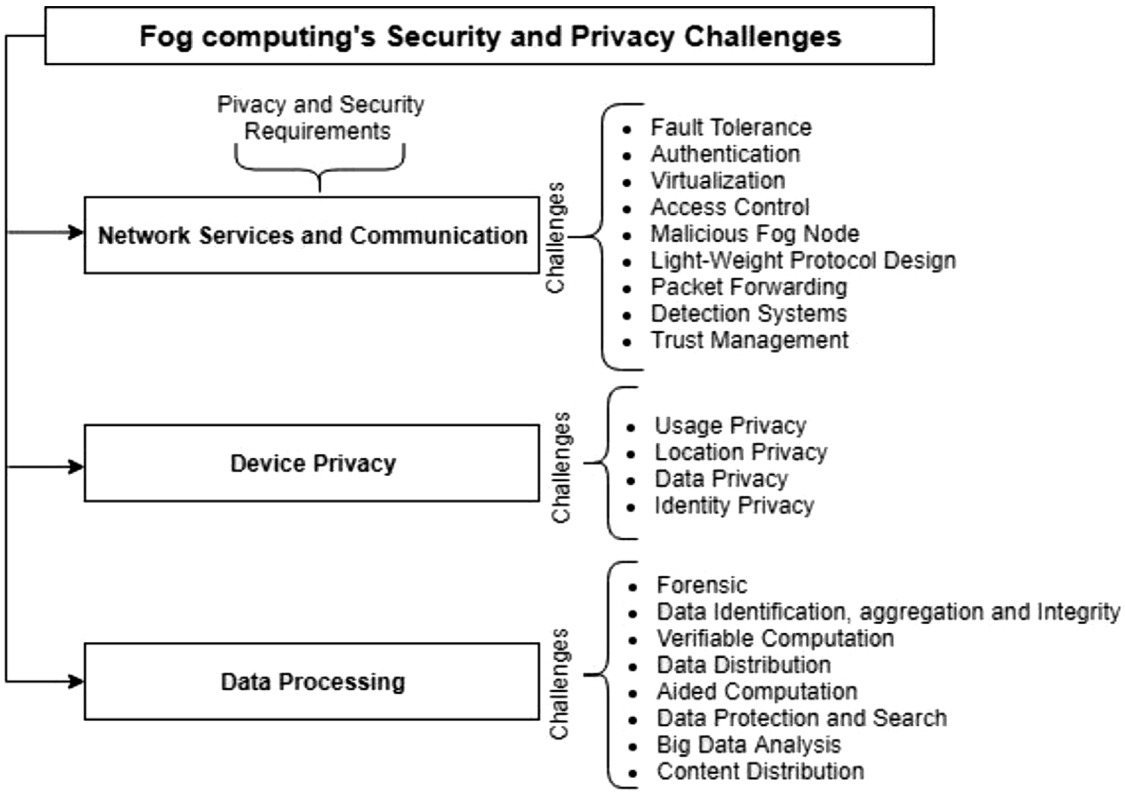


Fig. 44. FC's security and privacy challenge.

key IoT technology that offers the potential for intelligent data processing and inference, making it an essential component for creating the smart environments necessary for autonomous operations in diverse domains, such as smart grids, home automation, smart farming, and smart industries [412–414]. The integration of ML with cloud, edge, and fog computing is critical for the implementation of IoT, especially for resource-constrained devices as it addresses quality-of-service (QoS) issues while simultaneously improving performance, controllability, productivity, and reducing costs [414,418].

Fog computing extends the benefits of cloud computing to the network edge, where both IoT and human users are present. It can leverage human-driven or device-driven intelligence, and its capacities for sensing, processing, networking, and storage enable context awareness for decision-making and local resource management. By integrating machine learning with cloud, edge, and fog computing, it is possible to enhance the overall performance of the system [419]. As numerous sensing devices are used for monitoring and communication, IoT offers excellent opportunities for improving human life and industrial progress. In a human-driven model, people are the system's data source, and the network is shaped to best serve them based on their behavior. Together, these two strategies can assist in creating fog computing systems that can provide the QoS requirements of the IoT. Incorporating machine learning into an IoT environment allows for more profound insights and can facilitate the development of intelligent and effective IoT applications [413]. Moreover, it can be used to address networking issues with routing, distribution of resources, traffic engineering, and security [417,420–426]. To effectively analyze the enormous amounts of data generated by IoT, neural networks have been employed [427]. Moreover, advanced artificial intelligence (AI) utilizing deep learning has proved successful in making predictions., decision-making, and data analytics [428]. The integration of machine learning with AI has significantly expanded the capabilities of IoT. By leveraging cutting-edge machine intelligence techniques, it has become possible to make critical operational decisions based on the vast amount of IoT sensory data and gain valuable insights into a range of real-world scenarios. ML and IoT must collaborate to resolve complex real-world challenges and meet computational and communication requirements,

5.3. Deep learning enabling edge applications

Edge computing and deep learning have experienced rapid progress and considerable success in their respective sectors in recent years. The fusion of these two technologies has the potential to open up many more doors, enabling the development of a slew of new applications. The integration of edge computing with artificial intelligence (AI) has been increasingly utilized to achieve edge intelligence. In this section, we present a comprehensive review of current research on edge computing applications that incorporate deep learning. Specifically, we summarize these works across several areas, like intelligent media, transportation, cities, and smart enterprises, as illustrated in Fig. 54.

A range of businesses, including camera-based surveillance [429] and augmented reality, are increasingly using video analytics [430]. (AR). Due to limited processing power in cameras, conventional video analytics frequently utilize cloud computing for content processing. This method involves transmitting video content to backend cloud servers for processing, and then sending the results to frontend devices. However, this processing mode is associated with significant latency and high bandwidth usage, making it unsuitable for applications that require rapid responses, such as object tracking and identification [431]. For instance, Amazon has created the first deep learning-capable video camera in the world, complete with a locally executed DL function that enables real-time objection even without the use of the cloud. Adaptive video streaming [432] has become a critical issue in

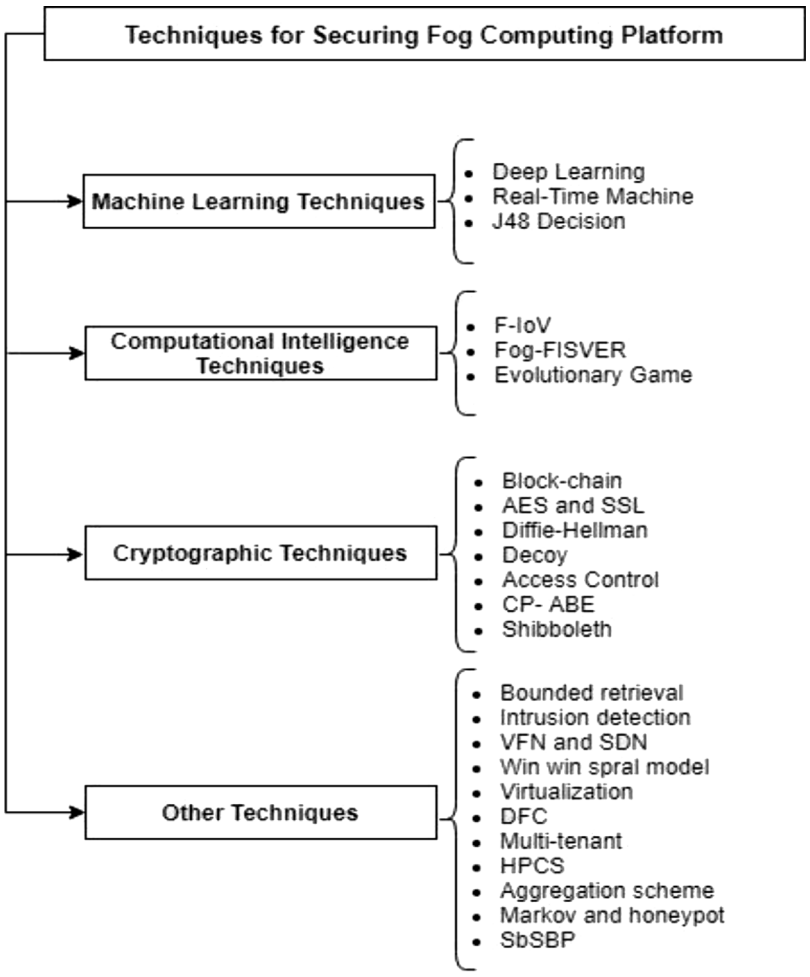


Fig. 45. Techniques for securing the FC platform.

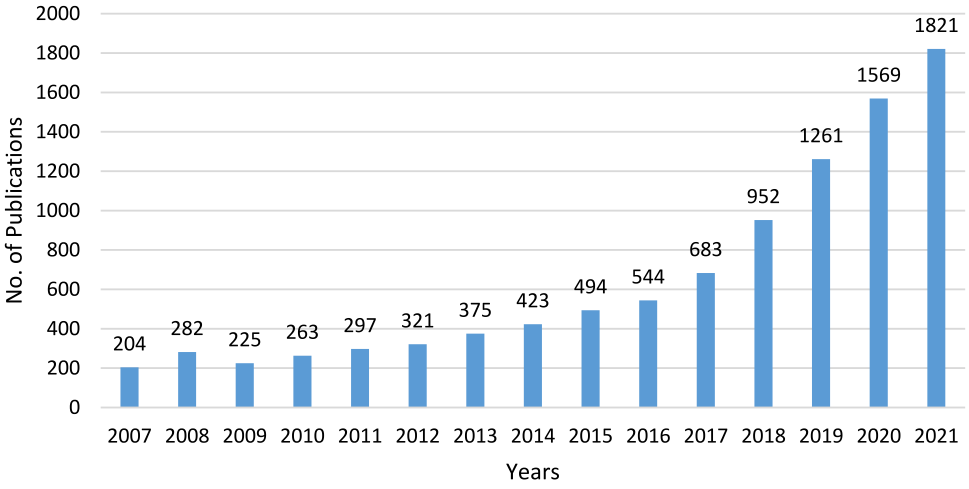


Fig. 46. No. of FC publications in the ScienceDirect database (2007–2021).

video content delivery today to ensure the best quality of experience (QoE) over the Internet. The primary concept is to choose the most suitable video bitrate based on various factors, such as network conditions, stability, fairness, and user preferences for video quality. The majority of existing adaptive streaming methods [433,434] focus on client-based adaptation, which aims to adjust to changes in bandwidth based on several observed or anticipated criteria, such as buffer size and bandwidth constraint. In recent times, deep reinforcement learning has been utilized at the client end to automatically learn the adaptation policy, resulting in improved QoE [435,436]. This approach is a recent development in deep learning. Video content caching is an essential application that has garnered significant academic attention due to its notable benefits in improving multimedia services [437].

With the advent of 5G and mobile edge computing (MEC), the car is being seen as the next intelligent information carrier

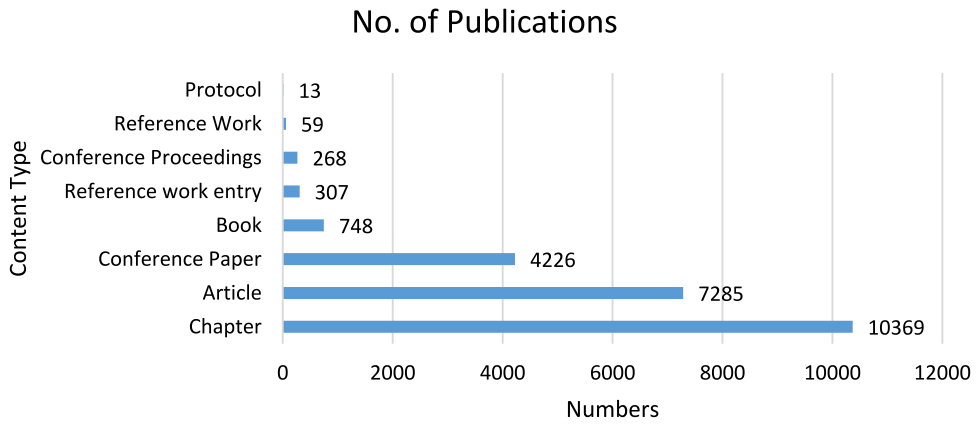


Fig. 47. Springer database depicts the No. of publications based on content types.

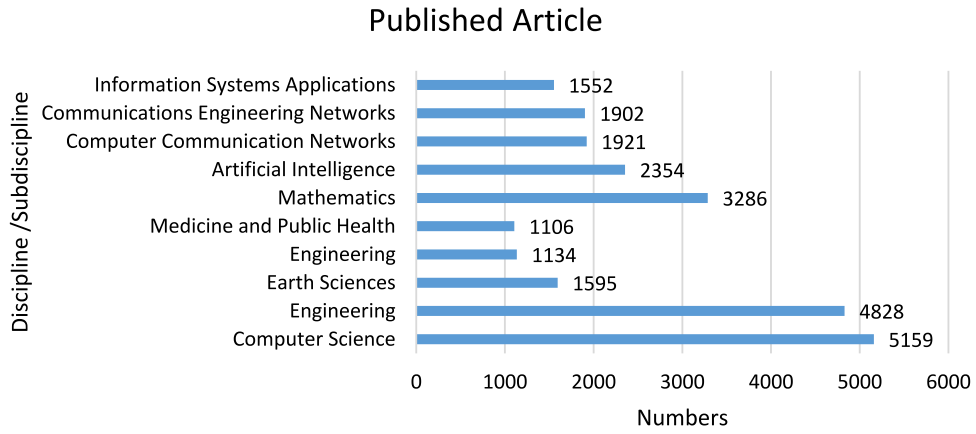


Fig. 48. Published articles based on discipline/subdiscipline in Springer databases.

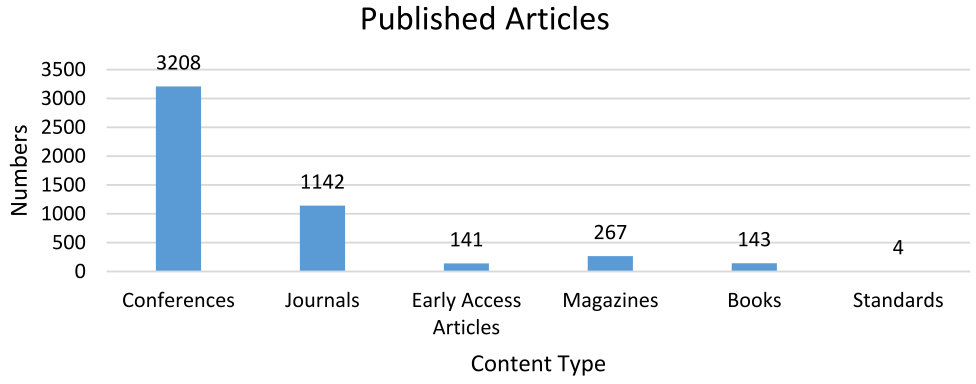


Fig. 49. Total published articles to date based on the content type in IEEE digital library.

after the smartphone. This has made it possible to access vehicle data everywhere and at any time, with very little delay, enabling the Internet of Vehicles (IoV) [25]. When combined with recent advances in deep learning, IoV will facilitate more intelligent transportation management, including autonomous driving [438], traffic prediction, and traffic signal control. Autonomous driving poses significant challenges in terms of intelligent sensing and perception [438]. Before making cognitive perception and decision-making, the cars first collect data from a variety of onboard sensors, including cameras and radars. The prerequisites for autonomous driving, which include "high computation capacity", "real-time feedback", "adequate redundancy", and "security", may not be met by relying solely on vehicle-based and cloud-based solutions.

Understanding the patterns of vehicular and pedestrian movement is crucial for urban congestion control, housing management, and service provision. Edge computing is well-suited for the analysis and prediction of vehicular traffic due to the distributed architecture of mobile edge servers [439].

By combining edge computing with deep learning, more effective intelligent transportation management, including traffic prediction and signal control, can be achieved. Intelligent traffic signal control [440] is a popular application that has been extensively studied. Average wait times, traffic jams, and accidents can all be reduced with effective traffic control measures. Early traffic signal control technologies frequently used fuzzy logic [441] or genetic algorithms [442].

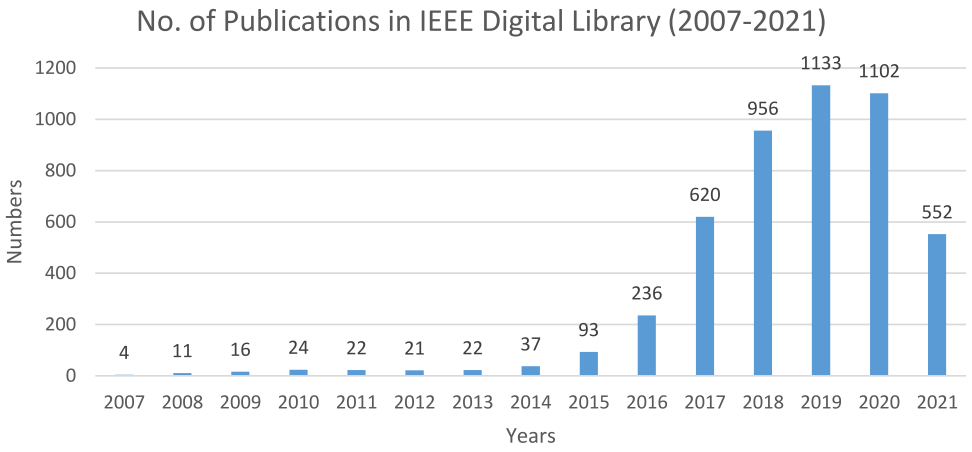


Fig. 50. Published articles of FC in IEEE digital library (2007–2021).

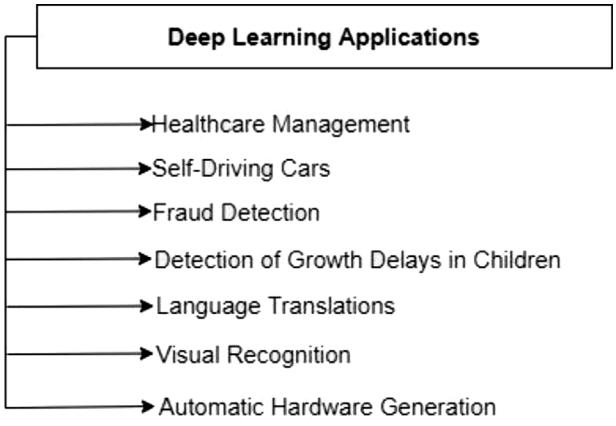


Fig. 51. Examples of different deep learning applications.

The integration of edge computing enabled with deep learning is crucial in the development of smart cities [443]. Managing the substantial amount of geo-distributed big data [444] in a city necessitates a distributed computing paradigm for efficient local processing and management. The fusion of edge computing and deep learning can enable computing intelligence to reach every corner of a city, resulting in a "smart city" that can deliver more efficient, cost-effective, energy-saving, and practical services [445,446]. The concept of Smart IoT in the context of smart homes has been widely studied to enable not only practical control but also intelligent sensing [447]. To achieve intelligent monitoring, sensing, and control in the building environment, a more comprehensive understanding and processing are required due to the complex architecture involved.

Recently, numerous initiatives [448–450] have been made towards smart buildings to reduce energy consumption, boost building security, and enhance building sensing capabilities, among other things. To lower building electricity consumption, Zheng et al. [451,452] addressed the challenge of chiller sequencing. Edge computing and fog computing have various applications in the smart grid, which is a distributed electricity distribution system that employs smart meters to monitor real-time conditions [453].

The Industry 4.0 era is characterized by a smart industrial revolution that is driven by two key concepts: intelligent data analysis and automated production processes [454]. The former is one of our primary aims and has the potential to significantly boost productivity, whilst the latter is one of our most effective methods for doing so. Edge computing has made significant strides in recent times, providing enhanced security and low-latency manufacturing by offloading considerable processing tasks from the cloud to the edge [455]. Furthermore, deep learning has significantly improved local analysis and prediction [456] at the industry's edge node, surpassing the cloud in terms of efficiency.

The smart industry heavily relies on edge computing's fast response and deep learning's intelligent processing. Manufacturing processes, including factory inspection and reliability prediction methods for assessing and minimizing faults, improve the reliability of the software technologies [457]. Weimer and colleagues [458] proposed an industrial inspection system using convolutional neural network (CNN) architecture, which can automatically generate features for inspection tasks from large amounts of raw data with minimal human intervention. Li and team [454] introduced a fog computing-based approach that offloads the computational burden to fog nodes for handling big data in industrial settings. Edge computing and deep learning have also enabled advanced analysis in industrial applications beyond process monitoring and inspection, as shown in Fig. 55.

5.4. Deep learning enabling IoT applications

Recent advancements in hardware and theoretical developments have led to the rapid expansion and growth of DL applications in IoT. In this section, we will give a detailed summary of the most recent advancements in four main areas of IoT. Applications in healthcare, such as illness analysis and monitoring, will be discussed first in our presentation. Afterward, we will examine how DL can be applied to process richer sensing inputs and upgrade our homes. Lastly, we discuss some DL applications in smart transportation, concentrating on "traffic prediction", "traffic monitoring", and "autonomous driving" (as shown in Fig. 56). Then, as a fascinating topic, we concentrate on factory inspection and related applications in the smart sector [459].

The healthcare sector is still encountering major obstacles when it comes to extracting valuable information and useful insights from intricate, multi-dimensional, and diverse biological data. delay distortion.

DL techniques have the potential to improve human health by converting big biomedical data into actionable insights [460]. Applications that track health and wellness are now possible through sensor-enabled smartphones and wearables, which utilize "Human Activity Recognition (HAR)" to identify and assess human activities [461–463]. In the field of medicine, DL methods have been successful in analyzing medical image data, particularly in medical image categorization and analysis [464,465]. For

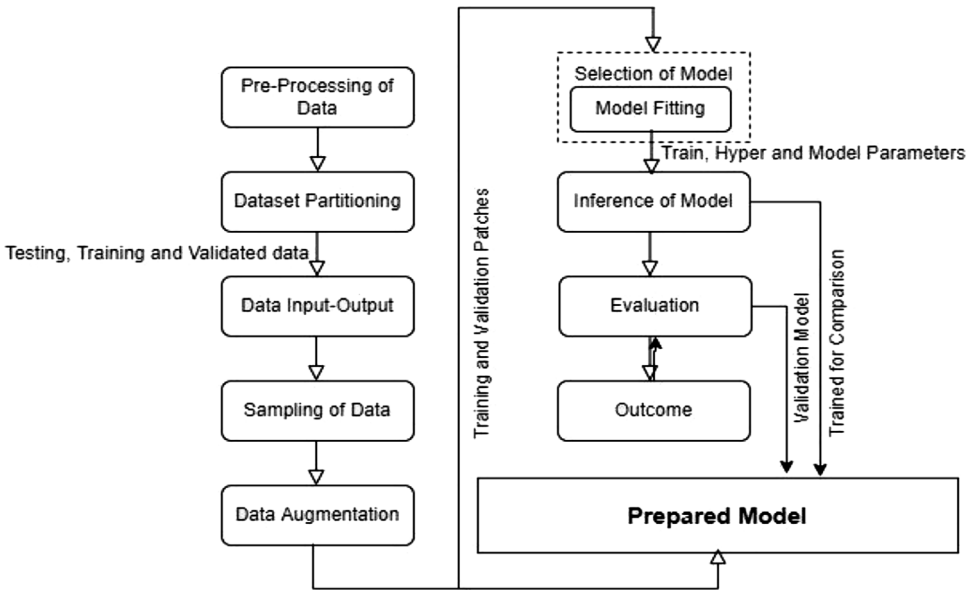


Fig. 52. Deep learning workflow.

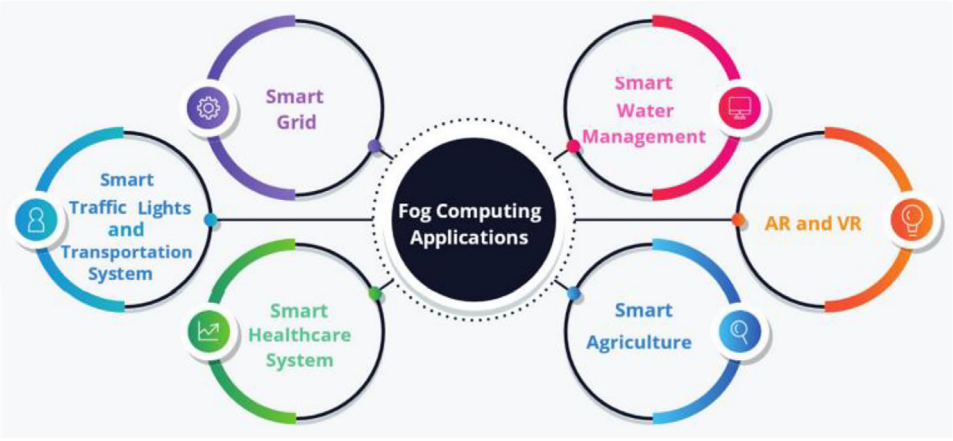


Fig. 53. Applications of fog computing.

example, CNNs have been used to automatically segment cartilage and predict the likelihood of osteoarthritis using low-field knee MRI data [466].

The concept of a "smart home" involves using home networking to improve quality of life by facilitating communication between smart home devices [447]. Deep learning techniques have been applied in many smart home applications, as illustrated in Fig. 56. However, indoor localization remains a significant challenge for many smart home applications, such as "wireless intruder detection", "elder monitoring", and "infant monitoring", due to propagation issues such as multipath effect, fading, and delay distortion, especially as GPS cannot be used indoors.

There is a growing trend of internet-connected smart home appliances that offer advanced services. Li and Lin [467] utilized power line communications (PLCs) and wireless sensor networks (WSNs) to construct a smart home control network. In addition, [468] proposed a gesture-based user interface for a smart home system, while digital personal assistants like "Microsoft Cortana", "Apple Siri", "Amazon Alexa", and "Google Assistant" are now frequently using deep learning algorithms [469] and may serve as the foundation for the next-generation smart home controller. Robots designed for home use have various capabilities such as sensing, actuation, and data storage. They can be utilized for a broad range of tasks including but not limited to map construction, human–robot interaction, object handling, and localization [470]. However, the navigation of robots in areas where GPS signals are not available necessitates tailored methods to guide the robot to its destination.

Intelligent transportation systems rely heavily on a combination of real-time and historical traffic data from various sources, such as inductive loops, cameras, crowdsourcing, and social media, to predict traffic flow. However, traditional machine learning techniques like SVM are not suitable for processing such large and diverse data sets, as they are computationally intensive. Automated traffic monitoring systems have emerged as one of the key research areas in smart transportation that can reduce human workload and alert drivers to dangerous situations [471,472]. Video analytics of traffic patterns and behavior are increasingly becoming an essential part of such systems.

Two paradigms are being used to develop vision-based autonomous driving systems, which are crucial for the automation of cities: mediated perception approaches and behavior reflex approaches. Mediated perception techniques require calculating a high-dimensional representation of the world to detect various driving-relevant objects, such as side lanes, road traffic signs,

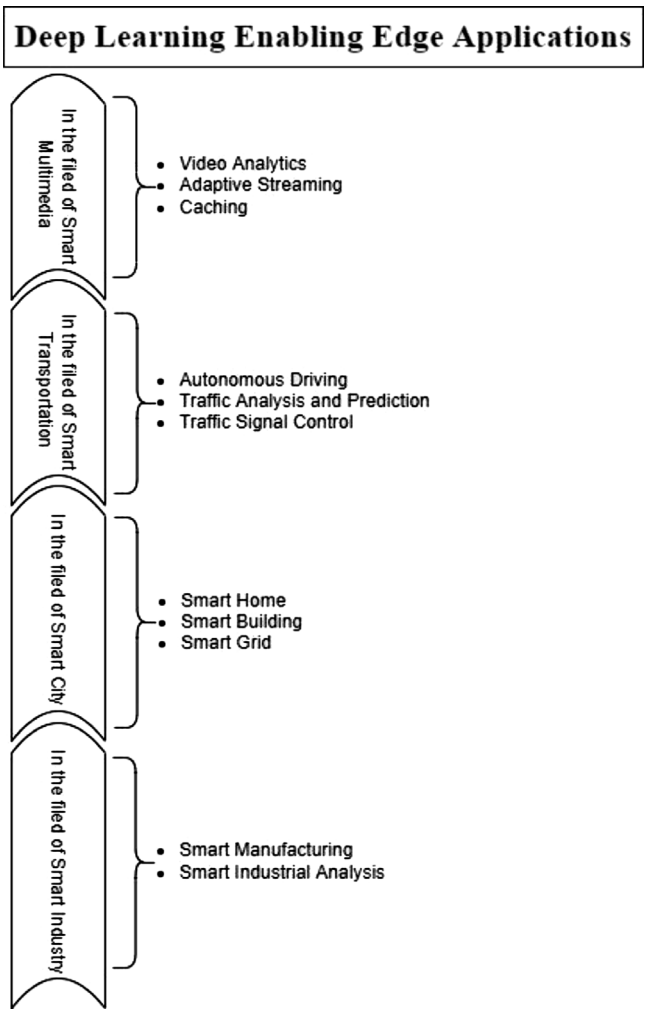


Fig. 54. Deep learning enabling edge application.

traffic lights, vehicles, and pedestrians crossing the road [473, 474].

The smart industry also referred to as Industry 4.0, is the latest stage in the evolution of manufacturing. By analyzing the vast amounts of data collected during production, computers, and manipulators can be endowed with human-like capabilities [454]. The primary focus of this section is on the various industrial applications of deep learning (as illustrated by Fig. 57). Different methods of visual inspection have been suggested, which utilize expert knowledge to identify relevant features for detecting defects in mass production. These techniques are crucial in ensuring that product quality is accurately evaluated [475]. Conventional machine learning methods form the basis of many of these approaches. However, deep learning has recently emerged as a practical method for visual inspection. The authors of the study [454] suggest a DL-based classification model for establishing a reliable inspection system. To implement smart manufacturing, a smart factory must monitor the health of its equipment, detect emerging issues, identify the root causes of failures, and integrate that knowledge into production output and control.

IoT is making significant impacts on various aspects of daily life, such as home automation, healthcare, and connected vehicles, as well as industries including logistics, automation, production processes, and security. Table 11 illustrates some of the recent developments in the IoT domain.

6. Challenges and opportunities

There are still several challenges in achieving secure, reliable, and practical usage, compelling continued efforts in this domain, even though the integration of different computing paradigms and deep learning has unveiled excellent potential and sparked the rapid growth of many applications. So let us now have a glance at the problems and difficulties that various computing models present.

Despite its existence for a while, cloud computing still faces several issues, including security, confidentiality, availability, and sustainability [488]. The dependability of cloud services is also a concern, as it could be catastrophic if one of the few data centers that provide essential services goes down [489]. The operation of cloud data centers requires large amounts of energy, making it essential to use resource provision optimization methods to reduce energy consumption. Difficulties faced by the cloud networking infrastructure include network utilization, data congestion, and cloud federation, among others, as outlined in [490]. With the emergence of IoT devices, the focus shifted to minimizing resource and energy consumption, and significant challenges included extending the battery life or improving the energy efficiency of smart devices [491]. The security of IoT devices and the protection of sensitive data collected by these devices present significant challenges [492]. IoT network challenges include availability, reliability, scalability, and interoperability.

Edge computing is a term used to describe the process of moving computing resources to the edge of a network. This shift presents a variety of challenges, including the need to ensure that edge devices can be programmed effectively, establish naming conventions for a large number of devices, and address security and privacy concerns, as well as issues surrounding data abstraction, service management, and optimization [493]. Fog computing is also in its nascent stages and has several unresolved challenges. It faces similar constraints to edge computing, such as “ensuring programmability, managing heterogeneous systems, addressing security, interoperability, mobility, scalability, federation, and improving resource and energy efficiency [35, 494]”.

Fog computing is a more comprehensive model compared to similar paradigms, as it operates in the Thing-to-Cloud continuum. It has a broader scope and can be considered a general model. Fig. 58 demonstrates the relationship between cloud, edge, and fog computing [495]. With the adoption of the OpenFog Reference Architecture by the IEEE Standard, it is evident that fog computing will soon be able to enhance connected devices operating under open standards [496]. Consequently, the focus of the remainder of the discussion will be on the issues and potential future scope in fog computing.

However, the main challenge in real-time applications of deep learning models in these computing paradigms is the need for low latency and high throughput. Real-time applications require quick response times, which can be affected by the time it takes for data to be transmitted to and processed by the deep learning model. To address this challenge, various techniques such as model compression, quantization, and pruning can be used to reduce the size and complexity of deep learning models, enabling them to be deployed on devices with limited processing power and memory. Additionally, hardware accelerators, such as GPUs and FPGAs, can be used to speed up the computation of deep learning models. The paper has previously highlighted the importance of distributed computing and edge computing in reducing the latency of deep learning applications. By distributing the computation of the deep learning model across multiple devices, the overall response time can be reduced, making real-time

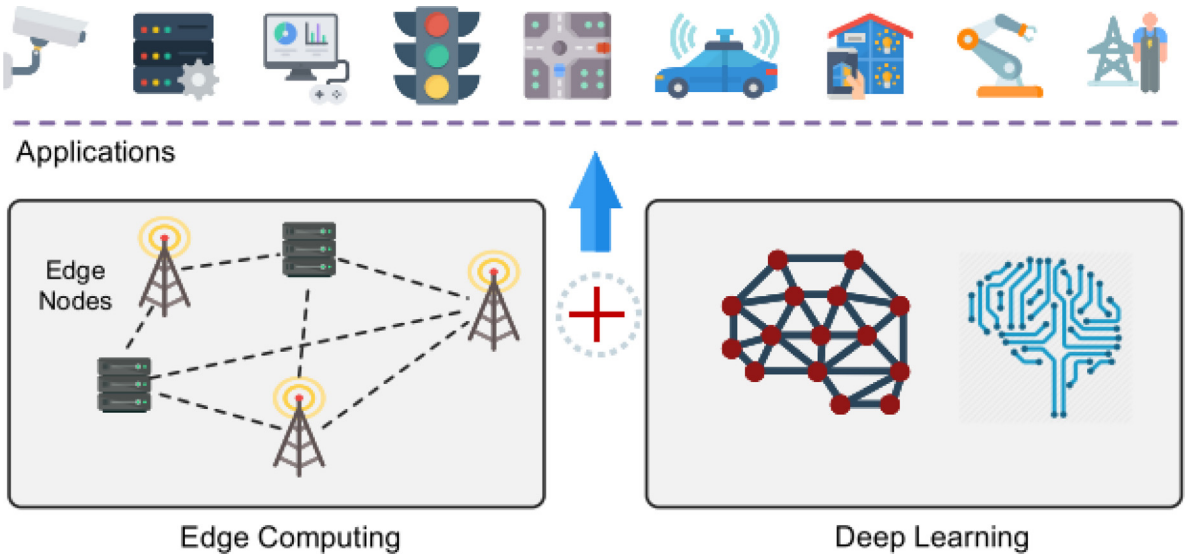


Fig. 55. Example of edge computing applications fused with deep learning.

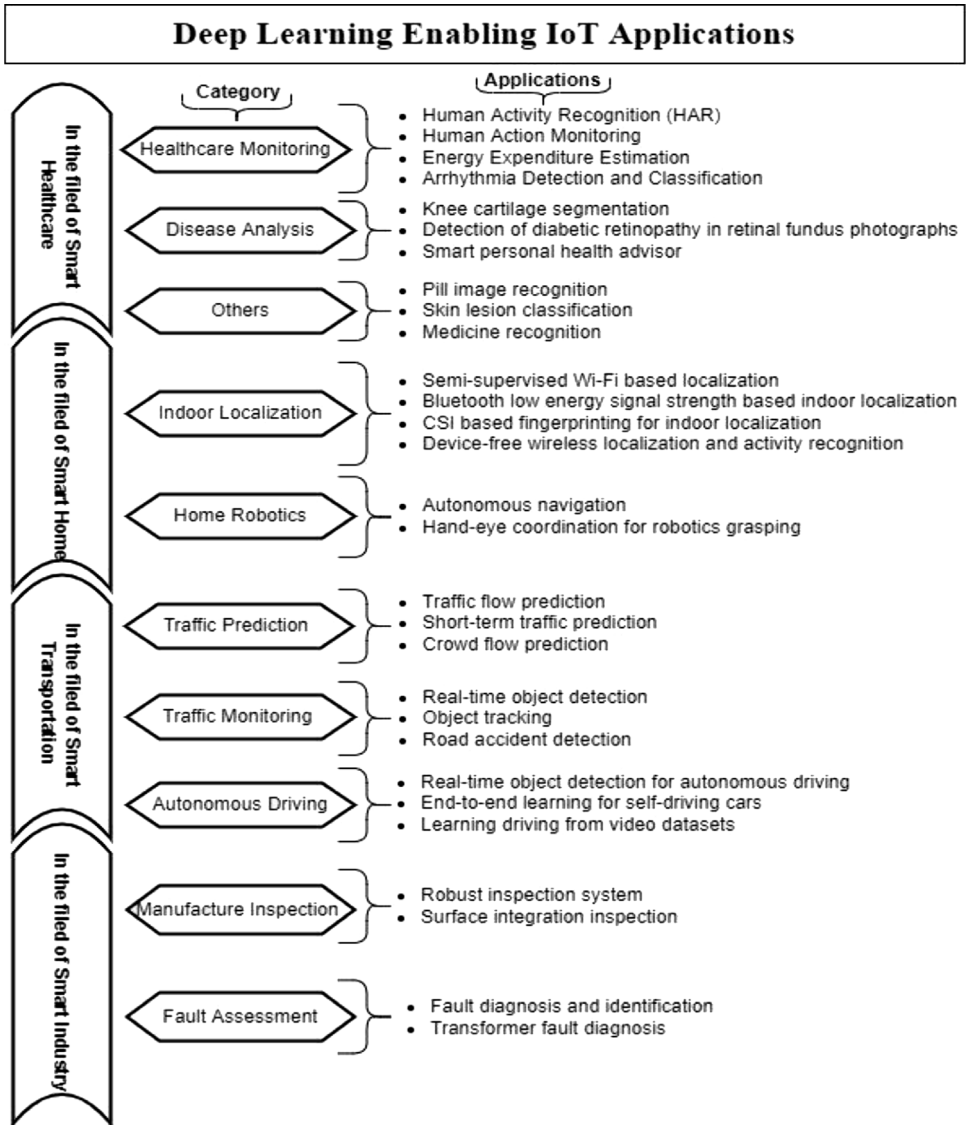


Fig. 56. Deep learning enabling IoT application.

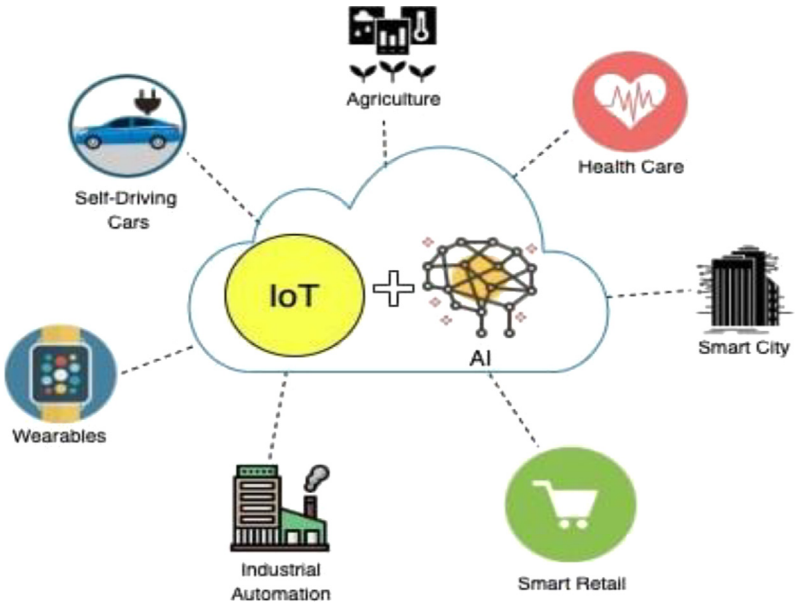


Fig. 57. Industrial applications of deep learning.

Table 11
Recent advancements in machine learning techniques for the internet of things.

Source	The machine Learning (ML) technique used	Parameters are taken into account	Focused field of interest	Application
[476]	Deep Learning	False alarm rate/2.2%.	Data Classification, Data Quality	IoT
[477]	Deep Learning, Convolution Neural Networks-Long–Short-Term Memory (CNN-LSTM)	Accuracy, Throughput gain, the Access rate	Cellular Networks	Wireless Communication
[478]	Long–short-term recurrent neural network	Error Rate Vs. Smart Bins, Processing Time, Accuracy	Waste Management	Smart City
[479]	Decision Tree, Random Forest, Naive Bayes.	Precision, Accuracy, Error Rate.	Cost Saving, Communication Quality	IoT Ecosystem
[480]	Artificial Neural Network	Time, Cost, Performance, and Stability	Extracting Features	IoT
[481]	Decision Trees	FPR, Recall, AUCPR, and F-Score	Anomaly Detection	IoT
[482]	Reinforcement Learning	No. of Alive Nodes vs. Time, Delay, Residual Energy.	Energy Optimization	Wireless Networks
[483]	Clustering	F1-Score	Supervised Techniques	IoT
[484]	SVM, DT, LSTM, KNN	True Positive Rate, False Positive Rate, Tree Confusion Matrix	Web documents Classification	Authentic News Data
[485]	Deep Learning	Delay, storage space, throughput, and accuracy.	Deep Boltzmann machine-based classification	IoT
[486]	DT, SVM, LR	Data Size, Time Cost, AUC	Social media messages	Social Media
[487]	Cell-free IoT	Energy Efficiency, Circuit, and Noise Power, Throughput.	Resource Optimization	IoT

applications feasible. Overall, the challenge in real-time applications of deep learning models in cloud, edge, fog, and IoT computing paradigms can be addressed through a combination of model optimization techniques, hardware accelerators, and distributed computing.

7. Future research directions

The potential technical advancements that could drive the future of cloud, fog, and edge computing paradigms, with the integration of deep learning, can be summarized as follows:

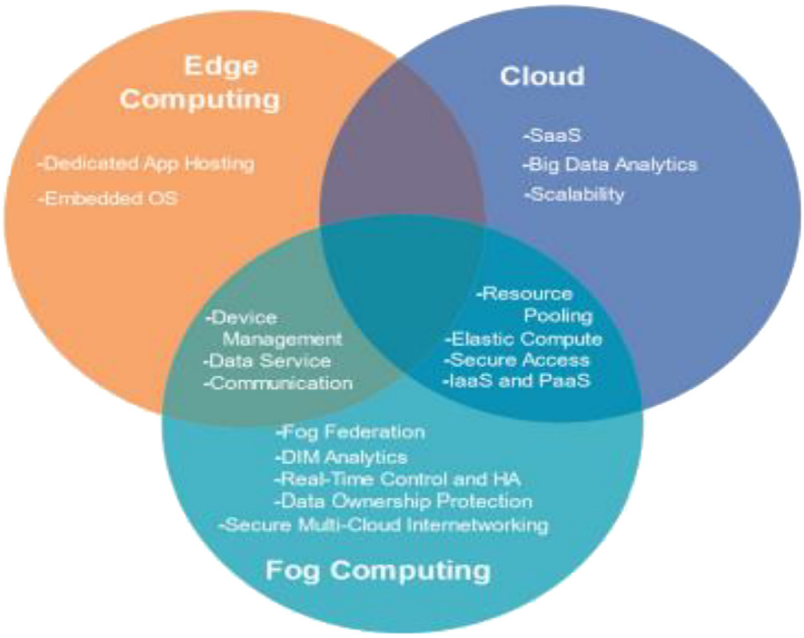


Fig. 58. Alliance between cloud, fog, and edge computing.

(a) **Hybrid cloud architectures:** Hybrid cloud architectures that combine public and private cloud resources will become more prevalent in the future. The research will focus on developing new hybrid cloud architectures that can support the needs of different types of applications and workloads [497,498].

(b) **Security and privacy:** As the amount of data being generated by cloud, fog, and edge computing systems continues to increase, security and privacy will become more important. Future research will focus on developing new security and privacy mechanisms that can protect data at rest and in transit [499–501].

(c) **Multi-access edge computing (MEC):** MEC is an emerging paradigm that brings cloud computing capabilities to the edge of the network [502]. Future research will focus on developing new MEC architectures that can support real-time applications and low-latency communication.

(d) **Intelligent resource management:** As cloud, fog, and edge computing systems become more complex, intelligent resource management will become more important [503]. Future research will focus on developing new algorithms and techniques that can optimize resource utilization and improve system performance [504].

(e) **Green computing**: Cloud, fog, and edge computing systems consume a lot of energy, and as the demand for these systems increases, the environmental impact will become more significant [505,506]. Future research will focus on developing new techniques for reducing energy consumption and making cloud, fog, and edge computing systems more environmentally friendly [507].

(f) **Data management:** As the amount of data being generated by cloud, fog, and edge computing systems continues to increase, data management will become more important [508–510]. Future research will focus on developing new data management techniques that can efficiently store, process, and analyze large datasets [511,512].

(g) **Edge intelligence:** Edge intelligence is a new paradigm that focuses on developing intelligent devices that can process data locally. Future research will focus on developing new algorithms and techniques that can enable intelligent devices to learn from their environment and make autonomous decisions [418,513–515].

8. Conclusion

In conclusion, the development of cloud computing has fundamentally changed how devices communicate with one another over the Internet, culminating in the rise of the Internet of Things and a myriad of linked devices capable of detecting and responding to human needs. The advent of computing paradigms like edge and fog computing was required by the issues given by the proliferation of networked IoT devices and the large data they generate for the traditional cloud computing architecture. By combining these paradigms with cloud computing, it may be possible to overcome some of the drawbacks of the current architecture and advance time-sensitive applications that need higher accuracy, lower latency, high-speed analytics, quicker response times, and increased reliability and availability. The fusion of fog/edge and cloud computing can provide several advantages for IoT, but these concepts are still in their early stages. This study explored the evolving computing paradigms, their challenges, and opportunities, highlighting the need for further research on this promising technology. In this study, the changing computer paradigms, together with the difficulties and opportunities, were examined. This thorough analysis can help budding researchers better understand recent developments in deep learning models for changing computing paradigms.

CRediT authorship contribution statement

Shahnawaz Ahmad: Conceptualization, Data curation, Investigation, Methodology, Resources, Writing – original draft. **Iman Shakeel:** Formal analysis, Investigation, Resources, Validation. **Shabana Mehfuz:** Conceptualization, Supervision, Validation, Visualization, Writing – review & editing. **Javed Ahmad:** Data curation, Formal analysis, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Consent for publication

All authors have read and agreed to the published version of the manuscript.

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